

during withdrawal from sedative, hypnotic, or anxiolytic taken as prescribed); **F15.921** amphetamine-type substance or other stimulant taken as prescribed; **F16.921** ketamine or other hallucinogen taken as prescribed or for medical reasons; **F19.921** for medications that do not fit into any of the classes (e.g., dexamethasone) and in cases in which a substance is judged to be an etiological factor but the specific class of substance is unknown (or **F19.931** if during withdrawal from medications that do not fit into any of the classes, taken as prescribed).

F05 Delirium due to another medical condition: There is evidence from the history, physical examination, or laboratory findings that the disturbance is attributable to the physiological consequences of another medical condition.

Coding note: Include the name of the other medical condition in the name of the delirium (e.g., F05 delirium due to hepatic encephalopathy). The other medical condition should also be coded and listed separately immediately before the delirium due to another medical condition (e.g., K72.90 hepatic encephalopathy; F05 delirium due to hepatic encephalopathy).

F05 Delirium due to multiple etiologies: There is evidence from the history, physical examination, or laboratory findings that the delirium has more than one etiology (e.g., more than one etiological medical condition; another medical condition plus substance intoxication or medication side effect).

Coding note: Use multiple separate codes reflecting specific delirium etiologies (e.g., K72.90 hepatic encephalopathy; F05 delirium due to hepatic failure; F10.231 alcohol withdrawal delirium). Note that the etiological medical condition both appears as a separate code that precedes the delirium code and is substituted into the delirium due to another medical condition rubric.

Recording Procedures

Substance intoxication delirium. The name of the substance intoxication delirium begins with the specific substance (e.g., cocaine) that is presumed to be causing the delirium. The diagnostic code is selected from the table included in the criteria set, which is based on the drug class and presence or absence of a comorbid substance use disorder. For substances that do not fit into any of the classes (e.g., dexamethasone), the code for “other substance” should be used; and in cases in which a substance is judged to be an etiological factor but the specific class of substance is unknown, the category “unknown substance” should be used.

When recording the name of the disorder, the comorbid substance use disorder (if any) is listed first, followed by the word “with,” followed by the name of the substance intoxication delirium, followed by the course (i.e., acute, persistent), followed by the specifier indicating level of psychomotor activity (i.e., hyperactive, hypoactive, mixed level of activity). For example, in the case of acute hyperactive intoxication delirium occurring in a man with a severe cocaine use disorder, the diagnosis is F14.221 severe cocaine use disorder with cocaine intoxication delirium, acute, hyperactive. A separate diagnosis of the comorbid severe cocaine use disorder is not given. If the intoxication delirium occurs without a comorbid substance use disorder (e.g., after a one-time heavy use of the substance), no accompanying substance use disorder is noted (e.g.,

F16.921 phencyclidine intoxication delirium, acute, hypoactive).

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Substance withdrawal delirium. The name of the substance withdrawal delirium begins with the specific substance (e.g., alcohol) that is presumed to be causing the withdrawal delirium. The diagnostic code is selected from substance-specific codes in the coding note included in the criteria set. When recording the name of the disorder, the comorbid substance use disorder (if any) is listed first, followed by the word “with,” followed by the substance withdrawal delirium, followed by the course (i.e., acute, persistent), followed by the specifier indicating level of psychomotor activity (i.e., hyperactive, hypoactive, mixed level of activity). For example, in the case of acute hyperactive withdrawal delirium occurring in a man with a severe alcohol use disorder, the diagnosis is F10.231 severe alcohol use disorder with alcohol withdrawal delirium, acute, hyperactive. A separate diagnosis of the comorbid severe alcohol use disorder is not given.

Medication-induced delirium. The name of the medication-induced delirium begins with the specific substance (e.g., dexamethasone) that is presumed to be causing the delirium. The name of the disorder is followed by the course (i.e., acute, persistent), followed by the specifier indicating level of psychomotor activity (i.e., hyperactive, hypoactive, mixed level of activity). For example, in the case of acute hyperactive medication-induced delirium occurring in a man using dexamethasone as prescribed, the diagnosis is F19.921 dexamethasone-induced delirium, acute, hyperactive.

Specifiers

Regarding course, in hospital settings, delirium usually lasts about 1 week, but some symptoms often persist even after individuals are discharged from the hospital.

Individuals with delirium may rapidly switch between hyperactive and hypoactive states. The hyperactive state may be more common or more frequently recognized and often is associated with medication side effects and drug withdrawal. The hypoactive state may be more frequent in older adults and is often unrecognized among older individuals in emergency departments and hospitals.

Diagnostic Features

The essential feature of delirium is an acute impairment of consciousness characterized by a disturbance in attention accompanied by reduced awareness of the environment, both core features of normal consciousness. Because these deficits reflect an altered state of consciousness affecting many higher cerebral cortical functions of the cerebral cortex, they are accompanied by a change from baseline in other cognitive functions that cannot be better explained by a preexisting or evolving neurocognitive disorder (NCD). The disturbance in attention (Criterion A) is manifested by reduced ability to direct, focus, sustain, and shift attention. Questions must be repeated because the individual’s attention wanders, or the individual may perseverate with an answer to a previous question rather than appropriately shift attention. The individual is easily distracted by irrelevant stimuli. The disturbance in awareness affects both internal thinking and insight as well as difficulty making sense of what is happening in the external environment.

The disturbance develops over a short period of time, usually hours to a few days, and tends

to fluctuate during the course of the day, often with worsening in the evening and night when external orienting stimuli decrease (Criterion B). There is evidence from the history, physical examination, or laboratory findings that the disturbance is a physiological consequence of an underlying medical condition, substance intoxication or withdrawal, use of a medication, or a toxin exposure, or a combination of these factors (Criterion E). The etiology should be coded according to the etiologically appropriate subtype (i.e., substance or medication intoxication, substance withdrawal, another medical condition, or multiple etiologies). Delirium often occurs in the context of an underlying NCD. The impaired brain function of individuals with mild and major NCD renders them more vulnerable to developing a delirium.

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There is an accompanying change in at least one other area that may include memory and learning (particularly recent memory), disorientation (particularly to time and place), alteration in language (particularly semantic comprehension), or perceptual distortion or a perceptual-motor disturbance (Criterion C). The perceptual disturbances accompanying delirium include misinterpretations, illusions, or hallucinations; these disturbances are typically visual, but may occur in other modalities as well, and range from simple and uniform to highly complex.

Normal attention/arousal, delirium, and coma lie on a continuum. *Coma* is defined as a state of unconsciousness with an absence of cognition or sleep-wake cycle, along with the lack of any meaningful response to verbal or physical stimuli. *Delirium* is an impaired state of consciousness in the setting of an aroused cortex. The ability to evaluate cognition to diagnose delirium depends on there being a level of cortical arousal and wakefulness sufficient for response to verbal stimulation; hence, delirium should not be diagnosed in the context of coma (Criterion D). Stuporous individuals also have a reduced level of brain arousal, but not to the extent of the complete unconsciousness of coma. Coma and stupor can be due to neurological conditions or drug-induced as with iatrogenic deep sedation in intensive care unit (ICU) settings or general anesthesia. Those individuals who show only minimal responses to verbal or physical stimulation are incapable of engaging with attempts at standardized testing or even interview. This inability to engage should be classified as a disorder of arousal such as coma or stupor, and not as delirium. However, delirium can be a stage that follows emergence from coma or stupor, especially when coma is the result of a neurological condition. Further, the sleep-wake cycle disturbance characteristic of the circadian rhythm disturbance in delirium can interfere with full assessment of the individual if in a sleep phase, which should be distinguished from a disorder of brain arousal.

Associated Features

Delirium is often associated with a disturbance in the sleep-wake cycle. This disturbance can include daytime sleepiness, nighttime agitation, difficulty falling asleep, excessive sleepiness throughout the day, or wakefulness throughout the night. In some cases, complete reversal of the night-day sleep-wake cycle can occur. Sleep-wake cycle disturbances are very common in delirium and have been proposed as a core criterion for the diagnosis.

The individual with delirium may exhibit emotional disturbances, such as anxiety, fear, depression, irritability, anger, euphoria, and apathy. There may be rapid and unpredictable shifts

from one emotional state to another. The disturbed emotional state may also be evident in calling out, screaming, cursing, muttering, moaning, or making other sounds. These behaviors are especially prevalent at night and under conditions in which stimulation and environmental cues are lacking.

Prevalence

The prevalence of delirium is highest among hospitalized older individuals and varies depending on the individuals' characteristics, setting of care, and sensitivity of the detection method. Data from the United States and Finland indicate that the prevalence of delirium in the community overall is low (1%–2%). The prevalence is 8%–17% in older individuals presenting to North American emergency departments, where the delirium often indicates a medical illness.

Based on data from various countries, the prevalence of delirium when individuals are admitted to the hospital ranges from 18% to 35%, and estimates of the occurrence of delirium arising during hospitalization range from 29% to 64% in general hospital populations. Internationally, delirium occurs in 11%–51% of older individuals postoperatively and in up to 81% of those in intensive care. The prevalence of delirium ranges from 20% to 22% in individuals in nursing homes or post-acute care settings and occurs in up to 88% of individuals with terminal illness at the end of life. Despite having higher risk factors for delirium, such as cardiovascular disease, sepsis, and respiratory failure, younger African Americans

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tended to have lower rates of the occurrence of delirium compared with White individuals of similar age in a large case series of ICU patients in the United States.

Development and Course

The majority of individuals with delirium have a full recovery with or without treatment, especially those who are not elderly. Delirium may progress to stupor, coma, seizures, or death, particularly if undetected and the underlying cause(s) remains untreated.

There is increasing evidence that delirium may be associated in long-term follow-up with cognitive decline or major NCD in the elderly, particularly in those with preexisting underlying cognitive impairment. Mortality among hospitalized individuals with delirium is high; as many as 38%–41% of individuals with delirium die within 1 year after diagnosis; the risk of death is particularly great among those with malignancies and other significant underlying medical illness.

Risk and Prognostic Factors

Delirium may be increased in the context of functional impairment, preexisting cognitive impairment, sensory impairment (e.g., vision/hearing), increasing age, illness severity or comorbidity, infection, depression, history of stroke, and history of alcohol use. Both major and mild NCDs can increase the risk for delirium and complicate the course. Falls may be an outcome of delirium but are not found to be a risk factor. In a meta-analysis of studies from 1990 through 2016, anticholinergic use was not a validated predictor of delirium.

Older individuals are especially susceptible to delirium compared with younger adults.

Among children, susceptibility to delirium in infancy and through childhood may be associated with significant childhood morbidity and mortality, whereas individuals in early adulthood through mid-adulthood may have less susceptibility to delirium and lower mortality risk.

Sex- and Gender-Related Diagnostic Issues

The symptoms associated with delirium may vary in men and women. Men more commonly manifest motor agitation and affective lability, whereas women more commonly manifest hypoactive delirium. Male sex is a risk factor for delirium, and sex- or gender-related factors may interact with other risk factors.

Diagnostic Markers

In addition to laboratory findings characteristic of underlying medical conditions (or intoxication or withdrawal states), there is often generalized irregular theta slowing on electroencephalography, and fast activity is occasionally found (e.g., in some cases of alcohol withdrawal delirium). However, electroencephalography is unable to detect slowing associated with delirium without comparison to premorbid baseline alpha rhythms unless the slowing is in the abnormal theta or delta frequency range.

Functional Consequences of Delirium

Delirium itself is associated with increased functional decline and risk of institutional placement. Hospitalized individuals 65 years or older with delirium are at greater risk for poor outcomes following discharge, including mortality, institutionalization, and dementia.

Differential Diagnosis

Psychotic disorders and bipolar and depressive disorders with psychotic features. Delirium that is characterized by vivid hallucinations, delusions, language disturbances, and agitation must be distinguished from brief psychotic disorder, schizophrenia,

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schizophreniform disorder, and other psychotic disorders, as well as from manic or major depressive episodes, with psychotic features.

Acute stress disorder. Delirium associated with fear, anxiety, and dissociative symptoms, such as depersonalization, must be distinguished from acute stress disorder, which is precipitated by exposure to a severely traumatic event.

Malingering and factitious disorder. Delirium can be distinguished from these disorders on the basis of the often atypical symptomatic presentation in malingering and factitious disorder and the absence of another medical condition or substance that is etiologically related to the apparent cognitive disturbance.

Other neurocognitive disorders. The most common differential diagnostic issue when evaluating confusion in older adults is disentangling symptoms of delirium and major NCD. The clinician must determine whether the individual has delirium; a delirium superimposed on a preexisting NCD, such as that due to Alzheimer's disease; or an NCD without delirium. The traditional

distinction between delirium and major NCD according to acuteness of onset and temporal course is particularly difficult in those elderly individuals who had a prior NCD that may not have been recognized, or who developed persistent cognitive impairment following an episode of delirium. When delirium and major NCD are comorbid, the management of the delirium should generally be given priority.

Other Specified Delirium

R41.0

This category applies to presentations in which symptoms characteristic of delirium that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for delirium or any of the disorders in the neurocognitive disorders diagnostic class. The other specified delirium category is used in situations in which the clinician chooses to communicate the specific reason that the presentation does not meet the criteria for delirium or any specific neurocognitive disorder. This is done by recording “other specified delirium” followed by the specific reason (e.g., “subsyndromal delirium”).

An example of a presentation that can be specified using the “other specified” designation is the following:

Subsyndromal delirium: A delirium-like presentation involving disturbances in attention, higher-level thought, and circadian rhythm, in which the severity of cognitive impairment falls short of that required for the diagnosis of delirium.

Unspecified Delirium

R41.0

This category applies to presentations in which symptoms characteristic of delirium that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for delirium or any of the disorders in the neurocognitive disorders diagnostic class. The unspecified delirium category is used in situations in which the clinician chooses *not* to specify the reason that the criteria are not met for delirium, and includes presentations for which there is insufficient information to make a more specific diagnosis (e.g., in emergency room settings).

Diagnostic Criteria

- A. Evidence of significant cognitive decline from a previous level of performance in one or more cognitive domains (complex attention, executive function, learning and memory, language, perceptual-motor, or social cognition) based on:
 - 1. Concern of the individual, a knowledgeable informant, or the clinician that there has been a significant decline in cognitive function; and
 - 2. A substantial impairment in cognitive performance, preferably documented by standardized neuropsychological testing or, in its absence, another quantified clinical assessment.
- B. The cognitive deficits interfere with independence in everyday activities (i.e., at a minimum, requiring assistance with complex instrumental activities of daily living such as paying bills or managing medications).
- C. The cognitive deficits do not occur exclusively in the context of a delirium.
- D. The cognitive deficits are not better explained by another mental disorder (e.g., major depressive disorder, schizophrenia).

Specify whether due to:

Note: Each subtype listed has specific diagnostic criteria and corresponding text, which follow the general discussion of major and mild neurocognitive disorders.

Alzheimer's disease

Frontotemporal degeneration

Lewy body disease

Vascular disease

Traumatic brain injury

Substance/medication use

HIV infection

Prion disease

Parkinson's disease

Huntington's disease

Another medical condition

Multiple etiologies

Unspecified etiology

Coding note: Code based on medical or substance etiology. In most cases of major neurocognitive disorder, there is need for an additional code for the etiological medical condition, which must immediately precede the diagnostic code for major neurocognitive disorder, as noted in the coding table on pp. 682–683.

Specify (see coding table for details):

Without behavioral disturbance: If the cognitive disturbance is not accompanied by any clinically significant behavioral disturbance.

With behavioral disturbance (*specify disturbance*): If the cognitive disturbance is accompanied by a clinically significant behavioral disturbance (e.g., psychotic symptoms, mood disturbance, agitation, apathy, or other behavioral symptoms).

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Coding note: Use additional code(s) to indicate clinically significant psychiatric symptoms due to the same medical condition causing the major neurocognitive disorder (e.g., **F06.2** psychotic disorder due to Alzheimer's disease, with delusions; **F06.32** depressive disorder due to Parkinson's disease, with major depressive-like episode). **Note:** Mental disorders due to another medical condition are included with disorders with which they share phenomenology (e.g., for depressive disorders due to another medical condition, see the chapter "Depressive Disorders").

Specify current severity:

Mild: Difficulties with instrumental activities of daily living (e.g., housework, managing money).

Moderate: Difficulties with basic activities of daily living (e.g., feeding, dressing).

Severe: Fully dependent.

Coding and Recording Procedures

The following are examples of coding and recording major neurocognitive disorders due to an etiological subtype (*for more information, see coding table on pp. 682–683 and coding notes in the specific diagnostic criteria for each major and mild neurocognitive disorder subtype*):

Major neurocognitive disorder due to probable Alzheimer's disease, without behavioral disturbance, mild: G30.9 Alzheimer's disease, F02.80 major neurocognitive disorder due to probable Alzheimer's disease, without behavioral disturbance, mild.

Major neurocognitive disorder due to traumatic brain injury, with behavioral disturbance, moderate: S06.2X9S diffuse traumatic brain injury with loss of consciousness of unspecified duration, sequela; F02.81 major neurocognitive disorder due to traumatic brain injury, with behavioral disturbance, moderate; F06.34 bipolar and related disorder due to traumatic brain injury, with mixed features.

Mild Neurocognitive Disorder

Diagnostic Criteria

- A. Evidence of modest cognitive decline from a previous level of performance in one or more cognitive domains (complex attention, executive function, learning and memory, language, perceptual-motor, or social cognition) based on:
 - 1. Concern of the individual, a knowledgeable informant, or the clinician that there has been a mild decline in cognitive function; and
 - 2. A modest impairment in cognitive performance, preferably documented by standardized neuropsychological testing or, in its absence, another quantified clinical assessment.
- B. The cognitive deficits do not interfere with capacity for independence in everyday activities (i.e., complex instrumental activities of daily living such as paying bills or managing medications are preserved, but greater effort, compensatory strategies, or accommodation may be required).
- C. The cognitive deficits do not occur exclusively in the context of a delirium.
- D. The cognitive deficits are not better explained by another mental disorder (e.g., major depressive disorder, schizophrenia).

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Specify whether due to:

Note: Each subtype listed has specific diagnostic criteria and corresponding text, which follow the general discussion of major and mild neurocognitive disorders.

Alzheimer's disease

Frontotemporal degeneration

Lewy body disease

Vascular disease

Traumatic brain injury

Substance/medication use

HIV infection

Prion disease

Parkinson's disease

Huntington's disease

Another medical condition

Multiple etiologies

Unspecified etiology

Coding note: For mild neurocognitive disorder due to any of the medical etiologies listed above, code **G31.84**. Do *not* use additional codes for the presumed etiological medical conditions. For substance/medication-induced mild neurocognitive disorder, code based on type of substance; see "Substance/Medication-Induced Major or Mild Neurocognitive Disorder." For unspecified mild neurocognitive disorder, code **R41.9**.

Specify (behavioral disturbance cannot be coded but should still be recorded):

Without behavioral disturbance: If the cognitive disturbance is not accompanied by any clinically significant behavioral disturbance.

With behavioral disturbance (specify disturbance): If the cognitive disturbance is accompanied by a clinically significant behavioral disturbance (e.g., psychotic symptoms, mood disturbance, agitation, apathy, or other behavioral symptoms).

Coding note: Use additional code(s) to indicate clinically significant psychiatric symptoms due to the same medical condition causing the mild neurocognitive disorder (e.g., **F06.2** psychotic disorder due to traumatic brain injury, with delusions; **F06.32** depressive disorder due to HIV disease, with major depressive-like episode). **Note:** Mental disorders due to another medical condition are included with disorders with which they share phenomenology (e.g., for depressive disorders due to another medical condition, see the chapter “Depressive Disorders”).

Coding and Recording Procedures

The following are examples of coding and recording mild neurocognitive disorders due to an etiological subtype (*for more information, see coding table on pp. 682–683 and coding notes in the specific diagnostic criteria for each major and mild neurocognitive disorder subtype*):

G31.84 Mild neurocognitive disorder due to Alzheimer’s disease, without behavioral disturbance.

G31.84 Mild neurocognitive disorder due to traumatic brain injury, with behavioral disturbance; F06.34 bipolar and related disorder due to traumatic brain injury, with mixed features.

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Etiological subtype	Associated etiological medical code for major neurocognitive disorder ^a	Major neurocognitive disorder code	Mild neurocognitive disorder code
Alzheimer’s disease	G30.9	F02.8x ^b	G31.84 ^c Do not use additional code for Alzheimer’s disease.
Frontotemporal degeneration	G31.09	F02.8x ^b	G31.84 ^c Do not use additional code for frontotemporal degeneration.
Lewy body disease	G31.83	F02.8x ^b	G31.84 ^c Do not use additional code for Lewy body disease.
Vascular disease	No additional medical code.	F01.5x ^b Do not use additional code for the vascular disease.	G31.84 ^c Do not use additional code for the vascular disease.
Traumatic brain injury	S06.2X9S	F02.8x ^b	G31.84 ^c Do not use additional code for the traumatic brain injury.
Substance/medication-induced	No additional medical code.	Code based on the type of substance causing the major	Code based on the type of substance causing the mild neurocognitive disorder. ^d

		causing the major neurocognitive disorder. ^d	
HIV infection	B20	F02.8x ^b	G31.84 ^c Do not use additional code for HIV infection.
Prion disease	A81.9	F02.8x ^b	G31.84 ^c Do not use additional code for prion disease.
Parkinson's disease	G20	F02.8x ^b	G31.84 ^c Do not use additional code for Parkinson's disease.
683	G10	F02.8x ^b	G31.84 ^c Do not use additional code for Huntington's disease.
Huntington's disease			
Due to another medical condition	Code the other medical condition first (e.g., G35 multiple sclerosis).	F02.8x ^b	G31.84 ^c Do not use additional codes for the presumed etiological medical conditions.
Due to multiple etiologies	Code all of the etiological medical conditions first (with the exception of vascular disease).	F02.8x ^b (code once for major neurocognitive disorder due to all etiologies that apply) Code also major vascular NCD (F01.5x), if present. Code also the relevant substance/medication-induced major neurocognitive disorders if substances or medications play a role in the etiology.	G31.84 ^c Do not use additional codes for the presumed etiological medical conditions. Code also the relevant substance/medication-induced mild neurocognitive disorders if substances or medications play a role in the etiology.
Unspecified neurocognitive disorder	No additional medical code.	R41.9 ^c	R41.9 ^c

^aCode first, before code for major neurocognitive disorder.

^bCode fifth character based on symptom specifier: .x0 without behavioral disturbance; .x1 with behavioral disturbance (e.g., psychotic symptoms, mood disturbance, agitation, apathy, or other behavioral symptoms). **Note:** The severity specifiers "mild," "moderate," and "severe" cannot be coded for major neurocognitive disorder but should still be recorded.

^c**Note:** "With behavioral disturbance" and "without behavioral disturbance" cannot be coded but should still be recorded.

^dSee coding table in "Substance/Medication-Induced Major or Mild Neurocognitive Disorder" for ICD-10-CM code. **Note:** The severity specifiers "mild," "moderate," and "severe" (for substance/medication-induced major neurocognitive disorder) and the accompanying symptom specifiers "with behavioral disturbance" and "without behavioral disturbance" (for substance/medication-induced major or mild neurocognitive disorder) cannot be coded but should still be recorded.

Subtypes

Major and mild neurocognitive disorders (NCDs) are primarily subtyped according to the known or presumed etiological/pathological entity or entities underlying the cognitive decline. These subtypes are distinguished on the basis of a combination of time course, characteristic domains affected, and associated symptoms. For certain etiological subtypes, the diagnosis depends substantially on the presence of a potentially causative entity, such as Parkinson's or

Huntington's disease, or a traumatic brain injury or stroke in the appropriate time period. For other etiological subtypes (generally the neurodegenerative diseases like Alzheimer's disease, frontotemporal degeneration, and Lewy body disease), the diagnosis is based primarily on the cognitive, behavioral, and functional symptoms. Typically, the differentiation among these syndromes that lack an independently recognized etiological entity is clearer at the level of major NCD than at the level of mild NCD, but sometimes characteristic symptoms and associated features are present at the mild level as well.

NCDs are frequently managed by clinicians in multiple disciplines. For many subtypes, multidisciplinary international expert groups have developed specialized consensus criteria based on clinicopathological correlation with underlying brain pathology. The subtype criteria here have been harmonized with those expert criteria.

Specifiers

Evidence for distinct behavioral features in NCDs has been recognized, particularly in the areas of psychotic symptoms and depression. Psychotic features are common in many NCDs, particularly at the mild-to-moderate stage of major NCDs due to Alzheimer's disease, Lewy body disease, and frontotemporal degeneration. If the psychotic symptoms are judged to be due to the Alzheimer's disease, Lewy body disease, or frontotemporal degeneration, an additional diagnosis of psychotic disorder due to Alzheimer's disease, psychotic disorder due to Lewy body disease, or psychotic disorder due to frontotemporal degeneration may be given. Paranoia and other delusions are common features, and often a persecutory theme may be a prominent aspect of delusional ideation. In contrast to psychotic disorders with onset in earlier life (e.g., schizophrenia), disorganized speech and disorganized behavior are not characteristic of psychosis in NCDs. Hallucinations may occur in any modality, although visual hallucinations are more common in NCDs than in depressive, bipolar, or psychotic disorders.

Mood disturbances, including depression, anxiety, and elation, may occur. Depression is common early in the course (including at the mild NCD level) of NCD due to Alzheimer's disease and Parkinson's disease, while elation may occur more commonly in frontotemporal degeneration. If the mood disturbance is judged to be due to the Alzheimer's disease, Parkinson's disease, or frontotemporal degeneration, an additional diagnosis of depressive disorder due to Alzheimer's disease, depressive disorder due to Parkinson's disease, or bipolar and related disorder due to frontotemporal degeneration may be given. Mood symptoms are increasingly recognized to be a significant feature in the earliest stages of mild NCDs such that clinical recognition and intervention may be important.

Agitation is common in a wide variety of NCDs, particularly in major NCD of moderate to severe severity, and often occurs in the setting of confusion or frustration. It may arise as combative behaviors, particularly in the context of resisting caregiving duties such as bathing and dressing. Agitation is characterized as disruptive motor or vocal activity and tends to occur with advanced stages of cognitive impairment across all of the NCDs.

Individuals with NCD can present with a wide variety of behavioral symptoms that are the focus of treatment. Sleep disturbance is a common symptom that can create a need for clinical attention and may include symptoms of insomnia, hypersomnia, and circadian rhythm disturbances.

Apathy is common in mild and major NCD. It is observed particularly in NCD due to Alzheimer's disease and may be a prominent feature of NCD due to frontotemporal degeneration. Apathy is typically characterized by diminished motivation and reduced goal-directed behavior accompanied by decreased emotional responsiveness. Symptoms of apathy may manifest early in the course of NCDs when a loss of motivation to pursue daily activities or hobbies may be observed.

Other important behavioral symptoms include wandering, disinhibition, hyperphagia, and hoarding. Some of these symptoms are characteristic of specific disorders, as discussed in the relevant sections. When more than one behavioral disturbance is observed, each type should be noted in writing with the specifier "with behavioral disturbance."

Diagnostic Features

Major and mild NCDs exist on a spectrum of cognitive and functional impairment. Major NCD roughly corresponds to the condition labeled in ICD-10 and ICD-11 (as well as in DSM-IV) as *dementia*. The core feature of NCDs is acquired cognitive decline in one or more cognitive domains (Criterion A) based on both 1) a concern about cognition on the part of the individual, a knowledgeable informant, or the clinician, and 2) performance on an objective assessment that falls below the expected level or that has been observed to decline over time. Both a concern and objective evidence are required because they are complementary. When there is an exclusive focus on objective testing, a disorder may go undiagnosed in high-functioning individuals whose currently "normal" performance actually represents a substantial decline in abilities, or an illness may be incorrectly diagnosed in individuals whose currently "low" performance does not represent a change from their own baseline or is a result of extraneous factors like test conditions or a passing illness. Alternatively, excessive focus on subjective symptoms may fail to diagnose illness in individuals with poor insight, or whose informants deny or fail to notice their symptoms, or it may be overly sensitive in the so-called worried well.

A cognitive concern differs from a complaint in that it may or may not be voiced spontaneously. Rather, it may need to be elicited by careful questioning about specific symptoms that commonly occur in individuals with cognitive deficits (see [Table 1](#) in the introduction to this chapter). For example, memory concerns include difficulty remembering a short grocery list or keeping track of the plot of a television program; executive concerns include difficulty resuming a task when interrupted, organizing tax records, or planning a holiday meal. At the mild NCD level, the individual is likely to describe these tasks as being more difficult or as requiring extra time or effort or compensatory strategies. At the major NCD level, such tasks may only be completed with assistance or may be abandoned altogether. At the mild NCD level, individuals and their families may not notice such symptoms or may view them as normal, particularly in the elderly; thus, careful history taking is of paramount importance. The difficulties must represent changes rather than lifelong patterns: the individual or informant may clarify this issue, or the clinician can infer change from prior experience with the individual or from occupational or other clues. It is also critical to determine that the difficulties are related to cognitive loss rather than to motor or sensory limitations.

Neuropsychological testing, with performance compared with norms appropriate to the

individual's age, sex, educational attainment, and cultural background, is part of the standard evaluation of NCDs and is particularly critical in the evaluation of mild NCD. The use of culturally validated assessment instruments is preferred, which are available for many racial/ethnic and linguistic populations. For major NCD, performance is typically 2 or more standard deviations below appropriate norms (3rd percentile or below). For mild NCD, performance typically lies in the 1–2 standard deviation range (between the 3rd and 16th percentiles). However, neuropsychological testing is not available in all settings, and neuropsychological thresholds are sensitive to the specific test(s) and norms employed, as well as to test conditions, sensory limitations, and intercurrent illness. A variety of brief

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office-based or “bedside” assessments, as described in [Table 1](#), can also supply objective data in settings where such testing is unavailable or infeasible. In any case, as with cognitive concerns, objective performance must be interpreted in light of the individual's prior performance. Optimally, this information would be available from a prior administration of the same test, but often it must be inferred based on appropriate norms, along with the individual's educational history, occupation, and other factors. Norms are more challenging to interpret in individuals with very high or very low levels of education and in individuals being tested outside their own language or cultural background.

Criterion B relates to the individual's level of independence in everyday functioning. Individuals with major NCD will have impairment of sufficient severity so as to interfere with independence, such that others will have to take over tasks that the individuals were previously able to complete on their own. Individuals with mild NCD will have preserved independence, although there may be subtle interference with function or a report that tasks require more effort or take more time than previously.

The distinction between major and mild NCD is inherently arbitrary, and the disorders exist along a continuum. Precise thresholds are therefore difficult to determine. Careful history taking, observation, and integration with other findings are required, and the implications of making a diagnosis should be considered when an individual's clinical manifestations lie at a boundary.

Associated Features

Typically the associated features that support a diagnosis of major or mild NCD will be specific to the etiological subtype (e.g., neuroleptic sensitivity and visual hallucinations in NCD due to Lewy body disease). Diagnostic features specific to each of the subtypes are found in the relevant sections.

Prevalence

The prevalence of NCD varies widely by age and by etiological subtype. Overall prevalence estimates are generally only available for older populations. Among individuals older than 60 years, prevalence increases steeply with age, so prevalence estimates are more accurate for narrow age bands than for broad categories such as “over 65” (where the mean age can vary greatly with the life expectancy of the given population). For those etiological subtypes occurring across the life span, prevalence estimates for NCD are likely to be available, if at all,

only as the fraction of individuals who develop NCD among those with the relevant condition (e.g., traumatic brain injury, HIV infection).

Female gender is associated with higher prevalence of dementia overall, and especially Alzheimer's disease, but this difference is largely, if not wholly, attributable to greater longevity in females.

Overall, international prevalence estimates for dementia (which is largely congruent with major NCD) are approximately 1%–2% at age 65 years and as high as 30% by age 85 years. The prevalence of mild NCD is very sensitive to the definition of the disorder, particularly in community settings, where evaluations are less detailed. In addition, in contrast with clinical settings, where cognitive concern must be high to seek and locate care, there may be a less clear decline from baseline functioning. Estimates of the prevalence of mild cognitive impairment (which is substantially congruent with mild NCD) among older individuals are fairly variable, ranging from 2% to 10% at age 65 and 5% to 25% by age 85.

Prevalence and incidence of dementia vary cross-nationally and among ethnic and racialized populations in the United States, although methodological differences complicate rate comparisons. Some U.S. studies found that incidence is highest in African Americans followed, in decreasing order, by American Indians/Alaska Natives, Latinx, Pacific Islanders, non-Latinx Whites, and Asian Americans. Among four Asian American populations, Filipino Americans had the highest incidence, followed by Japanese Americans, Chinese

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Americans, and Asian-Indian Americans. Latinx subpopulations in the United States have been found to vary considerably in prevalence and incidence of dementia; Caribbean Hispanics have much higher rates than Mexican Americans in some U.S. studies.

Development and Course

The course of NCD varies across etiological subtypes, and this variation can be useful in differential diagnosis. Some subtypes (e.g., those related to traumatic brain injury or stroke) typically begin at a specific time and (at least after initial symptoms related to inflammation or swelling subside) remain static. Others may fluctuate over time (although if this occurs, the possibility of delirium superimposed on NCD should be considered). NCDs due to neurodegenerative diseases like Alzheimer's disease or frontotemporal degeneration typically are marked by insidious onset and gradual progression, and the pattern of onset of cognitive deficits and associated features helps to distinguish among them.

NCDs with onset in childhood and adolescence may have broad repercussions for social and intellectual development, and in this setting intellectual developmental disorder (intellectual disability) or other neurodevelopmental disorders may also be diagnosed to capture the full diagnostic picture and ensure the provision of a broad range of services. In older individuals, NCDs often occur in the setting of medical illnesses, frailty, and sensory loss, which complicate the clinical picture for diagnosis and treatment.

When cognitive loss occurs in youth to midlife, individuals and families are likely to seek care. NCDs are typically easiest to identify at younger ages, although in some settings malingering or factitious disorder may be a concern. Very late in life, cognitive symptoms may

not cause concern or may go unnoticed. In late life, mild NCD must also be distinguished from the more modest deficits associated with “normal aging,” although a substantial fraction of what has been ascribed to normal aging likely represents prodromal phases of various NCDs. In addition, it becomes harder to recognize mild NCD with age because of the increasing prevalence of medical illness and sensory deficits. It becomes harder to differentiate among subtypes with age because there are multiple potential sources of neurocognitive decline.

Risk and Prognostic Factors

Risk factors vary not only by etiological subtype but also by age at onset within etiological subtypes. Some subtypes are distributed throughout the life span, whereas others occur exclusively or primarily in late life. Even within the NCDs of aging, the relative prevalence varies with age: Alzheimer’s disease is uncommon before age 60 years, and the prevalence increases steeply thereafter, while the overall less common frontotemporal degeneration has earlier onset and represents a progressively smaller fraction of NCDs with age. The strongest risk factor for major and mild NCDs is age, primarily because age increases the risk of neurodegenerative and cerebrovascular disease.

Risk of NCDs varies by ethnic and racialized background and is associated with variation in risk of underlying diseases (e.g., hypertension, diabetes), predisposing conditions (e.g., head injury), environment (e.g., access to nutritious food, safe spaces for exercise), and other factors. For example, in the United States, African Americans and Latinx tend to be at higher risk for vascular dementia than Whites. Lower education and literacy are risk factors for NCDs that also can vary by ethnoracial group because of differential exposure to adverse social determinants of health.

Culture-Related Diagnostic Issues

Individuals’ and families’ level of awareness and concern about neurocognitive symptoms may vary across ethnic, racialized, and occupational groups. Cultural differences regarding whether decreased cognitive ability is seen as a normal part of aging (“normalization”) and in dementia-related stigma can delay families’ recognition of a problem and

decrease help seeking for individuals in the early stages of cognitive loss. For example, social stigma appears to be associated with underutilization of services for cognitive impairment among some underserved ethnic and racialized groups (e.g., Chinese Americans, Korean Americans).

Neurocognitive symptoms are more likely to be noticed, particularly at the mild level, in individuals who engage in complex occupational, domestic, or recreational activities. In addition, norms for neuropsychological testing tend to be available only for broad populations, and thus they may not be easily applicable to individuals with less than high school education or those being evaluated outside their primary language or culture. Culturally related diagnostic challenges include accounting for intraethnic variation in the interpretation of assessments; evaluating the effect on neuropsychological testing of a) the test taker’s stereotype threat (i.e., anxiety from concerns that he or she will confirm the negative stereotype of the ethnic or racialized group by underperforming) and/or b) the clinician’s implicit (unconscious) bias on test

interpretation; and selecting the appropriate language when assessing bilingual individuals.

Bilingual individuals with dementia may lose their facility with acquired nonnative languages, which might affect their ability to communicate with caregivers. The caregiving environment may be influenced by cultural norms of family responsibility to care for the elderly, for example, by affecting the decision whether to care for the elder with NCD at home or in a care facility. In some cultures, adult children are expected to provide care for their older parents (e.g., filial piety) so that a functional limitation may not be as obvious to the dependent elder or the family.

Sex- and Gender-Related Diagnostic Issues

Some studies show that men and women experience major and mild NCD differently. Sex- and gender-related factors may influence incidence and prevalence, the etiology (risk and protective factors), and the clinical manifestations of major and mild NCD. More women than men experience major NCD because of their longer life span. Thus, a woman of a given age has a higher cumulative risk of developing major NCD before death than a man of the same age. The difference in incidence rates is less clear and may vary across populations and over time because of gender-related factors (e.g., education, occupation, family role, stress). For example, the incidence of dementia in several higher-income countries has declined in the past 30 years, and the decline was different in men and women across countries. Women tend to express a broader range of symptoms. In particular, women tend to manifest more psychiatric symptoms such as depression, anxiety, and delusions. Men tend to manifest more aggression, apathy, and vegetative symptoms.

Like age, culture, and occupation, sex and gender issues may affect the level of concern and awareness of cognitive symptoms. In addition, for late-life NCDs, women are likely to be older, to have more medical comorbidity, and to live alone, which can complicate evaluation and treatment. In addition, there are sex and gender differences in the frequency of some of the etiological subtypes.

Diagnostic Markers

In addition to a careful history, neuropsychological assessments are the key measures for diagnosis of NCDs, particularly at the mild level, where functional changes are minimal and symptoms more subtle. Ideally, individuals will be referred for formal neuropsychological testing, which will provide a quantitative assessment of all relevant domains and thus help with diagnosis; provide guidance to the family on areas where the individual may require more support; and serve as a benchmark for further decline or response to therapies. When such testing is unavailable or not feasible, the brief assessments in [Table 1](#) can provide insight into each domain. More global brief mental status tests may be helpful but may be

insensitive, particularly to modest changes in a single domain or in those with high premorbid abilities, and may be overly sensitive in those with low premorbid abilities.

In distinguishing among etiological subtypes, additional diagnostic markers may come into play, particularly neuroimaging studies such as magnetic resonance imaging scans and positron

emission tomography scans. In addition, specific markers may be involved in the assessment of specific subtypes and may become more important as additional research findings accumulate over time, as discussed in the relevant sections.

Association With Suicidal Thoughts or Behavior

Large-scale studies indicate elevated rates of suicidal behavior in individuals with NCD due to a variety of etiologies compared with persons without an NCD. A nationwide study in Taiwan reported that attempted suicide in late life is associated with subsequent dementia.

Functional Consequences of Major and Mild Neurocognitive Disorders

By definition, major and mild NCDs affect functioning, given the central role of cognition in human life. Thus, the criteria for the disorders, and the threshold for differentiating mild from major NCD, are based in part on functional assessment. Within major NCD there is a broad range of functional impairment, as implemented in the severity specifiers. In addition, the specific functions that are compromised can help identify the cognitive domains affected, particularly when neuropsychological testing is not available or is difficult to interpret.

Differential Diagnosis

Normal cognition. The differential diagnosis between normal cognition and mild NCD, as between mild and major NCD, is challenging because the boundaries are inherently arbitrary. Careful history taking and objective assessment are critical to these distinctions. A longitudinal evaluation using quantified assessments may be key in detecting mild NCD.

Delirium. Both mild and major NCD may be difficult to distinguish from a persistent delirium, which can co-occur. Careful assessment of attention and arousal will help to make the distinction.

Major depressive disorder. The distinction between mild NCD and major depressive disorder, which may co-occur with NCD, can also be challenging. Specific patterns of cognitive deficits may be helpful. For example, consistent memory and executive function deficits are typical of Alzheimer's disease, whereas nonspecific or more variable performance is seen in major depression. Alternatively, treatment of the depressive disorder with repeated observation over time may be required to make the diagnosis.

Specific learning disorder and other neurodevelopmental disorders. A careful clarification of the individual's baseline status will help distinguish an NCD from a specific learning disorder or other neurodevelopmental disorders. Additional issues may enter the differential for specific etiological subtypes, as described in the relevant sections.

Comorbidity

NCDs are common in older individuals and thus often co-occur with a wide variety of age-related diseases that may complicate diagnosis or treatment. Most notable of these is delirium, for which NCD increases the risk. In older individuals, a delirium during hospitalization is, in many cases, the first time that an NCD is noticed, although a careful history will often reveal evidence of earlier decline. Mixed NCDs are also common in older

individuals, as many etiological entities increase in prevalence with age. In younger individuals, NCD often co-occurs with neurodevelopmental disorders; for example, a head injury in a preschool child may also lead to significant developmental and learning issues. Additional comorbidity of NCD is often related to the etiological subtype, as discussed in the relevant sections.

Major or Mild Neurocognitive Disorder Due to Alzheimer's Disease

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. There is insidious onset and gradual progression of impairment in one or more cognitive domains (for major neurocognitive disorder, at least two domains must be impaired).
- C. Criteria are met for either probable or possible Alzheimer's disease as follows:

For major neurocognitive disorder:

Probable Alzheimer's disease is diagnosed if either of the following is present; otherwise, **possible Alzheimer's disease** should be diagnosed.

- 1. Evidence of a causative Alzheimer's disease genetic mutation from family history or genetic testing.
- 2. All three of the following are present:
 - a. Clear evidence of decline in memory and learning and at least one other cognitive domain (based on detailed history or serial neuropsychological testing).
 - b. Steadily progressive, gradual decline in cognition, without extended plateaus.
 - c. No evidence of mixed etiology (i.e., absence of other neurodegenerative or cerebrovascular disease, or another neurological, mental, or systemic disease or condition likely contributing to cognitive decline).

For mild neurocognitive disorder:

Probable Alzheimer's disease is diagnosed if there is evidence of a causative Alzheimer's disease genetic mutation from either genetic testing or family history.

Possible Alzheimer's disease is diagnosed if there is no evidence of a causative Alzheimer's disease genetic mutation from either genetic testing or family history, and all three of the following are present:

- 1. Clear evidence of decline in memory and learning.
- 2. Steadily progressive, gradual decline in cognition, without extended plateaus.

3. No evidence of mixed etiology (i.e., absence of other neurodegenerative or cerebrovascular disease, or another neurological or systemic disease or condition likely contributing to cognitive decline).
- D. The disturbance is not better explained by cerebrovascular disease, another neurodegenerative disease, the effects of a substance, or another mental, neurological, or systemic disorder.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to probable or possible Alzheimer's disease, with behavioral disturbance, code first **G30.9** Alzheimer's disease, followed by **F02.81**. For major neurocognitive disorder due to probable or possible Alzheimer's disease, without behavioral disturbance, code first **G30.9** Alzheimer's disease, followed by **F02.80**.

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Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild neurocognitive disorder due to Alzheimer's disease, code **G31.84**. (**Note:** Do *not* use the additional code for Alzheimer's disease. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder due to Alzheimer's disease: Use additional code(s) to indicate clinically significant psychiatric symptoms due to Alzheimer's disease (e.g., **F06.2** psychotic disorder due to Alzheimer's disease, with delusions; **F06.32** depressive disorder due to Alzheimer's disease, with major depressive-like episode).

Diagnostic Features

Beyond the neurocognitive disorder (NCD) syndrome (Criterion A), the core features of major or mild NCD due to Alzheimer's disease include an insidious onset and gradual progression of cognitive and behavioral symptoms (Criterion B). The typical presentation is amnesic (i.e., with impairment in memory and learning). Unusual nonamnesic presentations, particularly visuospatial and logopenic aphasic variants, also exist. A significant proportion of individuals, likely more than half, first present with behavioral symptoms before the onset of cognitive symptoms; the presence of behavioral disturbance should be noted using the appropriate specifier codes. At the mild NCD phase, Alzheimer's disease manifests typically with impairment in memory and learning, sometimes accompanied by deficits in executive function. At the major NCD phase, visuoconstructional/perceptual-motor ability and language (e.g., word retrieval) will also be impaired, particularly when the NCD is moderate to severe. Social cognition tends to be preserved until late in the course of the disease with the exception of individuals who have the less common variants with significant dysexecutive and behavioral disturbance.

A level of diagnostic certainty must be specified denoting Alzheimer's disease as the “probable” or “possible” etiology (Criterion C). *Probable Alzheimer's disease* is diagnosed in

both major and mild NCD if there is evidence of a causative Alzheimer's disease gene, either from genetic testing or from an autosomal dominant family history coupled with autopsy confirmation or a genetic test in an affected family member. At present, the designation of "probable" represents the highest level of diagnostic certainty within the current criteria framework. However, current developments in biomarkers continue to increase diagnostic certainty (e.g., when brain positron emission tomography [PET] scans may indicate the presence of Alzheimer's pathology, such as evidence of amyloid and/or tau deposition by imaging or cerebrospinal [CSF] analysis). For major NCD, a typical clinical picture, without extended plateaus or evidence of mixed etiology, can also be diagnosed as due to probable Alzheimer's disease. However, in some individuals there may be prolonged periods of very slow or minimal progression. For mild NCD, given the lesser degree of certainty that the deficits will progress, these features are only sufficient for a *possible* Alzheimer's etiology. As stated above, however, new biomarker methods may affect the use of "probable" and "possible" in mild NCD. If the etiology appears mixed, mild NCD due to multiple etiologies should be diagnosed. In any case, for both mild and major NCD due to Alzheimer's disease, the clinical features must not suggest another primary etiology for the NCD (Criterion D). As biomarker data continue to inform the nature of underlying pathologies, it is likely that the existence of multiple etiologies will be more systematically mapped in the future to better identify diagnostic variations in NCD due to multiple etiologies.

Associated Features

For individuals with NCD due to Alzheimer's disease, symptoms extend beyond cognitive deficits to include neuropsychiatric symptoms such as agitation, apathy, depression, delusions, and sleep disorders. Neuropsychiatric symptoms may also be described as

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behavioral and psychological symptoms of dementia and have been observed in neurocognitive disorders of all etiologies. These symptoms are nearly universal in Alzheimer's disease as confirmed in two U.S. population samples, with 5-year follow-up in one reporting that 98% of individuals with NCD due to Alzheimer's disease develop neuropsychiatric symptoms. Neuropsychiatric symptoms lead to disability, worsening quality of life, greater impairment in activities of daily living, faster cognitive and functional decline, greater caregiver burden, earlier institutionalization, and accelerated mortality. Neuropsychiatric symptoms are often more distressing than cognitive manifestations and are frequently the reason that health care assistance is sought. These symptoms are also frequently present at the mild NCD stage, with evidence suggesting that more than half of individuals who develop dementia begin with neuropsychiatric symptoms. At the mild NCD stage or the mildest level of major NCD, depression, irritability, and/or apathy are most often seen. With moderately severe major NCD, delusions, agitation, combativeness, and wandering are common. Late in the illness, gait disturbance, dysphagia, incontinence, myoclonus, and seizures are observed.

Prevalence

The prevalence of overall NCD due to Alzheimer's disease rises steeply with age. In high-

income countries, it ranges from 5% to 10% in individuals ages 60–69 years to at least 25% thereafter. An estimated 5.4 million Americans of all ages had dementia due to Alzheimer’s disease in 2016, including about 200,000 individuals with disease onset before age 65. Dementia due to Alzheimer’s disease is found in 11% of individuals age 65 and older and 32% of those age 85 and older. Estimates applying incidence rates of dementia due to Alzheimer’s disease to U.S. census data indicate that 81% of those with the disease are age 75 or older. The percentage of dementias attributable to Alzheimer’s disease ranges from about 60% to over 90%, depending on the setting and diagnostic criteria. Mild NCD due to Alzheimer’s disease is likely to represent a substantial fraction of mild cognitive impairment (MCI) as well.

Studies show that prevalence of dementia due to Alzheimer’s disease tends to vary by ethnoracial background; for example, in the United States, prevalence in individuals age 65 years and older ranges from 3.5% to 14.4%, depending on ethnoracial group, age, and assessment methodology. Higher prevalence has been found among African Americans and U.S. Latinx of Caribbean origin, after adjustment for gender and clinical comorbidities.

Development and Course

Major or mild NCD due to Alzheimer’s disease progresses gradually, at times with plateaus, through severe dementia to death. The mean duration of survival after diagnosis is approximately 10 years, reflecting the advanced age of the majority of individuals rather than the course of the disease; some individuals can live with the disease for as long as 20 years. Late-stage individuals are eventually mute and bedbound. Death most commonly results from aspiration in those who survive through the full course. In mild NCD due to Alzheimer’s disease, impairments increase over time, and functional status gradually declines until symptoms reach the threshold for the diagnosis of major NCD.

The onset of symptoms is usually at ages 70 through 89; early-onset forms seen in individuals ages 40–59 are often, but not always, related to known causative mutations. Symptoms and pathology do not differ much by onset ages. However, younger individuals are more likely to survive the full course of the disease, while older individuals are more likely to have numerous medical comorbidities that affect the course and management of the illness. Diagnostic complexity is higher in older adults because of the increased likelihood of comorbid medical illness and mixed pathology. Age at symptom onset, rate of cognitive decline, and survival rates appear to vary by ethnoracial background. For example, compared

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with non-Latinx Whites, U.S. Latinx can develop Alzheimer’s disease symptoms up to 4 years earlier, African Americans tend to show slower cognitive decline, and both underserved groups may have longer survival periods.

Risk and Prognostic Factors

A number of risk factors have been identified, including low educational status, midlife hypertension, obesity, and hearing loss, as well as late-life smoking, depression, physical inactivity, social isolation, and diabetes. The co-occurrence of multiple vascular risk factors also increases risk for Alzheimer’s disease and may act by increasing cerebrovascular pathology or

also through direct effects on Alzheimer's pathology. Traumatic brain injury, especially in men, may increase risk for major or mild NCD due to Alzheimer's disease, although this relationship remains controversial.

Genetic and physiological. Age is definitively the strongest risk factor for Alzheimer's disease, as the prevalence estimates demonstrate. A strong genetic predisposition (60%–80% of attributable risk) has been demonstrated. Rare mutations on chromosomes 1, 14, and 21 follow Mendelian inheritance, leading to autosomal dominant forms. Individuals with Down syndrome (trisomy 21) may develop Alzheimer's disease if they survive to midlife. The most common risk factors are polygenic, with more than 45 risk genes/loci having been identified, typically with small effects on risk. The strongest genetic susceptibility polymorphism, apolipoprotein E4 (*APOE*E4*), increases risk and decreases age at onset, particularly in homozygous individuals, although some homozygous individuals survive to advanced ages without developing symptoms.

Ethnoracial and national origin are related to the genetic susceptibility profile for Alzheimer's disease. While *APOE*E4* is associated with Alzheimer's disease risk, this association has not been consistently found across all ethnic and racialized groups. For example, some studies have identified a unique mutation in the *Gly206Ala* presenilin 1 gene among individuals of Puerto Rican descent with Alzheimer's disease, which is also related to early onset. Moreover, some studies have found a stronger association with *ABCA7*, a protein transporter gene, among individuals who identify as African American than among U.S. Whites.

Culture-Related Diagnostic Issues

Detection of an NCD may be more difficult in cultural and socioeconomic settings where memory loss is considered normal in old age, where older adults face fewer cognitive demands in everyday life, or where very low educational levels pose greater challenges to objective cognitive assessment.

Sex- and Gender-Related Diagnostic Issues

Women were found to have a higher incidence of Alzheimer's disease than men in several European studies, but the incidence was similar in men and women in most North American studies. Some studies suggested that the symptoms of dementia progress faster in women than in men. However, because women perform better than men of the same age on some verbal memory tests, it is also possible that gender differences reflect the cut-off scores of tests used to support a diagnosis. Different cut-off scores may be useful in men and women when assessing for mild cognitive impairment.

Diagnostic Markers

Amyloid-predominant neuritic plaques, tau-predominant neurofibrillary tangles, and neuronal loss observed microscopically or manifested in regional cortical atrophy (e.g., hippocampal, parietal, frontal) are hallmarks of the pathological diagnosis of Alzheimer's disease and may be confirmed via postmortem histopathological examination. For

early-onset cases with apparent autosomal dominant inheritance, a mutation in one of the known

causative Alzheimer's disease genes—amyloid precursor protein (*APP*), presenilin 1 (*PSEN1*), or presenilin 2 (*PSEN2*)—may be involved, and genetic testing for such mutations is commercially available, although usually without clinical utility. While *APOE E*4* cannot serve as a diagnostic marker because it is a risk factor (i.e., neither necessary nor sufficient for disease occurrence), in rare instances genetic testing at this locus may have utility in clinical settings.

Since amyloid beta-42 deposition in the brain occurs early in the pathophysiological cascade, amyloid-based diagnostic tests such as amyloid imaging on brain PET scans and reduced levels of amyloid beta-42 in the CSF may have diagnostic value. Similarly, tau PET imaging or CSF analyses for elevated total tau or phospho-tau levels are available for clinical use. Signs of neuronal injury, such as hippocampal and temporoparietal cortical atrophy on a magnetic resonance image scan and temporoparietal hypometabolism on a fluorodeoxyglucose PET scan, provide evidence of neuronal damage but are less specific for Alzheimer's disease. Most of these biomarkers have been validated and are widely available in tertiary care settings. Blood-derived biomarkers for Alzheimer's disease are being developed and are likely to become clinically available as diagnostic, prognostic, and theranostic indicators.

Association With Suicidal Thoughts or Behavior

Alzheimer's disease is associated with a moderate risk of suicide even many years after the diagnosis; thus, ongoing assessment of mood and suicidality is appropriate. A large population study in Denmark found that the risk of suicide in individuals with a hospital-determined diagnosis of dementia was three- to eightfold greater compared with persons without dementia. In contrast, several other studies found mixed results regarding suicide risk in individuals with Alzheimer's disease. A review of the neurobiology of suicide in the elderly found preliminary evidence of an association with cognitive deficits and elderly suicidal behavior, especially regarding impaired decision-making and reduced cognitive inhibition.

Functional Consequences of Major or Mild Neurocognitive Disorder Due to Alzheimer's Disease

Because of the effect on cognition, behavior, and functioning, NCD due to Alzheimer's disease has a serious and substantial impact on individuals, their caregivers, and families. Early in the disease course, memory loss, disorientation, and mood symptoms adversely impact independence, and create safety concerns (e.g., around driving). For individuals with onset at younger ages, NCD due to Alzheimer's disease can lead to early retirement. As the disease advances, individuals become increasingly disabled in instrumental and basic daily living activities, slowly becoming fully dependent on others. Caregivers for individuals with NCD due to Alzheimer's disease often see their social network deteriorate and develop a series of health and mental health problems that can adversely affect outcomes for both the caregiver and the individual with the NCD.

Differential Diagnosis

Other neurocognitive disorders. Major and mild NCDs due to other neurodegenerative processes (e.g., Lewy body disease, frontotemporal degeneration) share the insidious onset and gradual decline caused by Alzheimer's disease but have distinctive core features of their own (which are

not always present). For example, NCD with Lewy bodies is typically characterized by frequent fluctuations in cognition early in the disease, parkinsonian features, gait imbalances, and visual hallucinations. Individuals with frontotemporal NCD may present with a distinct behavioral or language variant. The behavioral variant typically

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first manifests with prominent changes in social behavior, such as disinhibition, apathy, or perseverative behavior, that may not infrequently lead to a primary psychiatric diagnosis. In contrast, the language variant of frontotemporal NCD may manifest with impairments in expressive language or word comprehension.

In major or mild vascular NCD, there is typically a history of stroke temporally related to the onset of cognitive impairment, and infarcts or hemosiderin deposits observed on brain imaging can be judged sufficient to account for the clinical picture. However, major or mild vascular NCD shares many clinical features with Alzheimer's disease; frequently Alzheimer's pathology is present alone or in combination with vascular pathologies. It should be noted that white matter change alone does not constitute enough evidence of cerebrovascular disease to propose a mixed etiology if the other diagnostic considerations support the diagnosis of NCD due to Alzheimer's disease. The presence of subcortical ischemic changes on neuroimaging must be interpreted carefully in view of whether concurrent Alzheimer's pathology is present.

Other concurrent, active neurological or systemic illness. Other neurological or systemic illness should be considered if there is an appropriate temporal relationship and severity to account for the clinical picture. At the mild NCD level, it may be difficult to distinguish an Alzheimer's disease etiology from that of another medical condition (e.g., thyroid disorders, vitamin B₁₂ deficiency).

Major depressive disorder. Particularly at the mild NCD level, the differential diagnosis also includes major depression. The presence of depression may be associated with reduced daily functioning and poor concentration that may resemble an NCD, but improvement with treatment of depression may be useful in making the distinction. If the symptoms meeting criteria for a major depressive episode are judged to be due to the physiological effects of Alzheimer's disease, a diagnosis of depressive disorder due to Alzheimer's disease, with major depressive-like episode, should be given instead of major depressive disorder.

Comorbidity

Most individuals with Alzheimer's disease are elderly and have multiple medical conditions that can complicate diagnosis and influence the clinical course. Major or mild NCD due to Alzheimer's disease commonly co-occurs with cerebrovascular disease, which contributes to the clinical picture. When a comorbid condition contributes to the NCD in an individual with Alzheimer's disease, then NCD due to multiple etiologies should be diagnosed.

Major or Mild Frontotemporal Neurocognitive Disorder

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. The disturbance has insidious onset and gradual progression.
- C. Either (1) or (2):

- 1. Behavioral variant:

- a. Three or more of the following behavioral symptoms:

- i. Behavioral disinhibition.
- ii. Apathy or inertia.
- iii. Loss of sympathy or empathy.

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- iv. Perseverative, stereotyped or compulsive/ritualistic behavior.

- v. Hyperorality and dietary changes.

- b. Prominent decline in social cognition and/or executive abilities.

- 2. Language variant:

- a. Prominent decline in language ability, in the form of speech production, word finding, object naming, grammar, or word comprehension.

- D. Relative sparing of learning and memory and perceptual-motor function.

- E. The disturbance is not better explained by cerebrovascular disease, another neurodegenerative disease, the effects of a substance, or another mental, neurological, or systemic disorder.

Probable frontotemporal neurocognitive disorder is diagnosed if either of the following is present; otherwise, **possible frontotemporal neurocognitive disorder** should be diagnosed:

- 1. Evidence of a causative frontotemporal neurocognitive disorder genetic mutation, from either family history or genetic testing.
- 2. Evidence of disproportionate frontal and/or temporal lobe involvement from neuroimaging.

Possible frontotemporal neurocognitive disorder is diagnosed if there is no evidence of a genetic mutation, and neuroimaging has not been performed.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to probable or possible frontotemporal degeneration, with behavioral disturbance, code first **G31.09** frontotemporal degeneration, followed by **F02.81**. For major neurocognitive disorder due to probable or possible frontotemporal degeneration, without behavioral disturbance, code first **G31.09** frontotemporal degeneration, followed by **F02.80**.

Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild neurocognitive disorder due to frontotemporal degeneration, code **G31.84**. (**Note:** Do *not* use the additional code for frontotemporal degeneration. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild frontotemporal neurocognitive disorder: Use additional code(s) to indicate clinically significant psychiatric symptoms due to frontotemporal degeneration (e.g., **F06.33** bipolar and related disorder due to frontotemporal degeneration, with manic features; **F07.0** personality change due to frontotemporal degeneration, disinhibited type).

Diagnostic Features

Major or mild frontotemporal neurocognitive disorder (NCD) comprises a number of syndromic variants characterized by the progressive development of behavioral and personality change and/or language impairment. The behavioral variant and two language variants (semantic and agrammatic/nonfluent) exhibit distinct patterns of brain atrophy and some distinctive neuropathology. The criteria must be met for either the behavioral or the language variant to make the diagnosis, but many individuals present with features of both.

Individuals with behavioral-variant major or mild frontotemporal NCD present with varying degrees of apathy or disinhibition. They may lose interest in socialization, self-care, and personal responsibilities, or display socially inappropriate behaviors. Insight is usually impaired, and this often delays medical consultation. The first referral is often to a psychiatrist. Individuals may develop changes in social style, and in religious and political

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beliefs, with repetitive movements, hoarding, changes in eating behavior, and hyperorality. In later stages, loss of sphincter control may occur. Cognitive decline is less prominent, and formal testing may show relatively few deficits in the early stages. Common neurocognitive symptoms are lack of planning and organization, distractibility, and poor judgment. Deficits in executive function, such as poor performance on tests of mental flexibility, abstract reasoning, and response inhibition, are present, but learning and memory are relatively spared, and perceptual-motor abilities are almost always preserved in the early stages.

Individuals with language-variant major or mild frontotemporal NCD present with primary progressive aphasia with gradual onset, with two subtypes commonly described: semantic variant and agrammatic/nonfluent variant; each variant has distinctive features and corresponding neuropathology. A third form of progressive language decline, called logopenic progressive aphasia, is associated with left temporoparietal dysfunction and is often caused by Alzheimer’s disease pathology.

“Probable” is distinguished from “possible” frontotemporal NCD by the presence of causative genetic factors (e.g., mutations in the gene coding for microtubule-associated protein tau) or by the presence of distinctive atrophy or reduced activity in frontotemporal regions on structural or functional imaging.

Associated Features

Extrapyramidal features may be prominent in some cases, with an overlap with syndromes such as progressive supranuclear palsy and corticobasal degeneration. Features of motor neuron disease may be present in some cases (e.g., muscle atrophy, weakness). A subset of individuals develop visual hallucinations.

Prevalence

Major or mild frontotemporal NCD is a common cause of early-onset NCD in individuals younger than 65 years. In international studies, population prevalence estimates are in the range of 2–31 per 100,000, with overall rates generally noted to be equal in men and women, although variation exists among studies. Approximately 20%–25% of cases of frontotemporal NCD occur in individuals older than 65 years. Frontotemporal NCD accounts for about 5% of all cases of dementia in unselected autopsy series. The behavioral variant is the most common presentation of NCD due to frontotemporal degeneration, occurring in approximately 60% of cases.

Development and Course

Individuals with major or mild frontotemporal NCD commonly present in their 50s, although the age at onset varies from the 20s through the 80s. The disease is gradually progressive, with median survival being 6–11 years after symptom onset and 3–4 years after diagnosis. Survival is shorter and decline is faster in major or mild frontotemporal NCD than in typical Alzheimer's disease.

Risk and Prognostic Factors

Genetic and physiological. Approximately 40% of individuals with major or mild frontotemporal NCD have a family history of early-onset NCD, and approximately 10% show an autosomal dominant inheritance pattern. A number of genetic factors have been identified, such as mutations in the gene encoding the microtubule-associated protein tau (*MAPT*), the granulin gene (*GRN*), and the C9ORF72 gene (*C9orf72*). A number of families with causative mutations have been identified (see the section “Diagnostic Markers” for this disorder), but many individuals with known familial transmission do not have a known mutation. The presence of motor neuron disease is associated with a more rapid deterioration.

Diagnostic Markers

Computed tomography (CT) or structural magnetic resonance imaging (MRI) may show distinct patterns of atrophy. In behavioral-variant major or mild frontotemporal NCD, both frontal lobes (especially the medial frontal lobes) and the anterior temporal lobes are atrophic. In semantic language-variant major or mild frontotemporal NCD, the middle, inferior, and anterior temporal lobes are atrophic bilaterally but asymmetrically, with the left side usually being more affected. Nonfluent language-variant major or mild frontotemporal NCD is associated with predominantly left posterior frontal-insular atrophy. Functional imaging demonstrates hypoperfusion and/or cortical hypometabolism in the corresponding brain regions, which may be present in the early stages in the absence of structural abnormality. Emerging biomarkers for Alzheimer's disease

(e.g., cerebrospinal fluid amyloid-beta and tau levels, and amyloid imaging) may help in the differential diagnosis, but the distinction from Alzheimer's disease can remain difficult.

In familial cases of frontotemporal NCD, the identification of genetic mutations may help confirm the diagnosis. Mutations associated with frontotemporal NCD include the genes encoding microtubule-associated protein tau (MAPT) and granulin (GRN), C9ORF72, transactive response DNA-binding protein of 43 kDa (TDP-43, or TARDBP), valosin-containing protein (VCP), chromatin modifying protein 2B (CHMP2B), and fused in sarcoma protein (FUS).

Functional Consequences of Major or Mild Frontotemporal Neurocognitive Disorder

Because of the relative early age at onset of the disorder, the disorder often affects workplace and family life. Because of the involvement of language and/or behavior, function is often more severely impaired relatively early in the course. For individuals with the behavioral variant, prior to diagnostic clarification there may be significant family disruption, legal involvement, and problems in the workplace because of socially inappropriate behaviors. The functional impairment attributable to behavioral change and language dysfunction, which can include hyperorality, impulsive wandering, and other disinhibited behaviors, may far exceed that attributable to the cognitive disturbance and may lead to nursing home placement or institutionalization. These behaviors can be severely disruptive, even in structured care settings, particularly when the individuals are otherwise healthy, nonfrail, and free of other medical comorbidities.

Differential Diagnosis

Other neurocognitive disorders. Other neurodegenerative diseases may be distinguished from major or mild frontotemporal NCD by their characteristic features. In major or mild NCD due to Alzheimer's disease, decline in learning and memory is an early feature. However, 10%–30% of individuals presenting with a syndrome suggestive of major or mild frontotemporal NCD are found at autopsy to have Alzheimer's disease pathology. This occurs more frequently in individuals who present with progressive dysexecutive syndromes in the absence of behavioral changes or movement disorder or in those with the logopenic variant.

In major or mild NCD with Lewy bodies, core and suggestive features of Lewy bodies must be present. In major or mild NCD due to Parkinson's disease, spontaneous parkinsonism emerges well before the cognitive decline. In major or mild vascular NCD, depending on affected brain regions, there may also be loss of executive function and behavioral changes such as apathy, and this disorder should be considered in the differential diagnosis. However, history of a cerebrovascular event is temporally related to the onset of cognitive impairment in major or mild vascular NCD, and neuroimaging reveals infarctions or white matter lesions sufficient to account for the clinical picture.

Other neurological conditions. Major or mild frontotemporal NCD overlaps with progressive supranuclear palsy, corticobasal degeneration, and motor neuron disease clinically as well as

pathologically. Progressive supranuclear palsy is characterized by supranuclear gaze palsies and axial-predominant parkinsonism. Pseudobulbar signs may be present, and retropulsion (losing balance in a backward direction) is often prominent. Neurocognitive assessment shows psychomotor slowing, poor working memory, and executive dysfunction. Corticobasal degeneration presents with asymmetric rigidity, limb apraxia, postural instability, myoclonus, alien limb phenomenon, and cortical sensory loss. Many individuals with behavioral-variant major or mild frontotemporal NCD show features of motor neuron disease, which tend to be mixed upper and predominantly lower motor neuron disease.

Other mental disorders and medical conditions. Behavioral-variant major or mild frontotemporal NCD may be mistaken for a primary mental disorder, such as major depression, bipolar disorders, or schizophrenia, and individuals with this variant often present initially to psychiatry. Over time, the development of progressive neurocognitive difficulties will help to make the distinction. A careful medical evaluation will help to exclude treatable causes of NCDs, such as metabolic disturbances, nutritional deficiencies, and infections. If the symptoms characteristic of a primary mental disorder (e.g., delusions) are judged to be due to the physiological effects of frontotemporal degeneration, a diagnosis of the appropriate mental disorder due to frontotemporal degeneration should be given instead of the primary psychotic disorder (e.g., psychotic disorder due to frontotemporal degeneration, with delusions).

Major or Mild Neurocognitive Disorder With Lewy Bodies

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. The disorder has an insidious onset and gradual progression.
- C. The disorder meets a combination of core diagnostic features and suggestive diagnostic features for either probable or possible neurocognitive disorder with Lewy bodies.

For probable major or mild neurocognitive disorder with Lewy bodies, the individual has two core features, or one suggestive feature with one or more core features. **For possible major or mild neurocognitive disorder with Lewy bodies,** the individual has only one core feature, or one or more suggestive features.

- 1. Core diagnostic features:
 - a. Fluctuating cognition with pronounced variations in attention and alertness.
 - b. Recurrent visual hallucinations that are well formed and detailed.
 - c. Spontaneous features of parkinsonism, with onset subsequent to the development of cognitive decline.
- 2. Suggestive diagnostic features:
 - a. Meets criteria for rapid eye movement sleep behavior disorder.

b. Severe neuroleptic sensitivity.

D. The disturbance is not better explained by cerebrovascular disease, another neurodegenerative disease, the effects of a substance, or another mental, neurological, or systemic disorder.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder with probable or possible Lewy bodies, with behavioral disturbance, code first **G31.83** Lewy body disease, followed by **F02.81**.

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For major neurocognitive disorder with probable or possible Lewy bodies, without behavioral disturbance, code first **G31.83** Lewy body disease, followed by **F02.80**. *Note:* The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild neurocognitive disorder with Lewy bodies, code **G31.84**. (**Note:** Do *not* use the additional code for Lewy body disease. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder with Lewy bodies: Use additional code(s) to indicate clinically significant psychiatric symptoms due to Lewy body disease (e.g., **F06.0** psychotic disorder due to Lewy body disease, with hallucinations; **F06.31** depressive disorder due to Lewy body disease, with depressive features).

Diagnostic Features

Major neurocognitive disorder with Lewy bodies corresponds to the condition known as dementia with Lewy bodies (DLB). The overall major or mild neurocognitive disorder with Lewy bodies (NCDLB) category includes not only progressive cognitive impairment (with early changes in attention, executive function, and visuospatial ability, rather than learning and memory) but also recurrent, complex, visual hallucinations; and concurrent symptoms of rapid eye movement (REM) sleep behavior disorder (which can be a very early manifestation); as well as hallucinations in other sensory modalities, apathy, anxiety, depression, and delusions. The cognitive symptoms may fluctuate in a pattern that can resemble a delirium, for which an adequate precipitating factor may or may not be found. The variable presentation of NCDLB symptoms reduces the likelihood of all symptoms being observed in a brief clinic visit and necessitates a thorough assessment, including caregiver observations. The use of assessment scales specifically designed to assess fluctuation may aid in diagnosis. Another core feature is spontaneous parkinsonism; this can often be relatively mild, and the degree of response to levodopa therapy is variable. Up to 25% of individuals with probable NCDLB may never develop extrapyramidal signs, and they are not essential for diagnosis. The parkinsonism must be distinguished from neuroleptic-induced extrapyramidal signs. Accurate diagnosis is essential to safe treatment planning, as up to 50% of individuals with NCDLB have severe sensitivity to neuroleptic drugs, and these medications should be used with extreme caution in individuals suspected of having an NCDLB diagnosis.

The diagnosis of mild NCDLB is appropriate for individuals who present with the core

clinical features at a stage when cognitive or functional impairments are not of sufficient severity to fulfill criteria for major NCD, particularly if nonamnestic cognitive deficits are prominent. However, as for all mild NCDs, there will often be insufficient evidence to justify any single etiology, and use of the unspecified diagnosis may be more appropriate.

Associated Features

Individuals with NCDLB frequently experience repeated falls, syncope, or other transient episodes of unresponsiveness. Autonomic dysfunction may be observed, including orthostatic hypotension, constipation, and urinary incontinence; hypersomnia and hyposmia may also be observed.

Prevalence

Limited data from several high-income and low- and middle-income countries show that the population-based prevalence estimates for NCDLB range from 0% to 1.2% of the general elderly population, and from 0% to 9.7% of all dementia cases. The mean prevalence of major NCDLB was 4.2% of all dementias in the community, and in clinic-based studies this increased to 7.5% of all dementias. The clinical prevalence of major NCDLB among

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individuals with dementia does not appear to be significantly affected by either age or sex. In studies from the United States and United Kingdom, the pathological lesions known as Lewy bodies are present in 20%–35% of cases of dementia. In a population-based study in Minnesota that relied on medical records, the incidence of NCDLB was approximately three times higher in men than in women age 65 or older.

Development and Course

NCDLB is a gradually progressive disorder with insidious onset. However, there is often a prodromal history of confusional episodes (delirium) of acute onset, which may be precipitated by illness or surgery. The distinction between NCDLB, in which Lewy bodies are primarily limbic in location (with or without neocortical involvement), and major or mild NCD due to Parkinson's disease, which starts in the brain stem, is the order in which the cognitive and motor symptoms emerge. In NCDLB, the cognitive decline is manifested early in the course of illness (see the section "Differential Diagnosis" for this disorder).

Onset of symptoms is typically observed in individuals ages 50–89, with most cases having onset in individuals in the mid-70s. Disease course may be characterized by occasional plateaus but eventually progresses through severe dementia to death. Average duration of survival is 5.5–7.7 years from the onset of cognitive decline.

Risk and Prognostic Factors

Genetic and physiological. Familial aggregation may occur, and several risk genes have been identified; but in most cases of NCDLB, there is no family history. The available studies suggest that genetic risk factors are as important in NCDLB as in Alzheimer's disease or Parkinson's

disease.

Diagnostic Markers

Biomarkers indicative of NCDLB may be considered to carry diagnostic weight equivalent to core clinical features; these include low striatal dopamine transporter uptake on single photon emission computed tomography (SPECT) or positron emission tomography (PET) scan, abnormal (low uptake) (MIBG) myocardial scintigraphy suggesting cardiac sympathetic denervation, and polysomnographic confirmation of REM sleep without atonia. The associated condition REM sleep behavior disorder may be diagnosed through a formal sleep study or identified by questioning the individual or informant about relevant symptoms. The underlying neurodegenerative disease is primarily associated with misfolding and aggregation of α -synuclein, which may be confirmed via postmortem histopathological examination. Neuropsychological testing beyond the use of a brief screening instrument may be necessary to define cognitive deficits clearly. Assessment scales developed to measure fluctuation can be useful.

Biomarkers supportive of NCDLB but with more limited evidence of diagnostic value include the following: preservation of medial temporal volume relative to Alzheimer's disease on magnetic resonance imaging (MRI), generalized low uptake on SPECT/PET perfusion scan with reduced occipital activity with or without the cingulate island sign (sparing of the posterior cingulate cortex relative to the precuneus plus cuneus on fluorodeoxyglucose-PET imaging), and prominent slow-wave activity on electroencephalogram with periodic fluctuations in the pre-alpha/theta range.

Functional Consequences of Major or Mild Neurocognitive Disorder With Lewy Bodies

Individuals with NCDLB are more functionally impaired than would be expected for their cognitive deficits when contrasted to individuals with other neurodegenerative diseases,

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such as Alzheimer's disease. This is largely a result of motor and autonomic impairments, which cause problems with toileting, transferring, and eating. Sleep disorders and prominent psychiatric symptoms may also add to functional difficulties. Consequently, the quality of life of individuals with NCDLB is often significantly worse than that of individuals with Alzheimer's disease.

Differential Diagnosis

Major or mild neurocognitive disorder due to Parkinson's disease. The distinction between NCDLB and NCD due to Parkinson's disease is based on the timing and sequence of motor symptoms and cognitive symptoms. Consensus criteria for DLB separate NCDLB from NCD due to Parkinson's disease by specifying that for dementia to be attributed to Parkinson's disease, the Parkinson's disease diagnosis is present for at least 1 year before cognitive decline has reached the level of major NCD, whereas for NCDLB, the cognitive symptoms may begin before, with, or in the absence of parkinsonism. By contrast, expert consensus criteria for Parkinson's disease propose that if cognitive decline occurs prior to a motor diagnosis, the diagnosis of Parkinson's

disease may still be made; therefore, a clinician may attribute the cognitive decline to the Parkinson's disease and diagnose NCD due to Parkinson's disease. Consequently, the clinician may choose to diagnose NCD due to Parkinson's disease or NCDLB for individuals with major NCD that starts either before or within 12 months of Parkinson's disease. In such circumstances, the clinician decides which diagnosis is more appropriate. If Parkinson's disease has been diagnosed for at least 1 year prior to the onset of cognitive symptoms, then both expert criteria agree that NCD due to Parkinson's disease would typically be the appropriate diagnosis. The timing and sequence of parkinsonism and mild NCD may be particularly difficult to determine, and unspecified NCD may need to be diagnosed until the order of clinical progression becomes evident.

Comorbidity

Lewy body pathology frequently coexists with Alzheimer's disease, transactive response DNA-binding protein 43 (TDP-43)-related pathology, and cerebrovascular disease pathology, particularly in the oldest age groups. TDP-43 is a protein that has been identified as a source of the proteinopathies across a range of neurodegenerative disorders, including amyotrophic lateral sclerosis and frontotemporal degeneration. The presence of multiple pathological lesions has implications for disease prognosis and may be associated with a more rapid cognitive decline and shorter survival time.

Major or Mild Vascular Neurocognitive Disorder

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. The clinical features are consistent with a vascular etiology, as suggested by either of the following:
 - 1. Onset of the cognitive deficits is temporally related to one or more cerebrovascular events.
 - 2. Evidence for decline is prominent in complex attention (including processing speed) and frontal-executive function.

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- C. There is evidence of the presence of cerebrovascular disease from history, physical examination, and/or neuroimaging considered sufficient to account for the neurocognitive deficits.
- D. The symptoms are not better explained by another brain disease or systemic disorder.

Probable vascular neurocognitive disorder is diagnosed if one of the following is present; otherwise **possible vascular neurocognitive disorder** should be diagnosed:

1. Clinical criteria are supported by neuroimaging evidence of significant parenchymal injury attributed to cerebrovascular disease (neuroimaging-supported).
2. The neurocognitive syndrome is temporally related to one or more documented cerebrovascular events.
3. Both clinical and genetic (e.g., cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy) evidence of cerebrovascular disease is present.

Possible vascular neurocognitive disorder is diagnosed if the clinical criteria are met but neuroimaging is not available and the temporal relationship of the neurocognitive syndrome with one or more cerebrovascular events is not established.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder probably or possibly due to vascular disease, with behavioral disturbance, code **F01.51**.

For major neurocognitive disorder probably or possibly due to vascular disease, without behavioral disturbance, code **F01.50**.

An additional medical code for the vascular disease is not used.

Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild vascular neurocognitive disorder, code **G31.84**. (**Note:** Do *not* use an additional code for the vascular disease. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild vascular neurocognitive disorder: Use additional code(s) to indicate clinically significant psychiatric symptoms due to the cerebrovascular disease (e.g., **F06.31** depressive disorder due to cerebrovascular disease, with depressive features).

Diagnostic Features

The diagnosis of major or mild vascular neurocognitive disorder (NCD) requires the establishment of an NCD (Criterion A) and the determination that cerebrovascular disease is the dominant if not exclusive pathology that accounts for the cognitive deficits (Criteria B and C). Vascular etiology may range from large vessel stroke to microvascular disease; the presentation is therefore very heterogeneous, stemming from the types of vascular lesions and their extent and location. The lesions may be focal, multifocal, or diffuse and occur in various combinations. Pathogenic mechanisms responsible for brain parenchymal injury include hypoperfusion and hypoxia, oxidative stress and inflammation leading to endothelial dysfunction, impairment of autoregulation, and disruption of neurovascular coupling.

Many individuals with major or mild vascular NCD present with multiple infarctions, with an acute stepwise or fluctuating decline in cognition, and intervening periods of stability and even some improvement. Others may have gradual onset with slow progression, a rapid development of deficits followed by relative stability, or another complex presentation. Major or mild vascular

NCD with a gradual onset and slow progression is generally attributable to small vessel disease leading to lesions in the white matter, basal ganglia, or thalamus. The gradual progression in these cases is often punctuated by acute

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events that leave subtle neurological deficits. The cognitive deficits in these cases can be attributed to disruption of cortical-subcortical circuits; complex attention, particularly speed of information processing, and executive function are likely to be affected. Clinical subtypes of vascular NCD have been described and include 1) poststroke NCD, manifesting immediately after a stroke; 2) subcortical ischemic vascular NCD; 3) multi-infarct (cortical) NCD; and 4) cortical-subcortical vascular NCD.

Assessing for the presence of sufficient cerebrovascular disease relies on history, physical examination, and neuroimaging (Criterion C). Etiological certainty requires the demonstration of abnormalities on neuroimaging. The lack of neuroimaging can result in significant diagnostic inaccuracy by overlooking “silent” brain infarction and white matter lesions. However, if the neurocognitive impairment is temporally associated with one or more well-documented strokes, a probable diagnosis can be made in the absence of neuroimaging. Clinical evidence of cerebrovascular disease includes documented history of stroke, with cognitive decline temporally associated with the event, or physical signs consistent with stroke (e.g., hemiparesis; pseudobulbar syndrome, visual field defect). Neuroimaging (magnetic resonance imaging [MRI] or computed tomography [CT]) evidence of cerebrovascular disease comprises one or more of the following: one or more large vessel infarcts or hemorrhages, a strategically placed single infarct or hemorrhage (e.g., in angular gyrus, thalamus, basal forebrain), two or more lacunes outside the brain stem, or extensive and confluent white matter lesions. The latter is often termed *small vessel disease* or *subcortical ischemic changes* on clinical neuroimaging evaluations. MRI is the preferred mode of neuroimaging, and there has been interest in using specialized MRI techniques to detect cerebral microbleeds, cortical microinfarcts, dilated perivascular spaces, and diffusion-based analyses of white matter tracts and network connectivity.

For mild vascular NCD, history of a single stroke or extensive white matter disease is generally sufficient. For major vascular NCD, two or more strokes, a strategically placed stroke, or a combination of white matter disease and one or more lacunes is generally necessary. However, the relationship between identifiable vascular pathology in the brain on neuroimaging and the cognitive symptoms is imperfect, and clinical judgment is generally needed to relate the vascular lesions to the cognitive syndrome.

The neurocognitive symptoms must not be better explained by another medical condition or mental disorder. For example, prominent memory deficit early in the course might suggest NCD due to Alzheimer’s disease, early and prominent parkinsonian features would suggest NCD due to Parkinson’s disease, and a close association between onset of cognitive and depressive symptoms would suggest cognitive impairment as a result of depression.

A number of expert international groups have similarly defined and categorized the vascular NCDs, with which DSM-5 criteria generally display good correspondence.

Associated Features

A neurological assessment often reveals history of stroke or transient ischemic episodes, and signs indicative of brain infarctions. Also commonly associated are personality and mood changes, abulia, depression, and emotional lability. The development of late-onset depressive symptoms accompanied by psychomotor slowing and executive dysfunction is a common presentation among older adults with progressive small vessel ischemic disease (so-called vascular depression).

Prevalence

Vascular disease is the second most common cause of NCD after Alzheimer's disease. In the United States, population prevalence estimates for vascular dementia are 0.98% for individuals ages 71–79 years, 4.09% for those ages 80–89 years, and 6.19% for those age 90 years or older. Within 3 months following stroke, 20%–30% of individuals are diagnosed with

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dementia. In a European autopsy series of decedents ages 60–103 years, the prevalence of pure vascular dementia was 12.3%. Among those ages 60–69 years, the prevalence was higher (15.0%) compared with those older than 90 years (8.7%). Mixed dementia (Alzheimer's plus vascular pathology) was present in 5.5% of the overall cohort, with a higher prevalence in those older than 90 years (10.6%) compared with those ages 60–69 years (5.2%). Higher prevalence of vascular dementia has been found among African Americans, Mexican Americans, and South Asian Americans compared with non-Latinx Whites, possibly because of higher rates of risk factors such as diabetes and cardiovascular disease. In Japan and several other Asian countries, the prevalence of dementia due to Alzheimer's disease has increased over time relative to vascular dementia. Currently, the prevalence of dementia due to Alzheimer's disease among Japanese Americans is 2.6 times higher than that of vascular dementia.

Stroke is more common in men through age 65 years, but more common in women after age 65 years. Overall, the rate of vascular NCD was higher in men in some studies.

Development and Course

Major or mild vascular NCD can occur at any age, although the prevalence increases exponentially after age 65 years. In older individuals, additional pathologies are almost always present and partly account for the neurocognitive deficits. The course may vary from acute onset with partial improvement to stepwise decline to progressive decline, with fluctuations and plateaus of varying durations. Pure subcortical major or mild vascular NCD can have a slowly progressive course that simulates major or mild NCD due to Alzheimer's disease. The risk of an ischemic stroke progressing to vascular NCD within 5 years was almost twice as high among African Americans as among non-Latinx Whites in the United States and occurred at younger ages. This is possibly a result of the impact of higher rates of hypertension, diabetes, and adverse social determinants of mental health known to worsen dementia risk, such as limited formal education and low socioeconomic status.

Risk and Prognostic Factors

Environmental. The neurocognitive outcomes of vascular brain injury are influenced by

neuroplasticity factors such as education, physical exercise, and mental activity.

Genetic and physiological. The major risk factors for major or mild vascular NCD are the same as those for cerebrovascular disease and stroke, including hypertension, diabetes, smoking, obesity, high cholesterol levels, high homocysteine levels, other risk factors for atherosclerosis and arteriolosclerosis, atrial fibrillation, and other conditions increasing the risk of cerebral emboli. Cerebral amyloid angiopathy, leading to cerebral hemorrhage, is an important risk factor in which amyloid deposits occur within arterial vessels. A genetic risk factor is the hereditary condition cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy, or CADASIL. Other rarer forms of genetic disorders linked to vascular NCD exist, but overall the contribution of genetics is small.

Diagnostic Markers

Structural neuroimaging, using MRI or CT, has an important role in the diagnostic process. There are no other established biomarkers of major or mild vascular NCD.

Functional Consequences of Major or Mild Vascular Neurocognitive Disorder

Major or mild vascular NCD is commonly associated with physical deficits that cause additional disability.

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Differential Diagnosis

Other neurocognitive disorders. Since incidental brain infarctions and white matter lesions are common in older individuals, it is important to consider other possible etiologies when an NCD is present in an individual with white matter lesions. A history of memory deficit early in the course, and progressive worsening of memory, language, executive function, and perceptual-motor abilities in the absence of corresponding focal lesions on brain imaging, are suggestive of Alzheimer's disease as the primary diagnosis. Potential biomarkers currently being validated for Alzheimer's disease, such as cerebrospinal fluid levels of β -amyloid and phosphorylated tau, and amyloid and tau imaging, may prove to be helpful in the differential diagnosis. NCD with Lewy bodies is distinguished from major or mild vascular NCD by its core features of fluctuating cognition, visual hallucinations, and spontaneous parkinsonism. While deficits in executive function and language occur in major or mild vascular NCD, the insidious onset and gradual progression of behavioral features or language impairment are characteristic of frontotemporal NCD and are not typical of vascular etiology.

Other medical conditions. A diagnosis of major or mild vascular NCD is not made if other diseases (e.g., brain tumor, multiple sclerosis, encephalitis, toxic or metabolic disorders) are present and are of sufficient severity to account for the cognitive impairment.

Other mental disorders. A diagnosis of major or mild vascular NCD is inappropriate if the symptoms can be entirely attributed to delirium, although delirium may sometimes be superimposed on a preexisting major or mild vascular NCD, in which case both diagnoses can be made. If the criteria for major depressive disorder are met and the cognitive impairment is

temporally related to the likely onset of the depression, major or mild vascular NCD should not be diagnosed. However, if the NCD preceded the development of the depression, or the severity of the cognitive impairment is out of proportion to the severity of the depression, depressive disorder due to cerebrovascular disease should be diagnosed instead of major depressive disorder.

Comorbidity

Major or mild NCD due to Alzheimer's disease commonly co-occurs with major or mild vascular NCD, in which case both diagnoses should be made. Major or mild vascular NCD and depression frequently co-occur.

Major or Mild Neurocognitive Disorder Due to Traumatic Brain Injury

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. There is evidence of a traumatic brain injury—that is, an impact to the head or other mechanisms of rapid movement or displacement of the brain within the skull, with one or more of the following:
 - 1. Loss of consciousness.
 - 2. Posttraumatic amnesia.
 - 3. Disorientation and confusion.
 - 4. Neurological signs (e.g., neuroimaging demonstrating injury; visual field cuts; anosmia; hemiparesis; hemisensory loss; cortical blindness; aphasia; apraxia; weakness;

loss of balance; other sensory loss that cannot be accounted for by peripheral or other causes).

- C. The neurocognitive disorder presents immediately after the occurrence of the traumatic brain injury or immediately after recovery of consciousness and persists past the acute post-injury period.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to traumatic brain injury, with behavioral disturbance: code first **S06.2X9S** diffuse traumatic brain injury with loss of consciousness of unspecified duration, sequela; followed by **F02.81** major neurocognitive disorder due to traumatic brain injury, with behavioral disturbance.

For major neurocognitive disorder due to traumatic brain injury, without behavioral

disturbance: code first **S06.2X9S** diffuse traumatic brain injury with loss of consciousness of unspecified duration, sequela; followed by **F02.80** major neurocognitive disorder due to traumatic brain injury, without behavioral disturbance.

Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild neurocognitive disorder due to traumatic brain injury, code **G31.84**. (*Note:* Do *not* use the additional code for traumatic brain injury. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder due to traumatic brain injury: Use additional code(s) to indicate clinically significant psychiatric symptoms due to the traumatic brain injury (e.g., **F06.34** bipolar and related disorder due to traumatic brain injury, with mixed features; **F07.0** personality change due to traumatic brain injury, apathetic type).

Specifiers

Rate the severity of the neurocognitive disorder (NCD), not the underlying traumatic brain injury (see the section “Development and Course” for this disorder).

Diagnostic Features

Major or mild NCD due to traumatic brain injury (TBI) denotes an acquired and persistent disorder of cognition resulting from a traumatic brain injury. *Traumatic brain injury* is defined as disruption of brain structure and/or function resulting from the application of biomechanical forces (including acceleration/deceleration forces and blast-related forces), as manifested immediately by one or more of the following clinical signs: loss of consciousness, loss of memory for events immediately before or after the injury (posttraumatic amnesia), alteration in mental state (e.g., confusion, disorientation, slowed thinking), or focal neurological signs (e.g., hemiparesis, hemisensory loss, cortical blindness, aphasia, apraxia, weakness, loss of balance, other sensory loss that cannot be accounted for by peripheral or other causes) (Criterion B). These manifestations of TBI must not be due to alcohol or other drugs or medications, other injuries or treatment(s) for other injuries (e.g., facial injuries, intubation, or bodily/systemic injuries), or psychological trauma, language barrier, or coexisting medical conditions.

The severity of a TBI is classified as mild, complicated mild, moderate, or severe according to the thresholds in [Table 2](#). An individual whose injury phenomenologically meets criteria for mild TBI but whose computed tomographic or magnetic resonance imaging in the acute period after TBI reveals traumatic intracranial abnormalities (i.e., traumatic epidural or subdural hematoma, subarachnoid or intracerebral hemorrhage, cerebral contusions or laceration) is classified as complicated mild TBI. Outcomes of individuals with complicated mild TBI are more like those with moderate TBI than those with uncomplicated mild TBI.

TABLE 2 Classification of traumatic brain injury (TBI) severity

TBI severity	Mild TBI	Complicated mild TBI	Moderate TBI	Severe TBI
Loss of consciousness duration	≤ 30 minutes	≤ 30 minutes	> 30 minutes to < 24 hours	≥ 24 hours
Posttraumatic amnesia duration (densely impaired new learning)	≤ 1 day	≤ 1 day	> 1 day to < 7 days	≥ 7 days
Alteration of consciousness duration (e.g., confusion, disorientation, slowed thinking)	≤ 1 day	≤ 1 day	> 1 day to < 7 days	≥ 7 days
Glasgow Coma Scale score (30 minutes after the event)	13–15	13–15	9–12	3–8
Computed tomography or magnetic resonance imaging of the brain	Normal	Abnormal	Normal or abnormal	Normal or abnormal

To be attributable to TBI, the NCD must manifest either immediately after the brain injury occurs or immediately after the individual recovers consciousness after the injury, and persist past the acute postinjury period (Criterion C).

While the specific cognitive impairments associated with major or mild NCD due to TBI are variable, impairments in complex attention, processing speed, learning and memory, and executive function are common, as are disturbances in social cognition. In more severe TBI in which there is brain contusion, intracranial hemorrhage, or penetrating injury, there may be additional neurocognitive impairments associated with the affected region of the brain and the volume of brain tissue lost (e.g., aphasia, apraxia, disturbances in perceptual-motor function).

Associated Features

The diagnosis may also be supported by subtle neurological signs (e.g., multiple primitive reflexes such as glabellar sign, snout response, palmomental reflex) or deficits in saccades and smooth-pursuit eye movements co-occurring with frontally mediated cognitive impairments such as complex attention problems, slow processing speed, impaired memory retrieval, or executive dysfunction. Particularly in some cases of penetrating TBI, the diagnosis of NCD due to TBI may be supported by posttraumatic epilepsy with focal onset in a location that corresponds to the anatomy of a cognitive domain in which an individual demonstrates impairment (e.g., medial temporal lobe—onset seizures and episodic memory impairment; frontal lobe seizures and executive dysfunction or social cognitive impairment).

Prevalence

The prevalence of major and mild NCD due to TBI varies with injury severity and time since injury, with the highest frequencies among individuals with more severe injury and during the acute/subacute post-injury period. In the United States, more than 2.87 million TBIs occur annually, including more than 837,000 TBIs in children. These TBIs account for 2.5 million emergency department visits, 288,000 hospitalizations, and more than 56,000

deaths annually. Among individuals presenting to an emergency department with TBI, the rates for men are 547.6 per 100,000 and for women are 385.9 per 100,000. The TBI rate is higher for men than women in every age group up to age 75 years, after which the rates of TBI between

men and women approach parity. The leading causes of TBI in the United States are falls (178.4 per 100,000), collision with a moving or stationary object (termed “struck by/against” events) (92.7 per 100,000), motor vehicle crashes (74.7 per 100,000), and assaults (50.6 per 100,000 persons). Concussion in sport is increasingly recognized as a cause of mild TBI.

Men are approximately 40% more likely to experience a TBI compared with women in the young and adult populations; however, women may have higher risk of TBI after age 65 years. It has been suggested that men with moderate or severe TBI may have a worse prognosis than women with the same level of severity; however, the findings have been mixed. The cause of TBI also differs by sex and gender. Men are more likely to experience injuries at work, in motor vehicle accidents, and during military activities, whereas women are more likely to experience injuries from assault and domestic violence.

Development and Course

The course of recovery from TBI is variable, depending not only on the specifics of the injury but also on pre-injury and postinjury factors. These factors may favor or impede recovery and include age; prior history of TBI; neurological, psychiatric, and substance use comorbidities and complications; genetics; the timeliness and effectiveness of medical and rehabilitative interventions; and psychosocial support, among others.

Neurocognitive impairments are most severe in the acute period following the TBI and may be accompanied by disturbances of emotion and behavior. Across the spectrum of TBI severity, substantial improvement in neurocognitive and associated psychiatric and neurological symptoms and signs is expected. The extent of recovery and the variability in neurocognitive outcomes tend to reflect the severity of TBI, with complete recovery being typical after mild TBI and more variable, and often incomplete, recovery following more severe TBI.

Neurocognitive impairments associated with mild TBI typically resolve within days to weeks after the injury, with complete resolution within 3–12 months post-injury. Other symptoms (e.g., depression, irritability, fatigue, headache, photosensitivity, sleep disturbance) that may potentially co-occur with the neurocognitive symptoms also tend to resolve in the weeks following mild TBI. Persistent symptoms after mild TBI or subsequent neurocognitive deterioration should trigger consideration of other potential causes of neurocognitive symptoms and functional limitations, including major depressive disorder, posttraumatic stress disorder (PTSD), anxiety disorders, substance use disorders, sleep disturbances, negative injury perceptions, and poor expectations for recovery. When neurocognitive symptoms and functional limitations persist after mild TBI (including repetitive mild TBI) despite treatment of their other potential causes, diagnosis of an NCD due to TBI may be appropriate.

Neurocognitive impairments and associated functional limitations produced by moderate and severe TBI typically improve over weeks to months after the injury, although long-term neurocognitive recovery is often incomplete among individuals with more severe injuries. Nonetheless, neurocognitive and functional improvement may continue for years after moderate or severe TBI, with more individuals cognitively improving than declining during the first 5 years postinjury. With moderate and severe TBI, in addition to persistence of neurocognitive deficits, there may be associated neurological, medical, emotional, and behavioral complications. These include seizures (particularly in the first year), photosensitivity, hyperacusis, irritability, aggression, depression, sleep disturbance, fatigue, apathy, inability to resume occupational and

social functioning at preinjury level, and deterioration in interpersonal relationships. Moderate and severe TBI have been

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associated with increased risk of depression, aggression, and possibly neurodegenerative diseases such as Alzheimer's disease, Lewy body disease, and frontotemporal degeneration.

The features of persisting major or mild NCD due to TBI will vary by age, specifics of the injury, and cofactors. Persisting TBI-related impairment in an infant or child may be reflected in delays in reaching developmental milestones (e.g., language acquisition), worse academic performance, and possibly impaired social development. Among older teenagers and adults, persisting symptoms may include various neurocognitive deficits, irritability, hypersensitivity to light and sound, easy fatigability, and mood changes, including depression, anxiety, hostility, or apathy. In older individuals, mild TBI may produce neurocognitive outcomes like those associated with moderate or severe TBI in younger adults.

Risk and Prognostic Factors

Risk factors for adverse cognitive outcomes after TBI include age older than 40 years, lower pre-injury cognitive abilities (especially as indexed by education or academic competence), preinjury depressive symptoms, possibly pre-injury unemployment, and injury severity. Other risk factors for adverse cognitive outcomes include a longer duration of posttraumatic amnesia, evidence of traumatic intracranial abnormalities on early computed tomography or magnetic resonance imaging (MRI) studies (i.e., traumatic epidural or subdural hematoma, subarachnoid or intracerebral hemorrhage, cerebral contusions or laceration, diffuse axonal injury) and neurogenetic profile (e.g., *APOE*E4* allele carrier status, catechol-O-methyltransferase genotype, *ANKK1* Taq1A allele status). Pre-injury alcohol or substance use disorders increase the risk of sustaining a TBI as well as the risk of adverse cognitive outcomes, including memory impairment and executive dysfunction.

Diagnostic Markers

The diagnosis of major or mild NCD due to TBI may be supported by contemporaneous computed tomographic or MRI findings (e.g., focal atrophy, encephalomalacia, gliosis, white matter abnormalities) in brain areas or networks subserving specific cognitive domains in which an individual demonstrates impairment. The diagnosis may also be supported by subtle neurological signs (e.g., multiple primitive reflexes such as glabellar sign, snout response, palmomental reflex) or deficits in saccades and smooth-pursuit eye movements co-occurring with frontally mediated cognitive impairments such as complex attention problems, slow processing speed, impaired memory retrieval, or executive dysfunction. Particularly in some cases of penetrating TBI, the diagnosis of NCD due to TBI may be supported by posttraumatic epilepsy with focal onset in a location that corresponds to the anatomy of a cognitive domain in which an individual demonstrates impairment (e.g., medial temporal lobe-onset seizures and episodic memory impairment; frontal lobe seizures and executive dysfunction or social cognitive impairment).

Performance on commonly used general cognitive screening measures, particularly when

interpreted using large-scale, population-based normative data, may usefully identify individuals in need of further neurodiagnostic assessment. However, the diagnosis of major or mild NCD due to TBI rests on performance on domain-specific cognitive assessment interpreted in light of the individual's prior performance (e.g., neuropsychological estimates of pre-injury cognitive ability or appropriate norms) and assessment of functional status.

While neuroimaging and other clinical assessments (e.g., subtle neurological signs) may provide supportive information, they cannot independently diagnose NCD due to TBI. At present, there are no other established biomarkers of major or mild NCD due to TBI.

Association With Suicidal Thoughts or Behavior

Individuals with TBI, including moderate or severe TBI, are at increased long-term risk for suicide. While depression is a substantial contributor to this risk, it does not fully account for it. Rates of suicidal ideation are as high as 10%, and rates of suicide attempt are 0.8%–1.7%

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over the first 20 years after TBI. The development of depression and/or suicidal behavior at 1-year post-injury is associated with consistently elevated rates of depression and suicidal behavior 5 years after TBI. While the relationship between cognitive impairments and suicide risk after TBI is complex, assessing suicide risk is an important element in the evaluation of individuals with major or mild NCD due to TBI.

Youth who have had concussions may be at higher risk for suicidal behavior. There is an increased risk of suicide among both veteran and civilian cohorts with TBI, and individuals seeking mental health care may have a history of TBI. Individuals seeking rehabilitative services for TBI are also at heightened risk for suicidal thoughts and behavior.

Functional Consequences of Major or Mild Neurocognitive Disorder Due to Traumatic Brain Injury

Approximately 3.17 million individuals in the United States (approximately 1.1% of the population) live with a TBI-related disability, including neurocognitive impairments that compromise the ability to work or perform daily activities and that are associated with the need for ongoing medical care, rehabilitation, support, and services. Cognitive impairments interfere with functional independence, productive employment, and community participation and may reduce satisfaction with life. The influence of cognitive impairments on functional status varies with the type and severity of those impairments; with the presence and severity of co-occurring psychiatric, substance use, neurological, and medical conditions; and with family, other psychosocial, and medical support.

With mild NCD due to TBI, individuals may report reduced cognitive efficiency, difficulty concentrating, and lessened ability to perform usual activities. With major NCD due to TBI, an individual may have difficulty in independent living and self-care. Prominent neuromotor features, such as severe incoordination, ataxia, and motor slowing, may be present in major NCD due to TBI and may add to functional difficulties.

Individuals with TBI histories report more depressive and anxious symptoms, and these can amplify cognitive complaints and worsen functional outcome. Additionally, loss of emotional

control, including aggressive or inappropriate affect and apathy, may be present after more severe TBI with greater neurocognitive impairment. These features may compound difficulties with functional independence and self-care.

Differential Diagnosis

Other mental disorders and medical conditions. Mental disorders (e.g., major depressive disorder, anxiety disorders, PTSD, alcohol and other substance use disorders, sleep disturbances), prescribed medications (e.g., typical antipsychotics, benzodiazepines, drugs with anticholinergic properties, antiepileptic drugs), and other medical conditions may contribute to or account for cognitive impairments among individuals with TBI, and need to be considered in the differential diagnosis of major or mild NCD due to TBI.

Factitious disorder and malingering. Alternative explanations for neurocognitive symptoms should be considered when the severity of neurocognitive symptoms and functional limitations are inconsistent with the cognitive outcomes expected after TBI—and particularly mild TBI—and when neuropsychological assessment reveals poor effort or is otherwise not valid for interpretation. In such circumstances, the possibility of factitious disorder or malingering (especially in situations in which there might be external incentives such as obtaining financial compensation) should be considered.

Comorbidity

Major or mild NCD due to TBI may be accompanied by other specified or unspecified depressive or anxiety disorders characterized by disturbances in emotional function (e.g., irritability, easy frustration, tension and anxiety, affective lability). Other specified or

unspecified personality disorders may also occur as a result of symptoms such as disinhibition, apathy, suspiciousness, or aggression. Medical comorbidities may occur with neurological and physical disturbances characterized by headache, fatigue, sleep disorders, vertigo or dizziness, tinnitus or hyperacusis, photosensitivity, anosmia, reduced tolerance to psychotropic medications, and, particularly in more severe TBI, neurological symptoms and signs (e.g., seizures, hemiparesis, visual disturbances, cranial nerve deficits) and evidence of orthopedic injuries. The most common medical and psychiatric comorbidities associated with moderate-to-severe TBI are (in order of frequency) back pain, depression, hypertension, anxiety, fractures, high blood cholesterol, sleep disorders, panic attacks, osteoarthritis, and diabetes.

Among individuals with substance use disorders, the neurocognitive effects of the substance contribute to or compound the TBI-associated cognitive disturbances, particularly among individuals with two or more TBIs.

PTSD can co-occur with TBI in civilian, military, and veterans populations. TBI and PTSD produce similar neurocognitive symptoms (e.g., disturbances of complex attention, processing speed, learning and memory, and executive function), and either or both conditions, as well as co-occurring depression and sleep disturbances, may explain neurocognitive symptoms in individuals with such comorbidities.

Substance/Medication-Induced Major or Mild Neurocognitive Disorder

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. The neurocognitive impairments do not occur exclusively during the course of a delirium and persist beyond the usual duration of intoxication and acute withdrawal.
- C. The involved substance or medication and duration and extent of use are capable of producing the neurocognitive impairment.
- D. The temporal course of the neurocognitive deficits is consistent with the timing of substance or medication use and abstinence (e.g., the deficits remain stable or improve after a period of abstinence).
- E. The neurocognitive disorder is not attributable to another medical condition or is not better explained by another mental disorder.

Coding note (see also coding table on pp. 682–683): The ICD-10-CM codes for the [specific substance/medication]-induced neurocognitive disorders are indicated in the table below. Note that the ICD-10-CM code depends on whether or not there is a comorbid substance use disorder present for the same class of substance. In any case, an additional separate diagnosis of a substance use disorder is not given.

Substance-induced major neurocognitive disorder: If a mild substance use disorder is comorbid with the substance-induced major neurocognitive disorder, the 4th position character is “1,” and the clinician should record “mild [substance] use disorder” before the substance-induced major neurocognitive disorder (e.g., “mild inhalant use disorder with inhalant-induced major neurocognitive disorder”). For alcohol and sedative, hypnotic, or anxiolytic substances, a mild substance use disorder is insufficient to cause a substance-induced major neurocognitive disorder; thus, there are no available ICD-10-CM codes for this combination. If a moderate or severe substance use disorder is comorbid with the substance-induced major neurocognitive disorder, the 4th position character is “2,” and the clinician should record “moderate [substance] use disorder” or “severe [substance] use disorder,” depending on the severity of the comorbid substance use disorder. If there is no comorbid substance use disorder, then the 4th position character is “9,” and the clinician should record only the substance-induced major neurocognitive disorder.

Substance-induced mild neurocognitive disorder: If a mild substance use disorder is comorbid with the substance-induced mild neurocognitive disorder, the 4th position character is “1,” and the clinician should record “mild [substance] use disorder” before the substance-induced mild neurocognitive disorder (e.g., “mild

cocaine use disorder with cocaine-induced mild neurocognitive disorder”). If a moderate or severe substance use disorder is comorbid with the substance-induced mild neurocognitive disorder, the 4th position character is “2,” and the clinician should record “moderate [substance] use disorder” or “severe [substance] use disorder,” depending on the severity of the comorbid substance use disorder. If there is no comorbid substance use disorder, then the 4th position character is “9,” and the clinician should record only the substance-induced mild neurocognitive disorder.

The severity specifiers “mild,” “moderate,” and “severe” (for major neurocognitive disorder) and the accompanying symptom specifiers “with behavioral disturbance” and “without behavioral disturbance” (for major or mild neurocognitive disorder) cannot be coded but should still be recorded.

ICD-10-CM			
	With mild use disorder	With moderate or severe use disorder	Without use disorder
Substance-induced major neurocognitive disorder (NCD)			
Alcohol (major NCD), nonamnesic-confabulatory type	NA	F10.27	F10.97
Alcohol (major NCD), amnesic-confabulatory type	NA	F10.26	F10.96
Inhalant (major NCD)	F18.17	F18.27	F18.97
Sedative, hypnotic, or anxiolytic (major NCD)	NA	F13.27	F13.97
Other (or unknown) substance (major NCD)	F19.17	F19.27	F19.97
Substance-induced mild neurocognitive disorder (NCD)			
Alcohol (mild NCD)	F10.188	F10.288	F10.988
Inhalant (mild NCD)	F18.188	F18.288	F18.988
Sedative, hypnotic, or anxiolytic (mild NCD)	F13.188	F13.288	F13.988
Amphetamine-type substance (or other stimulant) (mild NCD)	F15.188	F15.288	F15.988
Cocaine (mild NCD)	F14.188	F14.288	F14.988
Other (or unknown) substance (mild NCD)	F19.188	F19.288	F19.988

Specify if:

Persistent: Neurocognitive impairment continues to be significant after an extended period of abstinence.

Recording Procedures

The name of the substance/medication-induced neurocognitive disorder (NCD) begins with the specific substance (e.g., alcohol) that is presumed to be causing the neurocognitive symptoms. The ICD-10-CM code that corresponds to the applicable drug class is selected from the table included in the criteria set. For substances that do not fit into any of the classes (e.g., intrathecal methotrexate), the ICD-10-CM code for the other (or unknown) substance class should be used and the name of the specific substance recorded (e.g., F19.988 intrathecal methotrexate-induced

mild neurocognitive disorder). In cases in which a substance is judged to be an etiological factor but the specific substance is unknown, the ICD-10-CM code for the other (or unknown) substance class is used, and the fact that the substance is unknown is recorded (e.g., F19.97 unknown substance-induced major neurocognitive disorder).

When recording the name of the disorder, the comorbid substance use disorder (if any) is listed first, followed by the word “with,” followed by the name of the disorder (i.e., [specific substance]–induced major neurocognitive disorder or [specific substance]–induced mild neurocognitive disorder), followed by the type in the case of alcohol (i.e., nonamnesic-confabulatory type, amnesic-confabulatory type), followed by specification of duration (i.e., persistent). For example, in the case of persistent amnesic-confabulatory symptoms in a man with a severe alcohol use disorder, the diagnosis is F10.26 severe alcohol use disorder with alcohol-induced major neurocognitive disorder, amnesic-confabulatory type, persistent. A separate diagnosis of the comorbid severe alcohol use disorder is not given. If the substance-induced neurocognitive disorder occurs without a comorbid substance use disorder (e.g., after a sporadic heavy use of inhalants), no accompanying substance use disorder is noted (e.g., F18.988 [specific inhalant]–induced mild neurocognitive disorder).

Diagnostic Features

Substance/medication-induced major or mild NCD is characterized by neurocognitive impairments that persist beyond the usual duration of intoxication and acute withdrawal (Criterion B). Initially, these manifestations can reflect slow recovery of brain functions from a period of prolonged substance use, and improvements in neurocognitive as well as brain imaging indicators may be seen over many months. If the disorder continues for an extended period, *persistent* should be specified. The given substance and its use must be known to be capable of causing the observed impairments (Criterion C). While nonspecific decrements in a range of cognitive abilities can occur with nearly any substance of abuse and a variety of medications, some patterns occur more frequently with selected drug classes. For example, NCD due to sedative, hypnotic, or anxiolytic drugs (e.g., benzodiazepines, barbiturates) may show greater disturbances in memory than in other cognitive functions. NCD induced by alcohol frequently manifests with a combination of impairments in executive-function and memory and learning domains. The temporal course of the substance-induced NCD must be consistent with that of use of the given substance (Criterion D). Alcohol-induced, amnesic-confabulatory type (Korsakoff’s) NCD is characterized by an impairment in recent memory that is out of proportion to additional NCD symptoms. Features include prominent amnesia (severe difficulty learning new information with rapid forgetting) and a tendency to confabulate, although confabulation can be seen with any severe diminution of recent memory. These manifestations may co-occur with signs of thiamine encephalopathy (Wernicke’s encephalopathy) with associated features such as nystagmus and ataxia. Ophthalmoplegia of Wernicke’s encephalopathy is typically characterized by a lateral gaze paralysis. The neurocognitive deficits associated with inhalant misuse include diminished executive functioning, slower cognitive speed, and additional impaired performance on aspects of the Wisconsin Card Sorting and the Stroop tests. Neurocognitive symptoms associated with stimulant use include difficulties with learning and memory and executive function. Methamphetamine use can also be associated with

evidence of vascular injury (e.g., focal weakness, unilateral incoordination, asymmetrical reflexes). The most common neurocognitive profile approximates that seen in vascular NCD. Substances that cause NCD included in the other (or unknown) substance category include intrathecal methotrexate and organophosphate insecticides, as well as compounds that are misused and known to induce adverse cognitive effects but are less well characterized (e.g., kratom/*Mitragyna speciosa*).

When one is determining the relationship between NCD conditions and any group of drugs, it is important to consider whether the deficit was present before the use of the substance and consequently would not be attributable to the substance—and may have even contributed to poor judgment that resulted in the substance use. For example, evidence of decreased impulse control and related impairment of executive functions have been reported as associated with the onset of the use of stimulants and other drugs. In studies in which neurocognitive function is carefully assessed prior to substance use, and then subjects are followed up over several months or more, the ability of drugs other than alcohol, other depressants, and inhalants to cause clinically significant persistent NCDs is not clear.

Associated Features

Inhalant-induced NCD conditions may be associated with the smell of the inhalant on an individual's breath or a rash around the individual's nose or mouth from "huffing" the drug from a container. These are most often seen in individuals with limited access to other drugs who have histories of inhalant use as well as the early onset of use of multiple substances, especially if their symptoms fulfill criteria for conduct or antisocial personality disorders. A high risk is also seen in workers exposed to solvents in the workplace. Mild NCD induced by drugs with central nervous system depressant effects may manifest with added symptoms of increased irritability, anxiety, sleep disturbance, and dysphoria. NCD induced by stimulant drugs may manifest with rebound depression, hypersomnia, and apathy. In severe forms of substance/medication-induced major NCD (e.g., associated with long-term alcohol use), there may be prominent neuromotor features, such as incoordination, ataxia related to cerebellar damage, and motor slowing, as well as medical complications such as hypokalemia and cardiac arrhythmias. There may also be loss of emotional control, including aggressive or inappropriate affect, or apathy.

Prevalence

The prevalence of these conditions is not well known. Prevalence figures are more available for the use of these substances and for associated substance use disorders rather than for the neurocognitive conditions. Substance/medication-induced major or mild NCDs are more likely in those who are older, have longer duration of use, and have other risk factors such as nutritional deficits.

For alcohol use disorder, the rate of mild NCD is approximately 30%–40% in the first 2 months of abstinence. Mild NCD may persist, particularly in those who do not achieve stable abstinence until after age 50 years. Major NCD is rare and may result from concomitant nutritional deficits, as in alcohol-induced amnestic-confabulatory NCD. Alcohol-induced major NCD may be more common in men.

Few studies are available regarding the prevalence of NCD from other brain depressant drugs (i.e., sedatives, hypnotics, or anxiolytics), likely reflecting the relative rarity of studies of substance use disorders on these drugs and the relatively low level of heavy and persistent “recreational” use of sedative, hypnotic, or anxiolytic drugs compared with alcohol, cannabis, and many other drugs.

More data are available on the prevalence of inhalant use. Such exposure has been linked to both major and mild NCD of varied duration in both higher- and lower-income populations. However, persistent use to the point of developing an NCD is estimated to be less than 1% of the U.S. population.

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In the case of stimulants (methamphetamines and cocaine), cerebrovascular disease can also occur, resulting in diffuse or focal brain injury that can be of mild or major neurocognitive levels.

Development and Course

The onset of substance use disorders tends to occur during late adolescence and peak in the 20s and 30s. Although longer history of severe substance use disorder is associated with greater likelihood of NCD, the relationships are not straightforward, with substantial and even complete recovery of neurocognitive functions being common among persons who achieve stable abstinence prior to age 50 years. Substance/medication-induced major or mild NCD is most likely to become persistent in individuals who continue to have substance use disorders past the age of 50 years, presumably because of a combination of lessened neural plasticity and the onset of other age-related brain changes.

NCD conditions may involve a fairly rapid onset of neurocognitive impairment in individuals whose history includes the use of multiple types of drugs of abuse, especially with an early onset of substance use. Earlier commencement of heavy use, particularly of alcohol, may lead to defects in later neural development (e.g., later stages of maturation of frontal circuitries), which may have effects on social cognition as well as other neurocognitive abilities. For alcohol-induced NCD, there may be an additive effect of aging and alcohol-induced brain injury.

Risk and Prognostic Factors

Risk factors for substance/medication-induced NCDs include older age, longer duration of use, and persistent use past age 50 years.

For alcohol-induced NCD, long-term nutritional deficiencies, liver disease, vascular risk factors, and cardiovascular and cerebrovascular disease may contribute to risk. An increased risk for alcohol-induced, amnesic confabulatory-type NCD occurs in the context of a genetic transketolase deficiency as well as in the context of poor nutrition.

Sedative-, hypnotic-, or anxiolytic-induced NCDs have not been well studied, but these problems may be increased in individuals with long-term anxiety disorders or sleep impairment who have been taking benzodiazepines or other hypnotic medications in increasing amounts for months or years.

Diagnostic Markers

Magnetic resonance imaging (MRI) of individuals with chronic alcohol use disorder frequently reveals cortical thinning, white matter loss, and enlargement of sulci and ventricles. While neuroimaging abnormalities are more common in those with NCDs, it is possible to observe NCDs without neuroimaging abnormalities, and vice versa. Specialized techniques (e.g., diffusion tensor imaging) may reveal damage to specific white matter tracts. Magnetic resonance spectroscopy may reveal reduction in *N*-acetylaspartate, and increase in markers of inflammation (e.g., myoinositol) or white matter injury (e.g., choline). Many of these brain imaging changes and neurocognitive manifestations reverse following successful abstinence. In individuals with methamphetamine use disorder, MRI may also reveal hyperintensities suggestive of microhemorrhages or larger areas of infarction.

Functional Consequences of Substance/Medication-Induced Major or Mild Neurocognitive Disorder

The functional consequences of substance/medication-induced mild NCD are sometimes augmented by reduced cognitive efficiency and difficulty concentrating beyond that seen in many other NCDs. In addition, at both major and mild levels, substance/medication-induced NCDs may have associated motor syndromes that increase the level of functional impairment.

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Differential Diagnosis

Individuals with substance use disorders, substance intoxication, and substance withdrawal are at increased risk for other conditions that may independently, or through a compounding effect, result in neurocognitive disturbance. These include history of traumatic brain injury and infections that can accompany substance use disorder (e.g., HIV, hepatitis C virus, syphilis). Therefore, presence of substance/medication-induced major or mild NCD should be differentiated from NCDs arising outside the context of substance use, intoxication, and withdrawal, including these accompanying conditions (e.g., traumatic brain injury).

Comorbidity

Substance use disorders, substance intoxication, and substance withdrawal are highly comorbid with other mental disorders. In general, the higher the exposure to drugs of abuse, the greater the risk for a substance- or medication-induced NCD. Comorbid posttraumatic stress disorder, psychotic disorders, depressive and bipolar disorders, and neurodevelopmental disorders can contribute to neurocognitive impairment in substance users. Traumatic brain injury occurs more frequently with substance use, complicating efforts to determine the etiology of NCD in such cases. Severe, long-term alcohol use disorder can be associated with major organ system disease, including cerebrovascular disease and cirrhosis; inhalant use disorder is associated with higher rates of kidney and liver damage; and amphetamine- and cocaine-induced NCD may be accompanied by major or mild vascular NCD secondary to stimulant use.

Major or Mild Neurocognitive Disorder Due to HIV Infection

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. There is documented infection with human immunodeficiency virus (HIV).
- C. The neurocognitive disorder is not better explained by non-HIV conditions, including secondary brain diseases such as progressive multifocal leukoencephalopathy or cryptococcal meningitis.
- D. The neurocognitive disorder is not attributable to another medical condition and is not better explained by a mental disorder.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to HIV infection, with behavioral disturbance, code first **B20** HIV infection, followed by **F02.81** major neurocognitive disorder due to HIV infection, with behavioral disturbance.

For major neurocognitive disorder due to HIV infection, without behavioral disturbance, code first **B20** HIV infection, followed by **F02.80** major neurocognitive disorder due to HIV infection, without behavioral disturbance.

Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild neurocognitive disorder due to HIV infection, code **G31.84**. (*Note:* Do *not* use the additional code for HIV infection. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

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For major or mild neurocognitive disorder due to HIV infection: Use additional code(s) to indicate clinically significant psychiatric symptoms due to HIV infection (e.g., **F06.34** bipolar and related disorder due to HIV infection, with mixed features; **F07.0** personality change due to traumatic brain injury, apathetic type).

Diagnostic Features

HIV disease is caused by infection with human immunodeficiency virus type-1 (HIV-1), which is acquired through exposure to bodily fluids of an infected individual through injection substance use, unprotected sexual contact, or accidental or iatrogenic exposure (e.g., needle puncture injury to medical personnel). HIV infects several types of cells, most particularly “T-helper” (CD4) lymphocytes and monocytes. Over time, the infection can cause severe decreases in the CD4 count, resulting in severe immunocompromise, often leading to opportunistic infections and neoplasms. Infected monocytes can enter the central nervous system, leading to infection of macrophages and microglia. A small percentage of astrocytes may harbor productive

HIV infection. The advanced form of HIV infection is termed *acquired immune deficiency syndrome* (AIDS). Diagnosis of HIV is confirmed by established laboratory methods, such as the reverse-transcription polymerase chain reaction (RT-PCR) assay for HIV RNA and the antibody/antigen combination test. Of note, in-home self-testing for HIV is available.

Some individuals with HIV infection develop a neurocognitive disorder (NCD), which generally shows a “subcortical pattern” with prominently impaired executive function, slowing of processing speed, problems with more demanding attentional tasks, and difficulty in learning new information, but fewer problems with recall of learned information. In major NCD, slowing may be prominent. Language difficulties, such as aphasia, are uncommon, although reductions in fluency may be observed. HIV pathogenic processes can affect any part of the brain; therefore, other patterns are possible.

Associated Features

Major or mild NCD due to HIV infection is more prevalent in individuals of increasing age, lower educational level, or female sex, and among those with major depressive disorder, alcohol or other substance use disorders, and medical comorbidities (particularly diabetes and hypertension). The NCD risk due to HIV infection is also increased with any of the following: prior episodes of immunosuppression, high viral loads in the cerebrospinal fluid, and increased levels of tumor necrosis factor- α (TNF- α), interleukin-6 (IL-6), C-reactive protein, D-dimer, sCD14, sCD163, and neurofilament light chain in the peripheral blood or clinical laboratory indicators of advanced HIV disease, such as a low CD4 cell nadir, anemia, and hypoalbuminemia. Individuals with major NCD may show more prominent neuromotor features, such as severe incoordination, ataxia, and motor slowing. These features may become more prominent with NCD disease progression.

Prevalence

Depending on the clinical stage of HIV disease, approximately one-third to over one-half of HIV-infected individuals have at least some evidence of a neurocognitive disturbance, but most of these disturbances would not meet criteria for mild NCD and would instead represent individuals with asymptomatic neurocognitive impairment (ANI), who may have substandard performance on one or more tests of neurocognitive abilities but do not have any impairment in functional status. Rates in North America and Western Europe have largely shown that ANI accounts for the majority of neurocognitive disturbances, whereas mild NCD due to HIV accounts for approximately one-quarter of individuals, and major NCD criteria are met for typically less than 5% of individuals with HIV-related neurocognitive disturbances. In Germany, the overall prevalence of HIV-associated NCDs among HIV

clinic participants was 43%, 90% of whom were in treatment: 20% had ANI, 17% had mild NCD, and 6% had HIV-associated dementia. In low- and middle-income countries, prevalence of HIV-associated NCDs is higher among untreated individuals with HIV. In other parts of the world, and in cohorts composed mostly of individuals infected with HIV on effective antiretroviral treatment tested with comprehensive cognitive test batteries, the overall rates of

cognitive impairment were found to be around 25%–35%.

In the United States, the incidence of HIV infection is higher in men than in women in every ethnic group. However, evidence supports a sex difference in NCD due to HIV infection, with more frequent neurocognitive impairment in women, including when sex is maintained as a risk factor in a multivariate analysis. The higher rate of impairment in women may be associated with differences in educational quality.

Development and Course

In the development and course of NCD due to HIV, individuals may have neurocognitive impairment when the HIV infection is asymptomatic; the Centers for Disease Control and Prevention classifies the underlying HIV infection in three stages: asymptomatic, early symptomatic, and late symptomatic/AIDS. The course of NCD due to HIV infection can resolve, improve, remain stable, slowly worsen, rapidly worsen, or have a fluctuating course. Rapid progression of neurocognitive impairment is uncommon in the context of currently available combination antiretroviral treatment, although it may still occur in the context of a subgroup associated with older age, as well as in association with specific comorbidities promoting cognitive impairment. Nevertheless, for the predominant proportion of individuals with HIV, an abrupt change in mental status warrants an evaluation of other medical sources for the cognitive change, including secondary infections. Because HIV infection preferentially affects subcortical regions over the course of illness, including deep white matter, the progression of the disorder follows a subcortical pattern. The subcortical pattern of cognitive impairment is characterized by mental slowing associated with motor dysfunction, procedural learning deficits, and free recall deficits, with relative sparing of recognition memory, verbal abstraction, and naming.

Because HIV infection can affect a variety of brain regions and the illness can take on many different trajectories depending on associated comorbidities and the consequences of HIV infection, the overall course of an NCD due to HIV infection has considerable heterogeneity. A subcortical neurocognitive profile may interact with age over the life course, such that an interaction occurs between age and clinical stage of HIV disease in the domains of episodic memory and motor impairment (e.g., slowed gait). This interaction increases the overall prevalence of neurocognitive impairment and the likelihood that it will be more pronounced in later life.

Acquisition of HIV infection typically occurs in adults in high-income countries, via high-risk behaviors (e.g., unprotected sex; injection substance use) beginning in late adolescence and peaking during young and middle adulthood, with a significant contribution persisting into older age. In lower-income regions, where HIV testing and antiretroviral treatments for pregnant women are not readily available, perinatal transmission is common. The NCD in such infants and children may manifest primarily as neurodevelopmental delay. As individuals treated for HIV survive into older age, additive and interactive neurocognitive effects of HIV and aging, including other NCDs (e.g., due to Alzheimer's disease, due to Parkinson's disease), are possible. More than 50% of individuals with HIV in the United States are older than 50 years. Long-term antiretroviral therapy is indicated for the ongoing control of HIV infection. However, some antiretroviral therapy may be associated with inflammation, neurotoxic effects, and metabolic changes that can lead to vascular compromise and indirectly increase neurocognitive impairment in conjunction with aging and medical comorbidities that may worsen cognition.

Risk and Prognostic Factors

Paradoxically, NCD due to HIV infection has not declined significantly with the advent of effective antiretroviral therapy, although the most severe presentations (consistent with the diagnosis of major NCD) have decreased sharply. Contributory factors may include inadequate control of HIV in the central nervous system (CNS), the evolution of antiretroviral drug-resistant viral strains, the effects of chronic long-term systemic and brain inflammation, and the effects of comorbid factors such as aging, substance use disorder, hypertension, diabetes, past history of CNS trauma, and co-infections, such as with the hepatitis C virus. Chronic exposure to antiretroviral drugs has also been associated with neurotoxicity in its own right.

Diagnostic Markers

An HIV diagnosis may be made from a test conducted on the blood, oral fluids, or urine. In addition, HIV characterization of the cerebrospinal fluid may be helpful if it reveals a disproportionately high viral load in the cerebrospinal fluid versus in the plasma or if there are indicators of a high level of neuroinflammation. Neuroimaging (e.g., magnetic resonance imaging [MRI]) may reveal reduction in total brain volume, cortical thinning, reduction in white matter volume, and patchy areas of abnormal white matter (hyperintensities). MRI of the brain or lumbar puncture may be helpful to exclude a specific medical condition (e.g., cryptococcal meningitis, meningoencephalitis, herpes simplex virus type 1 or type 2 encephalitis, progressive multifocal leukoencephalopathy) that might contribute to CNS changes in the context of AIDS. Specialized techniques such as diffusion tensor imaging may reveal damage to specific white matter tracts. Arterial spin-labeling (ASL) developed as a new type of MRI (ASL-MRI) may reveal regional changes in brain perfusion in 3–5 minutes without infusion of extrinsic tracers, and translocator protein 18-kDa (TSPO) positron emission tomography scanning may reveal neuroinflammation.

Functional Consequences of Major or Mild Neurocognitive Disorder Due to HIV Infection

Functional consequences of major or mild NCD due to HIV infection are variable across individuals. Thus, impaired executive functions and slowed information processing may substantially interfere with adherence to the effective antiretroviral therapy regimens, although these regimens have been greatly simplified since their inception. Thus, functional status must be assessed and mapped directly to neurocognitive impairment in order to determine the severity of the NCD. Functional status related to neurocognitive impairment due to HIV should be separated from dysfunction attributable to other concomitant disorders that can affect neurocognitive function.

Differential Diagnosis

In the presence of comorbidities, such as other infections (e.g., hepatitis C virus, syphilis), substance use disorder (e.g., methamphetamine use disorder), prior traumatic brain injury, or neurodevelopmental conditions, major or mild NCD due to HIV infection can be diagnosed

provided there is evidence that infection with HIV has worsened any NCDs because of such preexisting or comorbid conditions. Among older adults, onset of neurocognitive decline related to cerebrovascular disease or primary neurodegeneration (e.g., major or mild NCD due to Alzheimer's disease) may need to be differentiated; these conditions may be suggested by a relatively more progressive course of decline than is seen in NCD due to HIV. HIV infection itself has been shown to increase the risk of cerebrovascular disease. Because more severe immunodeficiency can result in opportunistic infections of the brain (e.g., toxoplasmosis; cryptococcosis) and neoplasia (e.g., CNS lymphoma), sudden

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onset of an NCD or sudden worsening of an NCD demands active investigation of non-HIV etiologies. Delirium is important to consider because it occurs frequently over the disease course of individuals with HIV and may be due to multiple etiologies (including SARS-CoV-2 co-infection).

Comorbidity

HIV disease is accompanied by chronic systemic and CNS inflammation and diseases that can be associated with an NCD. These complications can be part of the pathogenesis of major or mild NCD as well as ANI due to HIV infection. HIV frequently co-occurs with conditions such as substance use disorders and other sexually transmitted infections. Both medical and psychiatric comorbidities have been identified that increase the likelihood of a diagnosis of NCD due to HIV infection. Women and members of underserved ethnic and racialized groups may show variation in the rates of the comorbidities associated with NCD due to HIV infection.

Major or Mild Neurocognitive Disorder Due to Prion Disease

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. There is insidious onset, and rapid progression of impairment is common.
- C. There are motor features of prion disease, such as myoclonus or ataxia, or biomarker evidence.
- D. The neurocognitive disorder is not attributable to another medical condition and is not better explained by another mental disorder.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to prion disease, with behavioral disturbance, code first **A81.9** prion disease, followed by **F02.81** major neurocognitive disorder due to prion disease, with behavioral disturbance.

For major neurocognitive disorder due to prion disease, without behavioral

disturbance, code first **A81.9** prion disease, followed by **F02.80** major neurocognitive disorder due to prion disease, without behavioral disturbance.

Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild neurocognitive disorder due to prion disease, code **G31.84**. (*Note:* Do *not* use the additional code for prion disease. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder due to prion disease: Use additional code(s) to indicate clinically significant psychiatric symptoms due to prion disease (e.g., **F06.2** psychotic disorder due to prion disease, with delusions; **F06.32** depressive disorder due to prion disease with major depressive–like episode).

Diagnostic Features

The classification of major or mild neurocognitive disorder (NCD) due to prion disease includes NCDs due to a group of subacute spongiform encephalopathies (including sporadic Creutzfeldt-Jakob disease, genetic Creutzfeldt-Jakob disease, iatrogenic Creutzfeldt-Jakob disease, variant Creutzfeldt-Jakob disease, variably protease-sensitive prionopathy,

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kuru [found among the Fore people in Papua New Guinea], Gerstmann-Sträussler-Scheinker syndrome, and fatal insomnia) caused by transmissible agents known as *prions*. Given that the most common type is sporadic Creutzfeldt-Jakob disease, it is typically referred to as simply Creutzfeldt-Jakob disease (CJD). Variant CJD is much rarer and is associated with transmission of bovine spongiform encephalopathy, also called “mad cow disease.” Typically, individuals with CJD present with neurocognitive deficits, ataxia, and abnormal movements such as myoclonus, chorea, or dystonia; a startle reflex is also common. The history often reveals rapid progression to major NCD over as little as 6 months, and thus the disorder is typically seen only at the major level. However, many individuals with the disorder may have atypical presentations, and the disease can be confirmed only by biopsy or at autopsy. For example, individuals with variant CJD may present with a greater preponderance of psychiatric symptoms than do individuals with other types of prion disease, characterized by low mood, withdrawal, and anxiety. Although biomarker evidence is not necessarily required for the diagnosis if the motor features of prion disease (e.g., myoclonus, ataxia) are present, confidence that the NCD is due to prion disease is greatly increased if characteristic biomarkers are present.

Prevalence

Prevalence is unknown but very low given the short survival. Based on data from nine high-income countries, the annual incidence of sporadic CJD is approximately one or two cases per million people. Incidence varies by age and is highest in those age 65 years or older (4.8/1,000,000 individuals) and is higher in Whites compared with Blacks. Incidence among ethnic Chinese in Taiwan is lower than general population rates in the United States and other reporting countries.

Development and Course

Prion disease may develop at any age in adults—the peak age for sporadic CJD is approximately 67 years—although it has been reported to occur in individuals spanning the teenage years to late life. Non-Latinx Whites were found to have an older mean age at onset compared with other ethnic and racialized populations in the United States. Prodromal symptoms of prion disease may include fatigue, anxiety, problems with appetite or sleeping, or difficulties with concentration. After several weeks, these symptoms may be followed by incoordination, altered vision, or abnormal gait or other movements that may be myoclonic, choreoathetoid, or ballistic, along with a rapidly progressive dementia. The disease typically progresses very rapidly to the major level of impairment over several months. More rarely, it can progress over 2 years and appear similar in its course to other NCDs.

Risk and Prognostic Factors

Environmental. Cross-species transmission of prion infections, with agents that are closely related to the human form, has been demonstrated (e.g., the outbreak of bovine spongiform encephalopathy inducing variant CJD in the United Kingdom during the mid-1990s). Transmission by corneal transplantation, cadaveric dura mater grafts, contaminated neurosurgical instruments, cadaver-derived human growth hormone and pituitary gonadotropin injections, and blood transfusion (only in the case of variant CJD) has been documented. Studies have not demonstrated an increased risk of sporadic CJD in health care professionals.

Genetic and physiological. In up to 15% of prion disease cases, there are autosomal dominant genetic mutations in the prion protein gene (*PRNP*), which encodes for a normal neuronal membrane-bound protein. The codon 129 polymorphism of *PRNP* mediates the risk

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of sporadic and acquired prion diseases as well as modifies clinical manifestation, age at disease onset, and disease duration.

Diagnostic Markers

Prion disease can be definitively confirmed only by brain biopsy or at autopsy. There are several cerebrospinal fluid (CSF) proteins that are markers of neuronal injury and are frequently elevated in prion disease; the ones most commonly used for diagnostic purposes are 14-3-3 and tau, which have high sensitivity but variable specificity. Real-time quaking induced conversion (RT-QuIC) is another CSF diagnostic test that is able to amplify minute amounts of disease-causing prion proteins and has extremely high specificity. Magnetic resonance brain imaging is currently considered the most sensitive diagnostic test when DWI (diffusion-weighted imaging) is performed, with the most common finding being multifocal gray matter hyperintensities in subcortical and/or cortical regions. In some individuals, the electroencephalogram reveals periodic sharp, often triphasic and synchronous discharges at a rate of 0.5–2 Hz at some point during the course of the disorder. It is important to note that the above diagnostic markers vary across prion disease type (e.g., sporadic CJD, genetic CJD, variant CJD).

Differential Diagnosis

Other major neurocognitive disorders. Major NCD due to prion disease may appear similar in its course to other NCDs, but prion diseases are typically distinguished by their rapid progression and prominent cerebellar and motor symptoms.

Major or Mild Neurocognitive Disorder Due to Parkinson's Disease

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. The disturbance occurs in the setting of established Parkinson's disease.
- C. There is insidious onset and gradual progression of impairment.
- D. The neurocognitive disorder is not attributable to another medical condition and is not better explained by another mental disorder.

Major or mild neurocognitive disorder probably due to Parkinson's disease should be diagnosed if 1 and 2 are both met. **Major or mild neurocognitive disorder possibly due to Parkinson's disease** should be diagnosed if 1 or 2 is met:

- 1. There is no evidence of mixed etiology (i.e., absence of other neurodegenerative or cerebrovascular disease or another neurological, mental, or systemic disease or condition likely contributing to cognitive decline).
- 2. The Parkinson's disease clearly precedes the onset of the neurocognitive disorder.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder probably or possibly due to Parkinson's disease, with behavioral disturbance, code first **G20** Parkinson's disease, followed by **F02.81**.

For major neurocognitive disorder probably or possibly due to Parkinson's disease, without behavioral disturbance, code first **G20** Parkinson's disease, followed by **F02.80**.

Note: The severity specifiers "mild," "moderate," and "severe" cannot be coded for major neurocognitive disorder but should still be recorded.

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For mild neurocognitive disorder due to Parkinson's disease, code **G31.84**. (**Note:** Do *not* use the additional code for Parkinson's disease. "With behavioral disturbance" and "without behavioral disturbance" cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder due to Parkinson's disease: Use additional code(s) to indicate clinically significant psychiatric symptoms due to Parkinson's

disease (e.g., **F06.0** psychotic disorder due to Parkinson's disease, with hallucinations; **F06.31** depressive disorder due to Parkinson's disease, with depressive features; **F07.0** personality change due to traumatic brain injury, apathetic type).

Diagnostic Features

The essential feature of major or mild neurocognitive disorder (NCD) due to Parkinson's disease is cognitive decline observed at the time of or following the onset of idiopathic Parkinson's disease. The disturbance must occur in the setting of established Parkinson's disease (Criterion B), and deficits must have developed gradually (Criterion C). The rate of progression of cognitive deficits may vary; for some individuals with mild deficits there may be very minimal change over time.

The NCD is viewed as *probably* due to Parkinson's disease when there is no evidence of another disorder that might be responsible for the cognitive decline *and* when the Parkinson's disease precedes onset of the NCD. The NCD is considered *possibly* due to Parkinson's disease when only one of these conditions is met but not both conditions. A diagnosis of Parkinson's disease prior to the onset of the cognitive change increases the diagnostic confidence that the NCD is attributable to Parkinson's disease, as denoted by the *probable* designation.

Associated Features

Frequently present features include apathy, depressed mood, anxious mood, hallucinations, delusions, personality changes, rapid eye movement (REM) sleep behavior disorder, excessive daytime sleepiness, freezing of gait, falls, bilateral involvement early in disease, postural instability and gait disturbance (PIGD) subtype, and hyposmia. The combination of postural and gait instability may occur early in the disease and may be described by the term *PIGD subtype* to distinguish from tremor-predominant Parkinson's disease.

Prevalence

The prevalence of Parkinson's disease in the United States steadily increases with age from approximately 0.4% between ages 60 and 69 years to 1.4% between ages 80 and 89 years. Parkinson's disease is more common in men than in women. Similarly, the prevalence of NCD due to Parkinson's disease is higher in men than in women. However, it is not clear if the incidence of NCD due to Parkinson's disease is higher in men than in women. Among individuals with Parkinson's disease, as many as 80% will eventually develop a major NCD. Among those without a major NCD, the prevalence of mild NCD in Parkinson's disease has been estimated at 25%–27%. For individuals with incident-untreated Parkinson's disease, a range of 9%–19% have mild NCD, whereas other studies have reported major NCD occurring in 24% of newly diagnosed untreated Parkinson's disease. Among African Americans, the risk of Parkinson's disease tends to be lower than among non-Latinx Whites, but the risk of dementia among those with the disease tends to be higher.

Development and Course

Onset of Parkinson's disease is typically between ages 50 and 89 years, with most expression in

the early 60s. Mild NCD often develops relatively early in the course of Parkinson's disease, whereas major impairment typically does not occur until individuals are older.

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Risk and Prognostic Factors

Environmental. Risk factors for Parkinson's disease include exposure to pesticides, solvents, and possibly traumatic brain injury.

Genetic and physiological. Potential risk factors for NCD among individuals with Parkinson's disease include older age at disease onset, increasing severity of disease, prominent gait symptoms, severe autonomic disturbance (particularly orthostatic hypotension), REM sleep behavior disorder, and possibly being a man and having fewer years of formal education. Individuals with Parkinson's disease with glucocerebrosidase gene (*GBA*) mutations and *APOE*E4* genotype have been demonstrated to have worse cognition in cross-sectional and longitudinal research.

Culture-Related Diagnostic Issues

Guam dementia is a late-onset NCD observed among 8.8% of Chamorros (the indigenous population of Guam) age 65 years and older. Characterized by neurofibrillary tangles but without the amyloid plaques found in Alzheimer's disease, it is thought to be possibly related to a unique parkinsonism-dementia complex and amyotrophic lateral sclerosis. An association has been found with processing and eating fadang made with cycad seeds.

Diagnostic Markers

Neuropsychological testing, with a focus on tests that are not affected by motor slowing (i.e., not timed or requiring use of hands), is critical in detecting the core cognitive deficits, particularly at the mild NCD phase. Characteristic features observed in neuropsychological testing early in the disorder may include reduced attention, executive dysfunction, slowed information processing, and deficits in memory and visuospatial function, whereas many language skills may remain intact.

Dopamine transporter scans, such as DaT scans, may differentiate Lewy body-related dementias (i.e., NCD due to Parkinson's disease, NCD with Lewy bodies) from non-Lewy body-related dementias (e.g., NCD due to Alzheimer's disease).

Differential Diagnosis

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Major or mild neurocognitive disorder with Lewy bodies (NCDLB). The distinction between NCDLB and NCD due to Parkinson's disease is based on the timing and sequence of motor symptoms and cognitive symptoms. Consensus criteria for dementia with Lewy bodies separate NCDLB from NCD due to Parkinson's disease by specifying that for dementia to be attributed to Parkinson's disease, the Parkinson's disease diagnosis must be present for at least 1 year before cognitive decline has reached the level of major NCD, whereas for NCDLB, the cognitive

symptoms may begin before, with, or in the absence of parkinsonism. By contrast, expert consensus criteria for Parkinson's disease propose that if cognitive decline occurs prior to a motor diagnosis, the diagnosis of Parkinson's disease may still be made; therefore, a clinician may attribute the cognitive decline to the Parkinson's disease and diagnose NCD due to Parkinson's disease. Consequently, the clinician may choose to diagnose NCD due to Parkinson's disease or NCDLB for individuals with major NCD that starts either before or within 12 months of Parkinson's disease. In such circumstances, the clinician decides which diagnosis is more appropriate. If Parkinson's disease has been diagnosed for at least 1 year prior to the onset of cognitive symptoms, then both expert criteria agree that NCD due to Parkinson's disease would typically be the appropriate diagnosis. The timing and sequence of parkinsonism and mild NCD may be particularly difficult to determine, and unspecified NCD may need to be diagnosed until the order of clinical progression becomes evident.

Major or mild neurocognitive disorder due to Alzheimer's disease. The motor features are the key to distinguishing major or mild NCD due to Parkinson's disease from major or mild NCD due to Alzheimer's disease. However, the two disorders can co-occur, and individuals with well-established Alzheimer's disease can develop mild parkinsonism.

Major or mild vascular neurocognitive disorder. Major or mild vascular NCD may manifest with parkinsonian features that may occur as a consequence of diffuse cortical or subcortical small vessel disease. However, the parkinsonian features typically are not sufficient for a diagnosis of Parkinson's disease, and the course of the NCD usually has a clear association with cerebrovascular changes.

Neurocognitive disorder due to another medical condition (e.g., neurodegenerative disorders). When a diagnosis of major or mild NCD due to Parkinson's disease is being considered, the distinction must also be made from other brain disorders, such as progressive supranuclear palsy, corticobasal degeneration, multiple system atrophy, tumors, and hydrocephalus.

Antipsychotic (or other dopamine receptor–blocking drug)–induced parkinsonism. Antipsychotic (or other dopamine receptor–blocking drug)–induced parkinsonism can occur in individuals with other NCDs, particularly when antipsychotic medications are prescribed for the behavioral manifestations of such disorders.

Comorbidity

Parkinson's disease may coexist with Alzheimer's disease and cerebrovascular disease, especially in older individuals. Individuals with NCD due to Parkinson's disease may display clinical or biomarker features that suggest the presence of both Parkinson's disease and other pathologies. Evidence for mixed etiology does not preclude the contribution of Parkinson's disease to an NCD. The compounding of multiple pathological features may diminish the functional abilities of individuals with Parkinson's disease. Motor symptoms and frequent co-occurrence of depression, psychosis, REM sleep behavior disorder, or apathy can make functional impairment worse.

Major or Mild Neurocognitive Disorder Due to Huntington's Disease

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. There is insidious onset and gradual progression.
- C. There is clinically established Huntington's disease, or risk for Huntington's disease based on family history or genetic testing.
- D. The neurocognitive disorder is not attributable to another medical condition and is not better explained by another mental disorder.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to Huntington's disease, with behavioral disturbance, code first **G10** Huntington's disease, followed by **F02.81** major neurocognitive disorder due to Huntington's disease, with behavioral disturbance.

For major neurocognitive disorder due to Huntington's disease, without behavioral disturbance, code first **G10** Huntington's disease, followed by **F02.80** major neurocognitive disorder due to Huntington's disease, without behavioral disturbance.

Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

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For mild neurocognitive disorder due to Huntington's disease, code **G31.84**. (*Note:* Do *not* use the additional code for Huntington's disease. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder due to Huntington's disease: Use additional code(s) to indicate clinically significant psychiatric symptoms due to Huntington's disease (e.g., **F06.31** depressive disorder due to Huntington's disease with depressive features; **F06.4** anxiety disorder due to Huntington's disease).

Diagnostic Features

Progressive cognitive impairment is a core feature of Huntington's disease, with early changes in executive function (i.e., processing speed, organization, and planning) typically being more prominent than decline in learning and memory. Cognitive and associated behavioral changes often precede the emergence of the typical motor abnormalities of bradykinesia (i.e., slowing of voluntary movement) and chorea (i.e., involuntary jerking movements). A diagnosis of definite Huntington's disease is given in the presence of unequivocal, extrapyramidal motor abnormalities in an individual with either a family history of Huntington's disease or genetic testing showing a CAG trinucleotide repeat expansion in the *HTT* gene, located on chromosome 4.

Associated Features

Irritability, apathy, anxiety, obsessive-compulsive symptoms, depression, and, more rarely,

psychosis can all be associated with Huntington's disease and often precede the onset of motor symptoms.

Prevalence

Neurocognitive deficits are an eventual outcome of Huntington's disease; the worldwide prevalence is estimated to be 2.7 per 100,000. The prevalence of Huntington's disease in North America, Europe, and Australia is 5.7 per 100,000, with a much lower prevalence of 0.40 per 100,000 in Asia.

Development and Course

The age at diagnosis of Huntington's disease varies widely, but symptoms are most often observed between ages 35 and 45 years. Age at onset is inversely correlated with CAG expansion length. Juvenile Huntington's disease (onset before age 20) may present more commonly with bradykinesia, dystonia, and rigidity than with the choreic movements characteristic of the adult-onset disorder. The disease is gradually progressive, with the average length of survival after clinical diagnosis estimated to be approximately 10–20 years, although affected individuals may demonstrate significant variability in disease progression.

Phenotypic expression of Huntington's disease varies by presence of motor, cognitive, and psychiatric symptoms. Psychiatric and cognitive abnormalities can predate the motor abnormality by a decade or more. Initial symptoms requiring care often include irritability, anxiety, or depressed mood. Other behavioral disturbances may include pronounced apathy, disinhibition, impulsivity, and impaired insight, with apathy often becoming more progressive over time. Early movement symptoms may involve the appearance of fidgetiness of the extremities as well as mild *apraxia* (i.e., difficulty with purposeful movements), particularly with fine motor tasks. As the disorder progresses, other motor problems include impaired gait (*ataxia*) and postural instability. Motor impairment eventually affects speech production (*dysarthria*) such that the speech becomes very difficult to understand, which may result in significant distress resulting from the communication barrier in the context of comparatively intact cognition. Advanced motor disease severely affects gait with

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progressive ataxia. Eventually individuals become nonambulatory. End-stage motor disease impairs motor control of eating and swallowing, typically a major contributor to the death of the individual from aspiration pneumonia.

Risk and Prognostic Factors

Genetic and physiological. The genetic basis of Huntington's disease is a fully penetrant autosomal dominant expansion of the CAG trinucleotide, often called a *CAG repeat* in the huntingtin gene. A repeat length of 40 or more is invariably associated with Huntington's disease, with longer repeat lengths associated with early age at onset. A CAG repeat length in the 36–39 range is considered to be partially penetrant, which means that this length could or could not lead to Huntington's disease. If Huntington's disease does occur with repeat lengths in this range, it is more often associated with onset late in life (diagnosis after age 70).

Diagnostic Markers

Genetic testing is the primary laboratory test for the determination of Huntington's disease, which is an autosomal dominant disorder with complete penetrance. The trinucleotide CAG is observed to have a repeat expansion in the gene that encodes huntingtin protein on chromosome 4. A diagnosis of Huntington's disease is not made in the presence of the gene expansion alone, but the diagnosis is made only after motor symptoms become manifest. Some individuals with a positive family history request genetic testing in a presymptomatic stage. Associated features may also include neuroimaging changes; volume loss in the basal ganglia, particularly the caudate nucleus and putamen, is well known to occur and progresses over the course of illness. Other structural and functional changes have been observed in brain imaging but remain research measures.

Association With Suicidal Thoughts or Behavior

In Huntington's disease, an elevated suicide risk compared with the general population has been well documented. A literature review and report of data from a large observational study found that suicide is among the leading causes of death in Huntington's disease. The elevated risk of suicidal thoughts in Huntington's disease has been shown in diagnosed individuals both prior to and after manifesting the motor symptoms of Huntington's disease. Risk factors for suicidal thoughts include depressive symptoms, anxiety, irritability, psychosis, and apathy—emphasizing the importance of treating depressive symptoms and assessing suicidal thoughts during clinical monitoring. A large European cohort study of Huntington's disease similarly found that the most frequent causes of death were pneumonia (19.5%), other infections (6.9%), and suicide (6.6%).

Functional Consequences of Major or Mild Neurocognitive Disorder Due to Huntington's Disease

In the prodromal phase of illness and at early diagnosis, occupational decline is most common, with most individuals reporting some loss of ability to engage in their typical work. The emotional, behavioral, and cognitive aspects of Huntington's disease, such as disinhibition and personality changes, are highly associated with functional decline. Cognitive deficits that contribute most to functional decline may include speed of processing, initiation, and attention rather than memory impairment. Given that Huntington's disease onset occurs in productive years of life, it may have a very disruptive effect on performance in the work setting as well as social life, family life, and important aspects of daily functioning such as driving. As the disease progresses, disability from problems such as

impaired gait, dysarthria, and impulsive or irritable behaviors may substantially add to the level of impairment and daily care needs, over and above the care needs attributable to cognitive decline. Severe choreic movements may substantially interfere with provision of care such as bathing, dressing, and toileting.

Differential Diagnosis

Other mental disorders. Early symptoms of Huntington’s disease may include instability of mood, irritability, or compulsive behaviors that may suggest another mental disorder. However, genetic testing or the development of motor symptoms will distinguish the presence of Huntington’s disease. In such cases, if the mood symptoms are a focus of clinical attention, they may be indicated by an additional diagnosis of depressive disorder due to Huntington’s disease, with depressive features.

Other neurocognitive disorders. The early symptoms of Huntington’s disease, particularly symptoms of executive dysfunction and impaired psychomotor speed, may resemble other neurocognitive disorders (NCDs), such as major or mild vascular NCD.

Other movement disorders. Huntington’s disease must also be differentiated from other disorders or conditions associated with chorea, such as Wilson’s disease, drug-induced tardive dyskinesia, Sydenham’s chorea, systemic lupus erythematosus, or senile chorea. Rarely, individuals may present with a course similar to that of Huntington’s disease but without positive genetic testing; this is considered to be a Huntington’s disease phenocopy that results from a variety of potential genetic factors.

Major or Mild Neurocognitive Disorder Due to Another Medical Condition

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. There is evidence from the history, physical examination, or laboratory findings that the neurocognitive disorder is the pathophysiological consequence of another medical condition (e.g., multiple sclerosis).
- C. The cognitive deficits are not better explained by another mental disorder (e.g., major depressive disorder) or another specific neurocognitive disorder (e.g., major neurocognitive disorder due to Alzheimer’s disease).

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to another medical condition, with behavioral disturbance, code first the other medical condition, followed by the major neurocognitive disorder due to another medical condition, with behavioral disturbance (e.g., **G35** multiple sclerosis, **F02.81** major neurocognitive disorder due to multiple sclerosis, with behavioral disturbance).

For major neurocognitive disorder due to another medical condition, without behavioral disturbance, code first the other medical condition, followed by the major neurocognitive disorder due to another medical condition, without behavioral disturbance (e.g., **G35** multiple sclerosis, **F02.80** major neurocognitive disorder due to multiple sclerosis, without behavioral disturbance).

Note: The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For mild neurocognitive disorder due to another medical condition, code **G31.84**. (**Note:** Do *not* use the additional code for the other medical condition. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder due to another medical condition: Use additional code(s) to indicate clinically significant psychiatric symptoms due to another medical condition (e.g., **F06.32** depressive disorder due to multiple sclerosis, with major depressive–like episode).

Diagnostic Features

A number of medical conditions can cause neurocognitive disorders (NCDs) other than those specific etiologies (e.g., Alzheimer’s disease) already included in prior NCD criteria sets in this chapter. These conditions include structural lesions (e.g., primary or secondary brain tumors, subdural hematoma, slowly progressive or normal-pressure hydrocephalus), hypoxia related to hypoperfusion from heart failure, endocrine conditions (e.g., hypothyroidism, hypercalcemia, hypoglycemia), nutritional conditions (e.g., deficiencies of thiamine or niacin), other infectious conditions (e.g., neurosyphilis, cryptococcosis), immune disorders (e.g., temporal arteritis, systemic lupus erythematosus), hepatic or renal failure, metabolic conditions (e.g., Kufs’ disease, adrenoleukodystrophy, metachromatic leukodystrophy, other storage diseases of adulthood and childhood), and other neurological conditions (e.g., epilepsy, multiple sclerosis). Unusual causes of central nervous system injury, such as electrical shock or intracranial radiation, are generally evident from the history. The temporal association between the onset or exacerbation of the medical condition and the development of the cognitive deficit offers the greatest support that the NCD is a pathophysiological consequence of the medical condition. Diagnostic certainty regarding this relationship may be increased if the neurocognitive deficits ameliorate partially or stabilize in the context of treatment of the medical condition.

Development and Course

Typically the course of the NCD progresses in a manner that is commensurate with progression of the underlying medical condition. In circumstances where the medical condition is treatable (e.g., hypothyroidism), the neurocognitive deficit may improve or at least not progress. When the medical condition has a deteriorative course (e.g., secondary progressive multiple sclerosis), the neurocognitive deficits will progress along with the temporal course of illness.

Diagnostic Markers

Associated physical examination and laboratory findings and other clinical features depend on the nature and severity of the medical condition.

Differential Diagnosis

Other major or mild neurocognitive disorder. The presence of an attributable medical condition does not entirely exclude the possibility of another etiological type of major or mild NCD. If cognitive

deficits persist following successful treatment of an associated medical condition, then another etiology may be responsible for the cognitive decline.

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Major or Mild Neurocognitive Disorder Due to Multiple Etiologies

Diagnostic Criteria

- A. The criteria are met for major or mild neurocognitive disorder.
- B. There is evidence from the history, physical examination, or laboratory findings that the neurocognitive disorder is the pathophysiological consequence of more than one etiological process, excluding substances (e.g., neurocognitive disorder due to Alzheimer's disease with subsequent development of vascular neurocognitive disorder).

Note: Refer to the diagnostic criteria for the various neurocognitive disorders due to specific medical conditions for guidance on establishing the particular etiologies.

- C. The cognitive deficits are not better explained by another mental disorder and do not occur exclusively during the course of a delirium.

Coding note (see coding table on pp. 682–683):

For major neurocognitive disorder due to multiple etiologies, code first all of the etiological medical conditions (with the exception of vascular disease, which is not coded), followed by either **F02.81** for major neurocognitive disorder due to multiple etiologies, with behavioral disturbance; or **F02.80** for major neurocognitive disorder due to multiple etiologies, without behavioral disturbance.

If vascular disease is among the multiple etiological medical conditions, code next either **F01.51** for major vascular neurocognitive disorder, with behavioral disturbance; or **F01.50** for major vascular neurocognitive disorder, without behavioral disturbance. *Note:* The severity specifiers “mild,” “moderate,” and “severe” cannot be coded for major neurocognitive disorder but should still be recorded.

For example, for a presentation of major neurocognitive disorder, moderate, with a behavioral disturbance, that is judged to be due to Alzheimer's disease, vascular disease, and HIV infection, and in which heavy chronic alcohol use is judged to be a contributing factor, code the following: **G30.9** Alzheimer's disease, **B20** HIV infection; **F02.81** major neurocognitive disorder due to Alzheimer's disease and HIV infection, moderate, with behavioral disturbance; **F01.51** major vascular neurocognitive disorder, moderate, with behavioral disturbance; and **F10.27** severe alcohol use disorder with alcohol-induced major neurocognitive disorder, moderate, nonamnestic-confabulatory type.

For mild neurocognitive disorder due to multiple etiologies, code **G31.84**. (**Note:** Do *not* use the additional codes for the etiologies. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

For major or mild neurocognitive disorder due to multiple etiologies: Use additional code(s) to indicate clinically significant psychiatric symptoms due to the various etiologies (e.g., **F06.2** psychotic disorder due to Alzheimer’s disease, with delusions; **F06.31** depressive disorder due to cerebrovascular disease, with depressive features).

This category is included to cover the clinical presentation of a neurocognitive disorder (NCD) for which there is evidence that multiple medical conditions have played a probable role in the development of the NCD. In addition to evidence indicative of the presence of multiple medical conditions that are known to cause NCD (i.e., findings from the history and physical examination, and laboratory findings), it may be helpful to refer to the diagnostic criteria and text for the various medical etiologies (e.g., NCD due to Parkinson’s disease) for more information on establishing the etiological connection for that particular medical condition.

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Unspecified Neurocognitive Disorder

R41.9

This category applies to presentations in which symptoms characteristic of a neurocognitive disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the neurocognitive disorders diagnostic class. The unspecified neurocognitive disorder category is used in situations in which the precise etiology cannot be determined with sufficient certainty to make an etiological attribution.

Coding note: For unspecified major or mild neurocognitive disorder, code **R41.9**. (**Note:** Do *not* use additional codes for any presumed etiological medical conditions. “With behavioral disturbance” and “without behavioral disturbance” cannot be coded but should still be recorded.)

Personality Disorders

This chapter begins with a general definition of personality disorder that applies to each of the 10 specific personality disorders. A *personality disorder* is an enduring pattern of inner experience and behavior that deviates markedly from the norms and expectations of the individual's culture, is pervasive and inflexible, has an onset in adolescence or early adulthood, is stable over time, and leads to distress or impairment.

With any ongoing review process, especially one of this complexity, different viewpoints emerge, and an effort was made to accommodate them. Thus, personality disorders are included in both Sections II and III. The material in Section II represents an update of text associated with the same criteria found in DSM-5 (which were carried over from DSM-IV-TR), whereas Section III includes the proposed model for personality disorder diagnosis and conceptualization developed by the DSM-5 Personality and Personality Disorders Work Group. As this field evolves, it is hoped that both versions will serve clinical practice and research initiatives, respectively.

The following personality disorders are included in this chapter.

- **Paranoid personality disorder** is a pattern of distrust and suspiciousness such that others' motives are interpreted as malevolent.
- **Schizoid personality disorder** is a pattern of detachment from social relationships and a restricted range of emotional expression.
- **Schizotypal personality disorder** is a pattern of acute discomfort in close relationships, cognitive or perceptual distortions, and eccentricities of behavior.
- **Antisocial personality disorder** is a pattern of disregard for, and violation of, the rights of others, criminality, impulsivity, and a failure to learn from experience.
- **Borderline personality disorder** is a pattern of instability in interpersonal relationships, self-image, and affects, and marked impulsivity.
- **Histrionic personality disorder** is a pattern of excessive emotionality and attention seeking.
- **Narcissistic personality disorder** is a pattern of grandiosity, need for admiration, and lack of empathy.
- **Avoidant personality disorder** is a pattern of social inhibition, feelings of inadequacy, and hypersensitivity to negative evaluation.
- **Dependent personality disorder** is a pattern of submissive and clinging behavior related to an excessive need to be taken care of.
- **Obsessive-compulsive personality disorder** is a pattern of preoccupation with orderliness, perfectionism, and control.
- **Personality change due to another medical condition** is a persistent personality disturbance that is judged to be the direct pathophysiological consequence of another medical condition (e.g., frontal lobe lesion).
- **Other specified personality disorder** is a category provided for two situations: 1) the individual's personality pattern meets

the general criteria for a personality disorder, and traits of several different personality disorders are present, but the criteria for any specific personality disorder are not met; or 2) the individual's personality pattern meets the general criteria for a personality disorder, but the individual is considered to have a personality disorder that is not included in the DSM-5 classification (e.g., passive-aggressive personality disorder). **Unspecified personality disorder** is for presentations in which symptoms characteristic of a personality disorder are present but there is insufficient information to make a more specific diagnosis.

The personality disorders are grouped into three clusters based on descriptive similarities. Cluster A includes paranoid, schizoid, and schizotypal personality disorders. Individuals with these disorders often appear odd or eccentric. Cluster B includes antisocial, borderline, histrionic, and narcissistic personality disorders. Individuals with these disorders often appear dramatic, emotional, or erratic. Cluster C includes avoidant, dependent, and obsessive-compulsive personality disorders. Individuals with these disorders often appear anxious or fearful. It should be noted that this clustering system, although useful in some research and educational situations, has serious limitations and has not been consistently validated. For instance, two or more disorders from different clusters, or traits from several of them, can often co-occur and vary in intensity and pervasiveness.

A review of epidemiological studies from several countries found a median prevalence of 3.6% for disorders in Cluster A, 4.5% for Cluster B, 2.8% for Cluster C, and 10.5% for any personality disorder. Prevalence appears to vary across countries and by ethnicity, raising questions about true cross-cultural variation and about the impact of diverse definitions and diagnostic instruments on prevalence assessments.

Dimensional Models for Personality Disorders

The diagnostic approach used in this manual represents the categorical perspective that personality disorders are qualitatively distinct clinical syndromes. An alternative to the categorical approach is the dimensional perspective that personality disorders represent maladaptive variants of personality traits that merge imperceptibly into normality and into one another. See Section III for a full description of a dimensional model for personality disorders. The DSM-5 personality disorder clusters (i.e., odd-eccentric, dramatic-emotional, and anxious-fearful) may also be viewed as dimensions representing spectra of personality dysfunction on a continuum with other mental disorders. The alternative dimensional models have much in common and together appear to cover the important areas of personality dysfunction. Their integration, clinical utility, and relationship with the personality disorder diagnostic categories and various aspects of personality dysfunction continue to be under active investigation. This includes research on whether the dimensional model can clarify the cross-cultural prevalence variations seen with the categorical model.

General Personality Disorder

Criteria

- A. An enduring pattern of inner experience and behavior that deviates markedly from the expectations of the individual's culture. This pattern is manifested in two (or more) of the following areas:
1. Cognition (i.e., ways of perceiving and interpreting self, other people, and events).
 2. Affectivity (i.e., the range, intensity, lability, and appropriateness of emotional response).
 3. Interpersonal functioning.
 4. Impulse control.
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- B. The enduring pattern is inflexible and pervasive across a broad range of personal and social situations.
- C. The enduring pattern leads to clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- D. The pattern is stable and of long duration, and its onset can be traced back at least to adolescence or early adulthood.
- E. The enduring pattern is not better explained as a manifestation or consequence of another mental disorder.
- F. The enduring pattern is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication) or another medical condition (e.g., head trauma).

Diagnostic Features

Personality traits are enduring patterns of perceiving, relating to, and thinking about the environment and oneself that are exhibited in a wide range of social and personal contexts. Only when personality traits are inflexible and maladaptive and cause significant functional impairment or subjective distress do they constitute personality disorders. The essential feature of a personality disorder is an enduring pattern of inner experience and behavior that deviates markedly from the norms and expectations of the individual's culture and is manifested in at least two of the following areas: cognition, affectivity, interpersonal functioning, or impulse control (Criterion A). This enduring pattern is inflexible and pervasive across a broad range of personal and social situations (Criterion B) and leads to clinically significant distress or impairment in social, occupational, or other important areas of functioning (Criterion C). The pattern is stable and of long duration, and its onset can be traced back at least to adolescence or early adulthood (Criterion D). The pattern is not better explained as a manifestation or consequence of another mental disorder (Criterion E) and is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication, exposure to a toxin) or another

medical condition (e.g., head trauma) (Criterion F). Specific diagnostic criteria are also provided for each of the personality disorders included in this chapter.

The diagnosis of personality disorders requires an evaluation of the individual's long-term patterns of functioning, and the particular personality features must be evident by early adulthood. The personality traits that define these disorders must also be distinguished from characteristics that emerge in response to specific situational stressors or more transient mental states (e.g., bipolar, depressive, or anxiety disorders; substance intoxication). The clinician should assess the stability of personality traits over time and across different situations. Although a single interview with the individual is sometimes sufficient for making the diagnosis, it is often necessary to conduct more than one interview and to space these over time. Assessment can also be complicated by the fact that the characteristics that define a personality disorder may not be considered problematic by the individual (i.e., the traits are often ego-syntonic). To help overcome this difficulty, supplementary information from other informants may be helpful.

Development and Course

The features of a personality disorder usually become recognizable during adolescence or early adult life. By definition, a personality disorder is an enduring pattern of thinking, feeling, and behaving that is relatively stable over time. Some types of personality disorder (notably, antisocial and borderline personality disorders) tend to become less evident or to remit with age, whereas this appears to be less true for some other types (e.g., obsessive-compulsive and schizotypal personality disorders).

Personality disorder categories may be applied with children or adolescents in those relatively unusual instances in which the individual's particular maladaptive personality traits appear to be pervasive, persistent, and unlikely to be limited to a particular

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developmental stage or attributable to another mental disorder. It should be recognized that the traits of a personality disorder that appear in childhood will often not persist unchanged into adult life. For a personality disorder to be diagnosed in an individual younger than 18 years, the features must have been present for at least 1 year. The one exception to this is antisocial personality disorder, which cannot be diagnosed in individuals younger than 18 years. Although, by definition, a personality disorder requires an onset no later than early adulthood, individuals may not come to clinical attention until relatively late in life. A personality disorder may be exacerbated following the loss of significant supporting persons (e.g., a spouse) or previously stabilizing social situations (e.g., a job). However, the development of a change in personality in middle adulthood or later life warrants a thorough evaluation to determine the possible presence of a personality change due to another medical condition or an unrecognized substance use disorder.

Culture-Related Diagnostic Issues

Core aspects of personality like emotion regulation and interpersonal functioning are influenced by culture, which also provides means of protection and assimilation and norms for acceptance and denunciation of specific behaviors and personality traits. Judgments about personality

functioning must take into account the individual's ethnic, cultural, and social background. Personality disorders should not be confused with problems associated with acculturation following migration or with the expression of habits, customs, or religious and political values based on the individual's cultural background or context. Behavioral patterns that appear to be rigid and dysfunctional aspects of personality disorder may reflect instead adaptive responses to cultural constraints. For example, reliance on an abusive relationship in a small community where divorce is proscribed may not reflect pathological dependence; conscientious political protest that puts friends and family members at risk with authorities or in conflict with legal norms does not necessarily reflect pathological callousness. There are marked variations in the recognition and diagnosis of personality disorders across cultural, ethnic, and racialized groups. Accuracy of diagnosis can be enhanced by attention to culturally patterned conceptions of self and attachment, assessment biases resulting from clinicians' own cultural backgrounds or use of diagnostic instruments that are not normed to the population being assessed, and the impact of social determinants such as poverty, acculturative stress, racism, and discrimination on feelings, cognitions, and behaviors. It is useful for the clinician, especially when evaluating someone from a different background, to obtain additional information from informants who are familiar with the person's cultural background.

Sex- and Gender-Related Diagnostic Issues

Certain personality disorders (e.g., antisocial personality disorder) are diagnosed more frequently in men. Others (e.g., borderline, histrionic, and dependent personality disorders) are diagnosed more frequently in women; however, in the case of borderline personality disorder, this may be due to higher help-seeking among women. Nonetheless, clinicians must be cautious not to overdiagnose or underdiagnose certain personality disorders in women or in men because of social stereotypes about typical gender roles and behaviors. There is currently insufficient evidence on differences between cis- and transgender individuals with respect to the epidemiology or clinical presentations of personality disorders to draw meaningful conclusions.

Differential Diagnosis

Other mental disorders and personality traits. Many of the specific criteria for the personality disorders describe features (e.g., suspiciousness, dependency, insensitivity) that are also characteristic of episodes of other mental disorders. A personality disorder should

be diagnosed only when the defining characteristics appeared before early adulthood, are typical of the individual's long-term functioning, and do not occur exclusively during an episode of another mental disorder. It may be particularly difficult (and not particularly useful) to distinguish personality disorders from persistent mental disorders such as persistent depressive disorder that have an early onset and an enduring, relatively stable course. Some personality disorders may have a "spectrum" relationship to other mental disorders (e.g., schizotypal personality disorder with schizophrenia; avoidant personality disorder with social anxiety disorder) based on phenomenological or biological similarities or familial aggregation.

Personality disorders must be distinguished from personality traits that do not reach the

threshold for a personality disorder. Personality traits are diagnosed as a personality disorder only when they are inflexible, maladaptive, and persisting and cause significant functional impairment or subjective distress.

Psychotic disorders. For the three personality disorders that may be related to the psychotic disorders (i.e., paranoid, schizoid, and schizotypal), there is an exclusion criterion stating that the pattern of behavior must not have occurred exclusively during the course of schizophrenia, a bipolar or depressive disorder with psychotic features, or another psychotic disorder. When an individual has a persistent mental disorder (e.g., schizophrenia) that was preceded by a preexisting personality disorder, the personality disorder should also be recorded, followed by “premorbid” in parentheses.

Anxiety and depressive disorders. The clinician must be cautious in diagnosing personality disorders during an episode of a depressive disorder or an anxiety disorder, because these conditions may have cross-sectional symptom features that mimic personality traits and may make it more difficult to evaluate retrospectively the individual’s long-term patterns of functioning.

Posttraumatic stress disorder. When personality changes emerge and persist after an individual has been exposed to extreme stress, a diagnosis of posttraumatic stress disorder should be considered.

Substance use disorders. When an individual has a substance use disorder, it is important not to make a personality disorder diagnosis based solely on behaviors that are consequences of substance intoxication or withdrawal or that are associated with activities in the service of sustaining substance use (e.g., antisocial behavior).

Personality change due to another medical condition. When enduring changes in personality arise as a result of the physiological effects of another medical condition (e.g., brain tumor), a diagnosis of personality change due to another medical condition should be considered.

Cluster A Personality Disorders

Paranoid Personality Disorder

Diagnostic Criteria	F60.0
A. A pervasive distrust and suspiciousness of others such that their motives are interpreted as malevolent, beginning by early adulthood and present in a variety of contexts, as indicated by four (or more) of the following:	
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1. Suspects, without sufficient basis, that others are exploiting, harming, or deceiving him or her.	

2. Is preoccupied with unjustified doubts about the loyalty or trustworthiness of friends or associates.
 3. Is reluctant to confide in others because of unwarranted fear that the information will be used maliciously against him or her.
 4. Reads hidden demeaning or threatening meanings into benign remarks or events.
 5. Persistently bears grudges (i.e., is unforgiving of insults, injuries, or slights).
 6. Perceives attacks on his or her character or reputation that are not apparent to others and is quick to react angrily or to counterattack.
 7. Has recurrent suspicions, without justification, regarding fidelity of spouse or sexual partner.
- B. Does not occur exclusively during the course of schizophrenia, a bipolar disorder or depressive disorder with psychotic features, or another psychotic disorder and is not attributable to the physiological effects of another medical condition.
- Note:** If criteria are met prior to the onset of schizophrenia, add “premorbid,” i.e., “paranoid personality disorder (premorbid).”

Diagnostic Features

The essential feature of paranoid personality disorder is a pattern of pervasive distrust and suspiciousness of others such that their motives are interpreted as malevolent. This pattern begins by early adulthood and is present in a variety of contexts.

Individuals with this disorder assume that other people will exploit, harm, or deceive them, even if no evidence exists to support this expectation (Criterion A1). They suspect on the basis of little or no evidence that others are plotting against them and may attack them suddenly, at any time and without reason. They often feel that they have been deeply and irreversibly injured by another person or persons even when there is no objective evidence for this. They are preoccupied with unjustified doubts about the loyalty or trustworthiness of their friends and associates, whose actions are minutely scrutinized for evidence of hostile intentions (Criterion A2). Any perceived deviation from trustworthiness or loyalty serves to support their underlying assumptions. They are so amazed when a friend or associate shows loyalty that they cannot trust or believe it. If they get into trouble, they expect that friends and associates will either attack or ignore them.

Individuals with paranoid personality disorder are reluctant to confide in or become close to others because they fear that the information they share will be used against them (Criterion A3). They may refuse to answer personal questions, saying that the information is “nobody’s business.” They read hidden meanings that are demeaning and threatening into benign remarks or events (Criterion A4). For example, an individual with this disorder may misinterpret an honest mistake by a store clerk as a deliberate attempt to shortchange, or view a casual humorous remark by a coworker as a serious character attack. Compliments are often misinterpreted (e.g., a compliment on a new acquisition is misinterpreted as a criticism for selfishness; a compliment on an accomplishment is misinterpreted as an attempt to coerce more and better performance). They may view an offer of help as a criticism that they are not doing well enough on their own.

Individuals with this disorder persistently bear grudges and are unwilling to forgive the insults, injuries, or slights that they think they have received (Criterion A5). Minor slights arouse major hostility, and the hostile feelings persist for a long time. Because they are constantly vigilant to the harmful intentions of others, they very often feel that their character or reputation has been attacked or that they have been slighted in some other way. They are quick to counterattack and react with anger to perceived insults (Criterion A6). Individuals with this disorder may be pathologically jealous, often suspecting that their spouse

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or sexual partner is unfaithful without any adequate justification (Criterion A7). They may gather trivial and circumstantial “evidence” to support their jealous beliefs. They want to maintain complete control of intimate relationships to avoid being betrayed and may constantly question and challenge the whereabouts, actions, intentions, and fidelity of their spouse or partner.

Paranoid personality disorder should not be diagnosed if the pattern of behavior occurs exclusively during the course of schizophrenia, a bipolar disorder or depressive disorder with psychotic features, or another psychotic disorder, or if it is attributable to the physiological effects of a neurological (e.g., temporal lobe epilepsy) or another medical condition (Criterion B).

Associated Features

Individuals with paranoid personality disorder are generally difficult to get along with and often have problems with close relationships. Their excessive suspiciousness and hostility may be expressed in overt argumentativeness, in recurrent complaining, or by hostile aloofness. They display a labile range of affect, with hostile, stubborn, and sarcastic expressions predominating. Their combative and suspicious nature may elicit a hostile response in others, which then serves to confirm their original expectations.

Because individuals with paranoid personality disorder lack trust in others, they need to have a high degree of control over those around them. They are often rigid, critical of others, and unable to collaborate, although they have great difficulty accepting criticism themselves. They may blame others for their own shortcomings. Because of their quickness to counterattack in response to the threats they perceive around them, they may be litigious and frequently become involved in legal disputes. Individuals with this disorder seek to confirm their preconceived negative notions regarding people or situations they encounter, attributing malevolent motivations to others that are projections of their own fears. They may exhibit thinly hidden, unrealistic grandiose fantasies, are often attuned to issues of power and rank, and tend to develop negative stereotypes of others, particularly those from population groups distinct from their own. Attracted by simplistic formulations of the world, they are often wary of ambiguous situations. They may be perceived as “fanatics” and form tightly knit “cults” or groups with others who share their paranoid belief systems.

Prevalence

The estimated prevalence of paranoid personality based on a probability subsample from Part II

of the National Comorbidity Survey Replication was 2.3%. The prevalence of paranoid personality disorder in the National Epidemiologic Survey on Alcohol and Related Conditions was 4.4%. A review of six epidemiological studies (four in the United States) found a median prevalence of 3.2%. In forensic settings, the estimated prevalence may be as high as 23%.

Development and Course

Paranoid personality disorder may be first apparent in childhood and adolescence with solitariness, poor peer relationships, social anxiety, underachievement in school, and interpersonal hypersensitivity. Adolescent onset of paranoid personality disorder is associated with a prior history of childhood maltreatment, externalizing symptoms, bullying of peers, and adult appearance of interpersonal aggression.

Risk and Prognostic Factors

Environmental. Exposure to social stressors such as socioeconomic inequality, marginalization, and racism is associated with decreased trust, which in some cases is adaptive.

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The combination of social stress and childhood maltreatment accounts for the increased prevalence of paranoid symptoms in social groups facing racial discrimination. Both longitudinal and cross-sectional studies confirm that childhood trauma is a risk factor for paranoid personality disorder.

Genetic and physiological. There is some evidence for an increased prevalence of paranoid personality disorder in relatives of probands with schizophrenia and for a more specific familial relationship with delusional disorder, persecutory type.

Culture-Related Diagnostic Issues

Some behaviors that are influenced by sociocultural contexts or specific life circumstances may be erroneously labeled paranoid and may even be reinforced by the process of clinical evaluation. Migrants, members of socially oppressed ethnic and racialized populations, and other groups facing social adversity, racism, and discrimination may display guarded or defensive behaviors because of unfamiliarity (e.g., language barriers or lack of knowledge of rules and regulations) or in response to the neglect, hostility, or indifference of the majority society. Some cultural groups develop low generalized trust, especially of outgroup members, which may lead to behaviors that can be misjudged as paranoid. These include guardedness, limited outward emotionality, cognitive rigidity, social distance, and hostility or defensiveness in situations experienced as unfair or discriminatory. These behaviors can, in turn, generate anger and frustration in others, including clinicians, thus setting up a vicious cycle of mutual mistrust, which should not be confused with paranoid traits or paranoid personality disorder.

Sex- and Gender-Related Diagnostic Issues

While paranoid personality disorder was found to be more common in men than in women in a meta-analysis relying on clinical and community samples, the National Epidemiologic Survey on

Alcohol and Related Conditions found it to be more common in women.

Differential Diagnosis

Other mental disorders with psychotic symptoms. Paranoid personality disorder can be distinguished from delusional disorder, persecutory type; schizophrenia; and a bipolar or depressive disorder with psychotic features because these disorders are all characterized by a period of persistent psychotic symptoms (e.g., delusions and hallucinations). For an additional diagnosis of paranoid personality disorder to be given, the personality disorder must have been present before the onset of psychotic symptoms and must persist when the psychotic symptoms are in remission. When an individual has another persistent mental disorder (e.g., schizophrenia) that was preceded by paranoid personality disorder, paranoid personality disorder should also be recorded, followed by “premorbid” in parentheses.

Personality change due to another medical condition. Paranoid personality disorder must be distinguished from personality change due to another medical condition, in which the traits that emerge are a direct physiological consequence of another medical condition.

Substance use disorders. Paranoid personality disorder must be distinguished from symptoms that may develop in association with persistent substance use.

Paranoid traits associated with physical handicaps. The disorder must also be distinguished from paranoid traits associated with the development of physical handicaps (e.g., a hearing impairment).

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Other personality disorders and personality traits. Other personality disorders may be confused with paranoid personality disorder because they have certain features in common. It is therefore important to distinguish among these disorders based on differences in their characteristic features. However, if an individual has personality features that meet criteria for one or more personality disorders in addition to paranoid personality disorder, all can be diagnosed. Paranoid personality disorder and schizotypal personality disorder share the traits of suspiciousness, interpersonal aloofness, and paranoid ideation, but schizotypal personality disorder also includes symptoms such as magical thinking, unusual perceptual experiences, and odd thinking and speech. Individuals with behaviors that meet criteria for schizoid personality disorder are often perceived as strange, eccentric, cold, and aloof, but they do not usually have prominent paranoid ideation. The tendency of individuals with paranoid personality disorder to react to minor stimuli with anger is also seen in borderline and histrionic personality disorders. However, these disorders are not necessarily associated with pervasive suspiciousness, and borderline personality disorder exhibits higher levels of impulsivity and self-destructive behavior. People with avoidant personality disorder may also be reluctant to confide in others, but more from fear of being embarrassed or found inadequate than from fear of others’ malicious intent. Although antisocial behavior may be present in some individuals with paranoid personality disorder, it is not usually motivated by a desire for personal gain or to exploit others as in antisocial personality disorder, but rather is more often attributable to a desire for revenge. Individuals with narcissistic personality disorder may occasionally display suspiciousness, social withdrawal, or alienation, but this derives primarily from fears of having their imperfections or flaws revealed.

Paranoid traits may be adaptive, particularly in threatening environments. Paranoid personality disorder should be diagnosed only when these traits are inflexible, maladaptive, and persisting and cause significant functional impairment or subjective distress.

Comorbidity

Particularly in response to stress, individuals with this disorder may experience very brief psychotic episodes (lasting minutes to hours). In some instances, paranoid personality disorder may appear as the premorbid antecedent of delusional disorder or schizophrenia. Individuals with paranoid personality disorder may develop major depressive disorder and may be at increased risk for agoraphobia and obsessive-compulsive disorder. Alcohol and other substance use disorders frequently occur. The most common co-occurring personality disorders appear to be schizotypal, schizoid, narcissistic, avoidant, and borderline.

Schizoid Personality Disorder

Diagnostic Criteria

F60.1

- A. A pervasive pattern of detachment from social relationships and a restricted range of expression of emotions in interpersonal settings, beginning by early adulthood and present in a variety of contexts, as indicated by four (or more) of the following:
1. Neither desires nor enjoys close relationships, including being part of a family.
 2. Almost always chooses solitary activities.
 3. Has little, if any, interest in having sexual experiences with another person.
 4. Takes pleasure in few, if any, activities.
 5. Lacks close friends or confidants other than first-degree relatives.
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6. Appears indifferent to the praise or criticism of others.
 7. Shows emotional coldness, detachment, or flattened affectivity.
- B. Does not occur exclusively during the course of schizophrenia, a bipolar disorder or depressive disorder with psychotic features, another psychotic disorder, or autism spectrum disorder and is not attributable to the physiological effects of another medical condition.

Note: If criteria are met prior to the onset of schizophrenia, add “premorbid,” i.e., “schizoid personality disorder (premorbid).”

Diagnostic Features

The essential feature of schizoid personality disorder is a pervasive pattern of detachment from social relationships and a restricted range of expression of emotions in interpersonal settings. This pattern begins by early adulthood and is present in a variety of contexts.

Individuals with schizoid personality disorder appear to lack a desire for intimacy, seem indifferent to opportunities to develop close relationships, and do not seem to derive much satisfaction from being part of a family or other social group (Criterion A1). They prefer spending time by themselves, rather than being with other people. They often appear to be socially isolated or “loners” and almost always choose solitary activities or hobbies that do not include interaction with others (Criterion A2). They prefer mechanical or abstract tasks, such as computer or mathematical games. They may have very little interest in having sexual experiences with another person (Criterion A3) and take pleasure in few, if any, activities (Criterion A4). There is usually a reduced experience of pleasure from sensory, bodily, or interpersonal experiences, such as walking on a beach at sunset or having sex. These individuals have no close friends or confidants, except possibly a first-degree relative (Criterion A5).

Individuals with schizoid personality disorder often seem indifferent to the approval or criticism of others and do not appear to be bothered by what others may think of them (Criterion A6). They may be oblivious to the normal subtleties of social interaction and often do not respond appropriately to social cues so that they seem socially inept or superficial and self-absorbed. They usually display a “bland” exterior without visible emotional reactivity and rarely reciprocate gestures or facial expressions, such as smiles or nods (Criterion A7). They claim that they rarely experience strong emotions such as anger and joy. They often display a constricted affect and appear cold and aloof. However, in those very unusual circumstances in which these individuals become at least temporarily comfortable in revealing themselves, they may acknowledge having painful feelings, particularly related to social interactions.

Schizoid personality disorder should not be diagnosed if the pattern of behavior occurs exclusively during the course of schizophrenia, a bipolar or depressive disorder with psychotic features, another psychotic disorder, or autism spectrum disorder, or if it is attributable to the physiological effects of a neurological (e.g., temporal lobe epilepsy) or another medical condition (Criterion B).

Associated Features

Individuals with schizoid personality disorder may have particular difficulty expressing anger, even in response to direct provocation, which contributes to the impression that they lack emotion. Their lives sometimes seem directionless, and they may appear to “drift” in their goals. Such individuals often react passively to adverse circumstances and have difficulty responding appropriately to important life events. Because of their lack of social skills and lack of desire for sexual experiences, individuals with this disorder have few friendships, date infrequently, and often do not marry. Occupational functioning may be impaired, particularly if interpersonal involvement is required, but individuals with this disorder may do well when they work under conditions of social isolation.

Prevalence

Schizoid personality disorder is uncommon in clinical settings. The estimated prevalence of schizoid personality disorder based on a probability subsample from Part II of the National Comorbidity Survey Replication was 4.9%. The prevalence of schizoid personality disorder in the National Epidemiologic Survey on Alcohol and Related Conditions was 3.1%. A review of six epidemiological studies (four in the United States) found a median prevalence of 1.3%.

Development and Course

Schizoid personality disorder may be first apparent in childhood and adolescence with solitariness, poor peer relationships, and underachievement in school, which mark these children or adolescents as different and make them subject to teasing.

Risk and Prognostic Factors

Genetic and physiological. Schizoid personality disorder may have increased prevalence in the relatives of individuals with schizophrenia or schizotypal personality disorder.

Culture-Related Diagnostic Issues

Individuals from a variety of cultural backgrounds sometimes exhibit defensive behaviors and interpersonal styles that may be erroneously labeled as “schizoid.” For example, those who have moved from rural to metropolitan environments may react with “emotional freezing” that may last for several months and manifest as solitary activities, constricted affect, and other deficits in communication. Immigrants from other countries are sometimes mistakenly perceived as cold, hostile, or indifferent, which may be a response to social ostracism from the host society.

Sex- and Gender-Related Diagnostic Issues

While some research suggests that schizoid personality disorder may be more common in men, other research suggests that there is no gender difference in prevalence.

Differential Diagnosis

Other mental disorders with psychotic symptoms. Schizoid personality disorder can be distinguished from delusional disorder, schizophrenia, and a bipolar or depressive disorder with psychotic features because these disorders are all characterized by a period of persistent psychotic symptoms (e.g., delusions and hallucinations). To give an additional diagnosis of schizoid personality disorder, the personality disorder must have been present before the onset of psychotic symptoms and must persist when the psychotic symptoms are in remission. When an individual has a persistent psychotic disorder (e.g., schizophrenia) that was preceded by schizoid personality disorder, schizoid personality disorder should also be recorded, followed by “premorbid” in parentheses.

Autism spectrum disorder. There may be great difficulty differentiating individuals with schizoid personality disorder from individuals with autism spectrum disorder, particularly with milder forms of either disorder, as both include a seeming indifference to companionship with others. However, autism spectrum disorder may be differentiated by stereotyped behaviors and interests.

Personality change due to another medical condition. Schizoid personality disorder must be

distinguished from personality change due to another medical condition, in which the traits that emerge are a direct physiological consequence of another medical condition.

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Substance use disorders. Schizoid personality disorder must also be distinguished from symptoms that may develop in association with persistent substance use.

Other personality disorders and personality traits. Other personality disorders may be confused with schizoid personality disorder because they have certain features in common. It is, therefore, important to distinguish among these disorders based on differences in their characteristic features. However, if an individual has personality features that meet criteria for one or more personality disorders in addition to schizoid personality disorder, all can be diagnosed. Although characteristics of social isolation and restricted affectivity are common to schizoid, schizotypal, and paranoid personality disorders, schizoid personality disorder can be distinguished from schizotypal personality disorder by the lack of cognitive and perceptual distortions and from paranoid personality disorder by the lack of suspiciousness and paranoid ideation. The social isolation of schizoid personality disorder can be distinguished from that of avoidant personality disorder, which is attributable to fear of being embarrassed or found inadequate and excessive anticipation of rejection. In contrast, people with schizoid personality disorder have a more pervasive detachment and limited desire for social intimacy. Individuals with obsessive-compulsive personality disorder may also show an apparent social detachment stemming from devotion to work and discomfort with emotions, but they do have an underlying capacity for intimacy.

Individuals who are “loners” or quite introverted may display personality traits that might be considered schizoid, consistent with the broader conceptualization of schizoid personality disorder as a disorder defined by pathological introversion/detachment. Only when these traits are inflexible and maladaptive and cause significant functional impairment or subjective distress do they constitute schizoid personality disorder.

Comorbidity

Particularly in response to stress, individuals with this disorder may experience very brief psychotic episodes (lasting minutes to hours). In some instances, schizoid personality disorder may appear as the premorbid antecedent of delusional disorder or schizophrenia. Individuals with this disorder may sometimes develop major depressive disorder. Schizoid personality disorder most often co-occurs with schizotypal, paranoid, and avoidant personality disorders.

Schizotypal Personality Disorder

Diagnostic Criteria

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- A. A pervasive pattern of social and interpersonal deficits marked by acute discomfort with, and reduced capacity for, close relationships as well as by

cognitive or perceptual distortions and eccentricities of behavior, beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:

1. Ideas of reference (excluding delusions of reference).
2. Odd beliefs or magical thinking that influences behavior and is inconsistent with subcultural norms (e.g., superstitiousness, belief in clairvoyance, telepathy, or “sixth sense”; in children and adolescents, bizarre fantasies or preoccupations).
3. Unusual perceptual experiences, including bodily illusions.
4. Odd thinking and speech (e.g., vague, circumstantial, metaphorical, overelaborate, or stereotyped).
5. Suspiciousness or paranoid ideation.
6. Inappropriate or constricted affect.

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7. Behavior or appearance that is odd, eccentric, or peculiar.
 8. Lack of close friends or confidants other than first-degree relatives.
 9. Excessive social anxiety that does not diminish with familiarity and tends to be associated with paranoid fears rather than negative judgments about self.
- B. Does not occur exclusively during the course of schizophrenia, a bipolar disorder or depressive disorder with psychotic features, another psychotic disorder, or autism spectrum disorder.

Note: If criteria are met prior to the onset of schizophrenia, add “premorbid,” e.g., “schizotypal personality disorder (premorbid).”

Diagnostic Features

The essential feature of schizotypal personality disorder is a pervasive pattern of social and interpersonal deficits marked by acute discomfort with, and reduced capacity for, close relationships as well as by cognitive or perceptual distortions and eccentricities of behavior. This pattern begins by early adulthood and is present in a variety of contexts.

Individuals with schizotypal personality disorder often have ideas of reference (i.e., incorrect interpretations of casual incidents and external events as having a particular and unusual meaning specifically for the person) (Criterion A1). These should be distinguished from delusions of reference, in which the beliefs are held with delusional conviction. These individuals may be superstitious or preoccupied with paranormal phenomena that are outside the norms of their subculture (Criterion A2). They may feel that they have special powers to sense events before they happen or to read others’ thoughts. They may believe that they have magical control over others, which can be implemented directly (e.g., believing that their spouse’s taking the dog out for a walk is the direct result of thinking an hour earlier it should be done) or indirectly through compliance with magical rituals (e.g., walking past a specific object three times to avoid a certain harmful outcome). Perceptual alterations may be present (e.g., sensing

that another person is present or hearing a voice murmuring their name) (Criterion A3). Their speech may include unusual or idiosyncratic phrasing and construction. It is often loose, digressive, or vague, but without actual derailment or incoherence (Criterion A4). Responses can be either overly concrete or overly abstract, and words or concepts are sometimes applied in unusual ways (e.g., the individual may state that he or she was not “talkable” at work).

Individuals with this disorder are often suspicious and may have paranoid ideation (e.g., believing their colleagues at work are intent on undermining their reputation with the boss) (Criterion A5). They are usually not able to negotiate the full range of affects and interpersonal cuing required for successful relationships and thus often appear to interact with others in an inappropriate, stiff, or constricted fashion (Criterion A6). These individuals are often considered to be odd or eccentric because of unusual mannerisms, an often unkempt manner of dress that does not quite “fit together,” and inattention to the usual social conventions (e.g., the individual may avoid eye contact, wear clothes that are ink stained and ill-fitting, and be unable to join in the give-and-take banter of co-workers) (Criterion A7).

Individuals with schizotypal personality disorder experience interpersonal relatedness as problematic and are uncomfortable relating to other people. Although they may express unhappiness about their lack of relationships, their behavior suggests a decreased desire for intimate contacts. As a result, they usually have no or few close friends or confidants other than a first-degree relative (Criterion A8). They are anxious in social situations, particularly those involving unfamiliar people (Criterion A9). They will interact with other individuals when they have to but prefer to keep to themselves because they feel that they are different and just do not “fit in.” Their social anxiety does not easily abate, even when they spend more time in the setting or become more familiar with the other people, because their anxiety tends to be associated with suspiciousness regarding others’ motivations. For example, when attending a dinner party, the individual with schizotypal

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personality disorder will not become more relaxed as time goes on, but rather may become increasingly tense and suspicious.

Schizotypal personality disorder should not be diagnosed if the pattern of behavior occurs exclusively during the course of schizophrenia, a bipolar or depressive disorder with psychotic features, another psychotic disorder, or autism spectrum disorder (Criterion B).

Associated Features

Individuals with schizotypal personality disorder often seek treatment for the associated symptoms of anxiety or depression rather than for the personality disorder features per se.

Prevalence

The estimated prevalence of schizotypal personality disorder based on a probability subsample from Part II of the National Comorbidity Survey Replication was 3.3%. The prevalence of schizotypal personality disorder in the National Epidemiologic Survey on Alcohol and Related Conditions data was 3.9%. A review of five epidemiological studies (three in the United States) found a median prevalence of 0.6%.

Development and Course

Schizotypal personality disorder has a relatively stable course, with only a small proportion of individuals going on to develop schizophrenia or another psychotic disorder. Schizotypal personality disorder may be first apparent in childhood and adolescence with solitariness, poor peer relationships, social anxiety, underachievement in school, hypersensitivity, peculiar thoughts and language, and bizarre fantasies. These children may appear “odd” or “eccentric” and attract teasing.

Risk and Prognostic Factors

Genetic and physiological. Schizotypal personality disorder appears to aggregate familially and is more prevalent among the first-degree biological relatives of individuals with schizophrenia than among the general population. There may also be a modest increase in schizophrenia and other psychotic disorders in the relatives of probands with schizotypal personality disorder. Twin studies indicate highly stable genetic factors and rather transient environmental factors for an increased risk for the schizotypal syndrome, and genetic risk variants for schizophrenia may be linked to schizotypal personality disorder. Neuroimaging studies detect group-level differences in the size and function of specific brain regions in individuals with schizotypal personality disorder in comparison with healthy persons, individuals with schizophrenia, and individuals with other personality disorders.

Culture-Related Diagnostic Issues

Cognitive and perceptual distortions must be evaluated in the context of the individual’s cultural milieu. Pervasive culturally determined characteristics, particularly those regarding supernatural and religious beliefs and practices (life beyond death, speaking in tongues, voodoo, shamanism, mind reading, sixth sense, evil eye, magical beliefs related to health and illness), can appear to be schizotypal to the uninformed clinician. Thus, observed cross-national and cross-ethnic variations in the prevalence and expression of schizotypal traits may be a true epidemiological finding or one due to differences in the cultural acceptance of these experiences.

Sex- and Gender-Related Diagnostic Issues

Schizotypal personality disorder appears to be slightly more common in men than in women.

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Differential Diagnosis

Other mental disorders with psychotic symptoms. Schizotypal personality disorder can be distinguished from delusional disorder, schizophrenia, and a bipolar or depressive disorder with psychotic features because these disorders are all characterized by a period of persistent psychotic symptoms (e.g., delusions and hallucinations). To give an additional diagnosis of schizotypal personality disorder, the personality disorder must have been present before the onset of psychotic symptoms and persist when the psychotic symptoms are in remission. When an individual has a persistent psychotic disorder (e.g., schizophrenia) that was preceded by schizotypal personality disorder, schizotypal personality disorder should also be recorded,

followed by “premorbid” in parentheses.

Neurodevelopmental disorders. There may be great difficulty differentiating children with schizotypal personality disorder from the heterogeneous group of solitary, odd children whose behavior is characterized by marked social isolation, eccentricity, or peculiarities of language and whose diagnoses would probably include milder forms of autism spectrum disorder or language communication disorders. Communication disorders may be differentiated by the primacy and severity of the disorder in language and by the characteristic features of impaired language found in a specialized language assessment. Milder forms of autism spectrum disorder are differentiated by the even greater lack of social awareness and emotional reciprocity and stereotyped behaviors and interests.

Personality change due to another medical condition. Schizotypal personality disorder must be distinguished from personality change due to another medical condition, in which the traits that emerge are a direct physiological consequence of another medical condition.

Substance use disorders. Schizotypal personality disorder must also be distinguished from symptoms that may develop in association with persistent substance use.

Other personality disorders and personality traits. Other personality disorders may be confused with schizotypal personality disorder because they have certain features in common. It is, therefore, important to distinguish among these disorders based on differences in their characteristic features. However, if an individual has personality features that meet criteria for one or more personality disorders in addition to schizotypal personality disorder, all can be diagnosed. Although paranoid and schizoid personality disorders may also be characterized by social detachment and restricted affect, schizotypal personality disorder can be distinguished from these two diagnoses by the presence of cognitive or perceptual distortions and marked eccentricity or oddness. Close relationships are limited in both schizotypal personality disorder and avoidant personality disorder; however, in avoidant personality disorder an active desire for relationships is constrained by a fear of rejection, whereas in schizotypal personality disorder there is a lack of desire for relationships and persistent detachment. Individuals with narcissistic personality disorder may also display suspiciousness, social withdrawal, or alienation, but in narcissistic personality disorder these qualities derive primarily from fears of having imperfections or flaws revealed. Individuals with borderline personality disorder may also have transient, psychotic-like symptoms, but these are usually more closely related to affective shifts in response to stress (e.g., intense anger, anxiety, disappointment) and are usually more dissociative (e.g., derealization, depersonalization). In contrast, individuals with schizotypal personality disorder are more likely to have enduring psychotic-like symptoms that may worsen under stress but are less likely to be invariably associated with pronounced affective symptoms. Although social isolation may occur in borderline personality disorder, it is usually secondary to repeated interpersonal failures due to angry outbursts and frequent mood shifts, rather than a result of a persistent lack of social contacts and desire for intimacy. Furthermore, individuals with schizotypal personality disorder do not usually demonstrate the impulsive or manipulative behaviors of the individual with

borderline personality disorder. However, there is a high rate of co-occurrence between the two disorders, so that making such distinctions is not always feasible. Schizotypal features during

adolescence may be reflective of transient emotional turmoil rather than an enduring personality disorder.

Comorbidity

Particularly in response to stress, individuals with this disorder may experience transient psychotic episodes (lasting minutes to hours), although they usually are insufficient in duration to warrant an additional diagnosis such as brief psychotic disorder or schizophreniform disorder. In some cases, clinically significant psychotic symptoms may develop that meet criteria for brief psychotic disorder, schizophreniform disorder, delusional disorder, or schizophrenia. There is considerable co-occurrence with schizoid, paranoid, avoidant, and borderline personality disorders.

Cluster B Personality Disorders

Antisocial Personality Disorder

Diagnostic Criteria	F60.2
A. A pervasive pattern of disregard for and violation of the rights of others, occurring since age 15 years, as indicated by three (or more) of the following: 1. Failure to conform to social norms with respect to lawful behaviors, as indicated by repeatedly performing acts that are grounds for arrest. 2. Deceitfulness, as indicated by repeated lying, use of aliases, or conning others for personal profit or pleasure. 3. Impulsivity or failure to plan ahead. 4. Irritability and aggressiveness, as indicated by repeated physical fights or assaults. 5. Reckless disregard for safety of self or others. 6. Consistent irresponsibility, as indicated by repeated failure to sustain consistent work behavior or honor financial obligations. 7. Lack of remorse, as indicated by being indifferent to or rationalizing having hurt, mistreated, or stolen from another. B. The individual is at least age 18 years. C. There is evidence of conduct disorder with onset before age 15 years. D. The occurrence of antisocial behavior is not exclusively during the course of schizophrenia or bipolar disorder.	

Diagnostic Features

The essential feature of antisocial personality disorder is a pervasive pattern of disregard for, and violation of, the rights of others that begins in childhood or early adolescence and continues into adulthood. This pattern has also been referred to as *psychopathy*, *sociopathy*, or *dysocial personality disorder*. Because deceit and manipulation are central features of antisocial personality disorder, it may be especially helpful to integrate information acquired from systematic clinical assessment with information collected from collateral sources.

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For this diagnosis to be given, the individual must be at least age 18 years (Criterion B) and must have had evidence of conduct disorder with onset before age 15 years (Criterion C). Conduct disorder involves a repetitive and persistent pattern of behavior in which the basic rights of others or major age-appropriate societal norms or rules are violated. The specific behaviors characteristic of conduct disorder fall into one of four categories: aggression to people and animals, destruction of property, deceitfulness or theft, or serious violation of rules.

The pattern of antisocial behavior continues into adulthood. Individuals with antisocial personality disorder fail to conform to social norms with respect to lawful behavior (Criterion A1). They may repeatedly perform acts that are grounds for arrest (whether they are arrested or not), such as destroying property, harassing others, stealing, or pursuing illegal occupations. Persons with this disorder disregard the wishes, rights, or feelings of others. They are frequently deceitful and manipulative in order to gain personal profit or pleasure (e.g., to obtain money, sex, or power) (Criterion A2). They may repeatedly lie, use an alias, con others, or malingering. A pattern of impulsivity may be manifested by a failure to plan ahead (Criterion A3). Decisions are made on the spur of the moment, without forethought and without consideration for the consequences to self or others; this may lead to sudden changes of jobs, residences, or relationships. Individuals with antisocial personality disorder tend to be irritable and aggressive and may repeatedly get into physical fights or commit acts of physical assault (including spouse beating or child beating) (Criterion A4). (Aggressive acts that are required to defend oneself or someone else are not considered to be evidence for this item.) These individuals also display a reckless disregard for the safety of themselves or others (Criterion A5). This may be evidenced in their driving behavior (i.e., recurrent speeding, driving while intoxicated, multiple accidents). They may engage in sexual behavior or substance use that has a high risk for harmful consequences. They may neglect or fail to care for a child in a way that puts the child in danger.

Individuals with antisocial personality disorder also tend to be consistently and extremely irresponsible (Criterion A6). Irresponsible work behavior may be indicated by significant periods of unemployment despite available job opportunities, or by abandonment of several jobs without a realistic plan for getting another job. There may also be a pattern of repeated absences from work that are not explained by illness either in themselves or in their family. Financial irresponsibility is indicated by acts such as defaulting on debts, failing to provide child support, or failing to support other dependents on a regular basis. Individuals with antisocial personality disorder show little remorse for the consequences of their acts (Criterion A7). They may be indifferent to, or provide a superficial rationalization for, having hurt, mistreated, or stolen from someone (e.g., “life’s unfair,” “losers deserve to lose”). These individuals may blame the victims for being foolish, helpless, or deserving their fate (e.g., “he had it coming anyway”); they may minimize the harmful consequences of their actions; or they may simply indicate complete

indifference. They generally fail to compensate or make amends for their behavior. They may believe that everyone is out to “help number one” and that one should stop at nothing to avoid being pushed around.

The antisocial behavior must not occur exclusively during the course of schizophrenia or bipolar disorder (Criterion D).

Associated Features

Individuals with antisocial personality disorder frequently lack empathy and tend to be callous, cynical, and contemptuous of the feelings, rights, and sufferings of others. They may have an inflated and arrogant self-appraisal (e.g., feel that ordinary work is beneath them or lack a realistic concern about their current problems or their future) and may be excessively opinionated, self-assured, or cocky. Some antisocial individuals may display a glib, superficial charm and can be quite voluble and verbally facile (e.g., using technical terms or jargon that might impress someone who is unfamiliar with the topic). Lack of

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empathy, inflated self-appraisal, and superficial charm are features that have been commonly included in traditional conceptions of psychopathy that may be particularly distinguishing of the disorder and more predictive of recidivism in prison or forensic settings, where criminal, delinquent, or aggressive acts are likely to be nonspecific. These individuals may also be irresponsible and exploitative in their sexual relationships. They may have a history of many sexual partners and may never have sustained a monogamous relationship. They may be irresponsible as parents, as evidenced by malnutrition of a child, an illness in the child resulting from a lack of minimal hygiene, a child’s dependence on neighbors or nonresident relatives for food or shelter, a failure to arrange for a caretaker for a young child when the individual is away from home, or repeated squandering of money required for household necessities. These individuals may receive dishonorable discharges from the armed services, may fail to be self-supporting, may become impoverished or even homeless, or may spend many years in penal institutions. Individuals with antisocial personality disorder are more likely than individuals in the general population to die prematurely from natural causes and suicide.

Prevalence

The estimated prevalence of antisocial personality disorder based on a probability subsample from Part II of the National Comorbidity Survey Replication was 0.6%. The prevalence of antisocial personality disorder in the National Epidemiologic Survey on Alcohol and Related Conditions data was 3.6%. A review of seven epidemiological studies (six in the United States) found a median prevalence of 3.6%. The highest prevalence of antisocial personality disorder (greater than 70%) is among samples of men with the most severe alcohol use disorders and from substance abuse clinics, prisons, or other forensic settings. Lifetime prevalence appears to be similar across non-Latinx White and Black individuals and lower in Latinx and Asian Americans. Prevalence may be higher in samples affected by adverse socioeconomic (i.e., poverty) or sociocultural (i.e., migration) factors.

Development and Course

Antisocial personality disorder has a chronic course but may become less evident or remit as the individual grows older, often by age 40. Although this remission tends to be particularly evident with respect to engaging in criminal behavior, there is likely to be a decrease in the full spectrum of antisocial behaviors and substance use. By definition, antisocial personality cannot be diagnosed before age 18 years.

Risk and Prognostic Factors

Environmental. Child abuse or neglect, unstable or erratic parenting, or inconsistent parental discipline may increase the likelihood that conduct disorder will evolve into antisocial personality disorder.

Genetic and physiological. Antisocial personality disorder is more common among the first-degree biological relatives of those with the disorder than in the general population. Biological relatives of individuals with this disorder are also at increased risk for somatization disorder (a diagnosis that was replaced in DSM-5 with somatic symptom disorder) and substance use disorders. Within a family that has a member with antisocial personality disorder, males more often have antisocial personality disorder and substance use disorders, whereas females more often have somatization disorder.

Culture-Related Diagnostic Issues

Antisocial personality disorder has been associated with low socioeconomic status and urban settings. The diagnosis may at times be misapplied to individuals in settings in which

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seemingly antisocial behavior may be part of a protective survival strategy (e.g., formation of youth gangs in urban areas with high rates of violence and discrimination). Sociocultural contexts with high rates of child maltreatment or exposure to violence also tend to have elevated prevalence of antisocial behaviors, suggesting either a potential risk factor for the development of antisocial personality disorder or an adverse environment that evokes reactive and contextual antisocial behaviors that do not represent pervasive and enduring traits consistent with a personality disorder. In assessing antisocial traits, it is helpful for the clinician to consider the social and economic context in which the behaviors occur. In the National Epidemiologic Survey on Alcohol and Related Conditions, prevalence appears to vary across U.S. ethnic and racialized groups, possibly because of a combination of true prevalence differences, measurement artifacts, and the impact of adverse environments that generate behaviors that resemble those of antisocial personality disorder but are instead reactive and contextual. Individuals from some socially oppressed groups may be at higher risk for misdiagnosis or overdiagnosis of antisocial personality disorder because they are more likely to be misdiagnosed with conduct disorder in adolescence, which is a requirement for a diagnosis of antisocial personality disorder.

Sex- and Gender-Related Diagnostic Issues

Antisocial personality disorder is three times as common in men than in women. Women with

antisocial personality disorder are more likely to have experienced childhood and adult adverse experiences such as sexual abuse compared with men. Clinical presentation may vary, with men more often presenting with irritability/aggression and reckless disregard for the safety of others compared with women. Comorbid substance use disorders are more common in men, while comorbid mood and anxiety disorders are more common in women. There has been some concern that antisocial personality disorder may be underdiagnosed in females, particularly because of the emphasis on aggressive items in the definition of conduct disorder.

Differential Diagnosis

The diagnosis of antisocial personality disorder is not given to individuals younger than 18 years and is given only if there is evidence of conduct disorder before age 15 years. For individuals older than 18 years, a diagnosis of conduct disorder is given only if the criteria for antisocial personality disorder are not met.

Substance use disorders. When antisocial behavior in an adult is associated with a substance use disorder, the diagnosis of antisocial personality disorder is not made unless the signs of antisocial personality disorder were also present in childhood and have continued into adulthood. When substance use and antisocial behavior both began in childhood and continued into adulthood, both a substance use disorder and antisocial personality disorder should be diagnosed if the criteria for both are met, even though some antisocial acts may be a consequence of the substance use disorder (e.g., illegal selling of drugs, thefts to obtain money for drugs).

Schizophrenia and bipolar disorders. Antisocial behavior that occurs exclusively during the course of schizophrenia or a bipolar disorder should not be diagnosed as antisocial personality disorder.

Other personality disorders. Other personality disorders may be confused with antisocial personality disorder because they have certain features in common. It is therefore important to distinguish among these disorders based on differences in their characteristic features. However, if an individual has personality features that meet criteria for one or more personality disorders in addition to antisocial personality disorder, all can be diagnosed. Individuals with antisocial personality disorder and narcissistic personality disorder share a tendency to be tough-minded, glib, superficial, exploitative, and lack empathy. However,

narcissistic personality disorder does not include characteristics of impulsivity, aggression, and deceit. In addition, individuals with antisocial personality disorder may not be as needy of the admiration and envy of others, and persons with narcissistic personality disorder usually lack the history of conduct disorder in childhood or criminal behavior in adulthood. Individuals with antisocial personality disorder and histrionic personality disorder share a tendency to be impulsive, superficial, excitement seeking, reckless, seductive, and manipulative, but persons with histrionic personality disorder tend to be more exaggerated in their emotions and do not characteristically engage in antisocial behaviors. Individuals with histrionic and borderline personality disorders are manipulative to gain nurturance, whereas those with antisocial personality disorder are manipulative to gain profit, power, or some other material gratification. Individuals with antisocial personality disorder tend to be less emotionally unstable and more aggressive than those with borderline personality disorder. Although antisocial behavior may be

present in some individuals with paranoid personality disorder, it is not usually motivated by a desire for personal gain or to exploit others as in antisocial personality disorder, but rather is more often attributable to a desire for revenge.

Criminal behavior not associated with a mental disorder. Antisocial personality disorder must be distinguished from antisocial behavior not due to a mental disorder, for example, criminal behavior undertaken for gain that is not accompanied by the personality features characteristic of this disorder. In these cases, the condition adult antisocial behavior may be coded (see “Other Conditions That May Be a Focus of Clinical Attention”).

Comorbidity

Individuals with antisocial personality disorder may also experience dysphoria, including complaints of tension, inability to tolerate boredom, and depressed mood. They may have associated anxiety disorders, mood disorders, substance use disorders, somatic symptom disorder, and gambling disorder. Individuals with antisocial personality disorder also often have personality features that meet criteria for other personality disorders, particularly borderline, histrionic, and narcissistic personality disorders. The likelihood of developing antisocial personality disorder in adult life is increased if the individual experienced childhood onset of conduct disorder (before age 10 years) and accompanying attention-deficit/hyperactivity disorder.

Borderline Personality Disorder

Diagnostic Criteria

F60.3

A pervasive pattern of instability of interpersonal relationships, self-image, and affects, and marked impulsivity, beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:

1. Frantic efforts to avoid real or imagined abandonment. (**Note:** Do not include suicidal or self-mutilating behavior covered in Criterion 5.)
2. A pattern of unstable and intense interpersonal relationships characterized by alternating between extremes of idealization and devaluation.
3. Identity disturbance: markedly and persistently unstable self-image or sense of self.
4. Impulsivity in at least two areas that are potentially self-damaging (e.g., spending, sex, substance abuse, reckless driving, binge eating). (**Note:** Do not include suicidal or self-mutilating behavior covered in Criterion 5.)
5. Recurrent suicidal behavior, gestures, or threats, or self-mutilating behavior.

6. Affective instability due to a marked reactivity of mood (e.g., intense episodic dysphoria, irritability, or anxiety usually lasting a few hours and only rarely more

than a few days).

7. Chronic feelings of emptiness.
 8. Inappropriate, intense anger or difficulty controlling anger (e.g., frequent displays of temper, constant anger, recurrent physical fights).
 9. Transient, stress-related paranoid ideation or severe dissociative symptoms.
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Diagnostic Features

The essential feature of borderline personality disorder is a pervasive pattern of instability of interpersonal relationships, self-image, and affects, and marked impulsivity that begins by early adulthood and is present in a variety of contexts.

Individuals with borderline personality disorder make frantic efforts to avoid real or imagined abandonment (Criterion 1). The perception of impending separation or rejection, or the loss of external structure, can lead to profound changes in self-image, affect, cognition, and behavior. These individuals are very sensitive to environmental circumstances. They experience intense abandonment fears and inappropriate anger even when faced with a realistic time-limited separation or when there are unavoidable changes in plans (e.g., sudden despair in reaction to a clinician's announcing the end of the hour; panic or fury when someone important to them is just a few minutes late or must cancel an appointment). They may believe that this "abandonment" implies they are "bad." These abandonment fears are related to an intolerance of being alone and a need to have other people with them. Their frantic efforts to avoid abandonment may include impulsive actions such as self-mutilating or suicidal behaviors, which are described separately in Criterion 5 (see also "Association With Suicidal Thoughts or Behavior").

Individuals with borderline personality disorder have a pattern of unstable and intense relationships (Criterion 2). They may idealize potential caregivers or lovers at the first or second meeting, demand to spend a lot of time together, and share the most intimate details early in a relationship. However, they may switch quickly from idealizing other people to devaluing them, feeling that the other person does not care enough, does not give enough, or is not "there" enough. These individuals can empathize with and nurture other people, but only with the expectation that the other person will "be there" in return to meet their own needs on demand. These individuals are prone to sudden and dramatic shifts in their view of others, who may alternatively be seen as beneficent supports or as cruelly punitive. Such shifts often reflect disillusionment with a caregiver whose nurturing qualities had been idealized or whose rejection or abandonment is expected.

There may be an identity disturbance characterized by markedly and persistently unstable self-image or sense of self (Criterion 3). There are sudden and dramatic shifts in self-image (e.g., suddenly changing from the role of a needy supplicant for help to that of a righteous avenger of past mistreatment). Although they usually have a self-image that is based on the feeling of being bad or evil, individuals with this disorder may at times have feelings that they do not exist at all. This can be both painful and frightening to those with this disorder. Such experiences usually occur in situations in which the individual feels a lack of a meaningful relationship, nurturing, and support. These individuals may show worse performance in unstructured work or school situations. This lack of a full and enduring identity makes it difficult for the individual with borderline personality disorder to identify maladaptive patterns of behavior and can lead to repetitive patterns of troubled relationships.

Individuals with borderline personality disorder display impulsivity in at least two areas that are potentially self-damaging (Criterion 4). They may gamble, spend money irresponsibly, binge eat, abuse substances, engage in unsafe sex, or drive recklessly.

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Individuals with this disorder display recurrent suicidal behavior, gestures, or threats, or self-mutilating behavior (Criterion 5). Recurrent suicidal thoughts or behavior are often the reason that these individuals present for help. These self-destructive acts are usually precipitated by threats of separation or rejection or by expectations that the individual assume increased responsibility. Self-mutilative acts (e.g., cutting or burning) are very common and may occur during periods in which the individual is experiencing dissociative symptoms. These acts often bring relief by reaffirming the individual's ability to feel or by expiating the individual's sense of being evil.

Individuals with borderline personality disorder may display affective instability that is due to a marked reactivity of mood (e.g., intense episodic dysphoria, irritability, or anxiety usually lasting a few hours and only rarely more than a few days) (Criterion 6). The basic dysphoric mood of those with borderline personality disorder is often disrupted by periods of anger, panic, or despair and is rarely relieved by periods of well-being or satisfaction. These episodes may reflect the individual's extreme reactivity to interpersonal stresses.

Individuals with borderline personality disorder may be troubled by chronic feelings of emptiness, which can co-occur with painful feelings of aloneness (Criterion 7). Easily bored, they may frequently seek excitement to avoid their feelings of emptiness.

Individuals with this disorder frequently express inappropriate, intense anger or have difficulty controlling their anger (Criterion 8). They may display extreme sarcasm, enduring bitterness, or verbal outbursts. The anger is often elicited when a caregiver or lover is seen as neglectful, withholding, uncaring, or abandoning. Such expressions of anger are often followed by shame and guilt and contribute to the feeling they have of being evil.

During periods of extreme stress, transient paranoid ideation or dissociative symptoms (e.g., depersonalization) may occur (Criterion 9), but these are generally of insufficient severity or duration to warrant an additional diagnosis. These episodes occur most frequently in response to a real or imagined abandonment. Symptoms tend to be transient, lasting minutes or hours. The real or perceived return of the caregiver's nurturance may result in a remission of symptoms.

Associated Features

Individuals with borderline personality disorder may have a pattern of undermining themselves at the moment a goal is about to be realized (e.g., dropping out of school just before graduation; regressing severely after a discussion of how well therapy is going; destroying a good relationship just when it is clear that the relationship could last). Some individuals develop psychotic-like symptoms (e.g., hallucinations, body-image distortions, ideas of reference, hypnagogic phenomena) during times of stress. Individuals with this disorder may feel more secure with transitional objects (i.e., a pet or inanimate possession) than in interpersonal relationships. Premature death from suicide may occur in individuals with borderline personality disorder, especially in those with co-occurring depressive disorders or substance use disorders.

However, deaths from other causes, such as accidents or illness, are more than twice as common as deaths by suicide in individuals with borderline personality disorder. Physical handicaps may result from self-inflicted abuse behaviors or failed suicide attempts. Recurrent job losses, interrupted education, and separation or divorce are common. Physical and sexual abuse, neglect, hostile conflict, and early parental loss are more common in the childhood histories of those with borderline personality disorder.

Prevalence

The estimated prevalence of borderline personality disorder based on a probability subsample from Part II of the National Comorbidity Survey Replication was 1.4%. The prevalence of borderline personality disorder in the National Epidemiologic Survey on Alcohol and Related Conditions data was 5.9%. A review of seven epidemiological studies (six in

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the United States) found a median prevalence of 2.7%. The prevalence of borderline personality disorder is about 6% in primary care settings, about 10% among individuals seen in outpatient mental health clinics, and about 20% among psychiatric inpatients.

Development and Course

Borderline personality disorder has typically been thought of as an adult-onset disorder. However, it has been found in treatment settings that symptoms in adolescents as young as age 12 or 13 years can meet full criteria for the disorder. It is not yet known what percentage of adults first entering treatment actually have such an early onset of borderline personality disorder.

Borderline personality disorder has long been thought of as a disorder with a poor symptomatic course, which tended to lessen in severity as those with borderline personality disorder entered their 30s and 40s. However, prospective follow-up studies have found that stable remissions of 1–8 years are very common. Impulsive symptoms of borderline personality disorder remit the most rapidly, while affective symptoms remit at a substantially slower rate. In contrast, recovery from borderline personality disorder (i.e., concurrent symptomatic remission and good psychosocial functioning) is more difficult to achieve and less stable over time. Lack of recovery is associated with supporting oneself on disability benefits and suffering from poor physical health.

Risk and Prognostic Factors

Environmental. Borderline personality disorder has also been found to be associated with high rates of various forms of reported childhood abuse and emotional neglect. However, reported rates of sexual abuse are higher in inpatients than in outpatients with this disorder, suggesting that a history of sexual abuse is as much a risk factor for severity of borderline psychopathology as it is for the disorder itself. In addition, an empirically based consensus has arisen that suggests that a childhood history of reported sexual abuse is neither necessary nor sufficient for the development of borderline personality disorder.

Genetic and physiological. Borderline personality disorder is about five times more common

among first-degree biological relatives of those with the disorder than in the general population. There is also an increased familial risk for substance use disorders, anxiety disorders, antisocial personality disorder, and depressive or bipolar disorders.

Culture-Related Diagnostic Issues

The pattern of behavior seen in borderline personality disorder has been identified in many settings around the world. Sociocultural contexts characterized by social demands that evoke attempts at self-affirmation and acceptance by others, ambiguous or conflictual relationships with authority figures, or marked uncertainties in adaptation can foster impulsivity, emotional instability, explosive or aggressive behaviors, and dissociative experiences that are associated with borderline personality disorder or with transient and contextual reactions to those environments that can be confused with borderline personality disorder. Given that psychodynamic, cognitive, behavioral, and mindfulness aspects of models of mind and self vary cross-culturally, symptoms or traits that suggest the presence of borderline personality disorder (e.g., number of sexual partners, shifting between relationships, substance use) must be evaluated in light of cultural norms to make a valid diagnosis.

Sex- and Gender-Related Diagnostic Issues

While borderline personality disorder is more common among women than men in clinical samples, community samples demonstrate no difference in prevalence between men and women. This discrepancy may reflect a higher degree of help-seeking among women,

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leading them to clinical settings. Clinical characteristics of men and women with borderline personality disorder appear to be similar, with potentially a higher degree of externalizing behaviors in boys and men and internalizing behaviors in girls and women.

Association With Suicidal Thoughts or Behavior

In a longitudinal study, impulsive and antisocial behaviors of individuals with borderline personality disorder were associated with increased suicide risk. In a sample of hospitalized patients with borderline personality disorder followed prospectively for 24 years, around 6% died by suicide, compared with 1.4% in a comparison sample of individuals with personality disorders other than borderline personality disorder. A study of individuals with borderline personality disorder followed for 10 years found that recurrent suicidal behavior was a defining characteristic of borderline personality disorder, associated with declining rates of suicide attempts from 79% to 13% over time.

Differential Diagnosis

Depressive and bipolar disorders. Borderline personality disorder often co-occurs with depressive or bipolar disorders, and when criteria for both are met, both should be diagnosed. Because the cross-sectional presentation of borderline personality disorder can be mimicked by an episode of depressive or bipolar disorder, the clinician should avoid giving an additional diagnosis of

borderline personality disorder based only on cross-sectional presentation without having documented that the pattern of behavior had an early onset and a long-standing course.

Separation anxiety disorder in adults. Separation anxiety disorder and borderline personality disorder are characterized by fear of abandonment by loved ones, but problems in identity, self-direction, interpersonal functioning, and impulsivity are additionally central to borderline personality disorder.

Other personality disorders. Other personality disorders may be confused with borderline personality disorder because they have certain features in common. It is therefore important to distinguish among these disorders based on differences in their characteristic features. However, if an individual has personality features that meet criteria for one or more personality disorders in addition to borderline personality disorder, all can be diagnosed. Although histrionic personality disorder can also be characterized by attention seeking, manipulative behavior, and rapidly shifting emotions, borderline personality disorder is distinguished by self-destructiveness, angry disruptions in close relationships, and chronic feelings of deep emptiness and loneliness. Paranoid ideas or illusions may be present in both borderline personality disorder and schizotypal personality disorder, but these symptoms are more transient, interpersonally reactive, and responsive to external structuring in borderline personality disorder. Although paranoid personality disorder and narcissistic personality disorder may also be characterized by an angry reaction to minor stimuli, the relative stability of self-image, as well as the relative lack of physical self-destructiveness, repetitive impulsivity, and profound abandonment concerns, distinguishes these disorders from borderline personality disorder. Although antisocial personality disorder and borderline personality disorder are both characterized by manipulative behavior, individuals with antisocial personality disorder are manipulative to gain profit, power, or some other material gratification, whereas the goal in borderline personality disorder is directed more toward gaining the concern of caretakers. Both dependent personality disorder and borderline personality disorder are characterized by fear of abandonment; however, the individual with borderline personality disorder reacts to abandonment with feelings of emotional emptiness, rage, and demands, whereas the individual with dependent personality disorder reacts with increasing appeasement and submissiveness and urgently seeks a replacement relationship to provide caregiving and support. Borderline personality disorder can

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further be distinguished from dependent personality disorder by the typical pattern of unstable and intense relationships.

Personality change due to another medical condition. Borderline personality disorder must be distinguished from personality change due to another medical condition, in which the traits that emerge are a direct physiological consequence of another medical condition.

Substance use disorders. Borderline personality disorder must also be distinguished from symptoms that may develop in association with persistent substance use.

Identity problems. Borderline personality disorder should be distinguished from an identity problem, which is reserved for identity concerns related to a developmental phase (e.g., adolescence) and does not qualify as a mental disorder. Adolescents and young adults with identity problems (especially when accompanied by substance use) may transiently display

behaviors that misleadingly give the impression of borderline personality disorder. Such situations are characterized by emotional instability, existential dilemmas, uncertainty, anxiety-provoking choices, conflicts about sexual orientation, and competing social pressures to decide on careers.

Comorbidity

Common co-occurring disorders include depressive and bipolar disorders, substance use disorders, anxiety disorders (particularly panic disorder and social anxiety disorder), eating disorders (notably bulimia nervosa and binge-eating disorder), posttraumatic stress disorder, and attention-deficit/hyperactivity disorder. Borderline personality disorder also frequently co-occurs with the other personality disorders.

Histrionic Personality Disorder

Diagnostic Criteria

F60.4

A pervasive pattern of excessive emotionality and attention seeking, beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:

1. Is uncomfortable in situations in which he or she is not the center of attention.
2. Interaction with others is often characterized by inappropriate sexually seductive or provocative behavior.
3. Displays rapidly shifting and shallow expression of emotions.
4. Consistently uses physical appearance to draw attention to self.
5. Has a style of speech that is excessively impressionistic and lacking in detail.
6. Shows self-dramatization, theatricality, and exaggerated expression of emotion.
7. Is suggestible (i.e., easily influenced by others or circumstances).
8. Considers relationships to be more intimate than they actually are.

Diagnostic Features

The essential feature of histrionic personality disorder is pervasive and excessive emotionality and attention-seeking behavior. This pattern begins by early adulthood and is present in a variety of contexts.

Individuals with histrionic personality disorder are uncomfortable or feel unappreciated when they are not the center of attention (Criterion 1). Often lively and dramatic, they tend to draw attention to themselves and may initially charm new acquaintances by their enthusiasm, apparent openness, or flirtatiousness. These qualities wear thin, however, as these individuals continually demand to be the center of attention. They commandeer the

role of “the life of the party.” If they are not the center of attention, they may do something

dramatic (e.g., make up stories, create a scene) to draw the focus of attention to themselves. This need is often apparent in their behavior with a clinician (e.g., being flattering, bringing gifts, providing dramatic descriptions of physical and psychological symptoms that are replaced by new symptoms each visit).

The appearance and behavior of individuals with this disorder are often inappropriately sexually provocative or seductive (Criterion 2). This behavior not only is directed toward persons in whom the individual has a sexual or romantic interest but also occurs in a wide variety of social, occupational, and professional relationships beyond what is appropriate for the social context. Emotional expression may be shallow and rapidly shifting (Criterion 3). Individuals with this disorder consistently use physical appearance to draw attention to themselves (Criterion 4). They are overly concerned with impressing others by their appearance and expend an excessive amount of time, energy, and money on clothes and grooming. They may “fish for compliments” regarding appearance and may be easily and excessively upset by a critical comment about how they look or by a photograph that they regard as unflattering.

These individuals have a style of speech that is excessively impressionistic and lacking in detail (Criterion 5). Strong opinions are expressed with dramatic flair, but underlying rationales are usually vague and diffuse, without supporting facts and details. For example, an individual with histrionic personality disorder may comment that a certain individual is a wonderful human being, yet be unable to provide any specific examples of good qualities to support this opinion. Individuals with this disorder are characterized by self-dramatization, theatricality, and an exaggerated expression of emotion (Criterion 6). They may embarrass friends and acquaintances by an excessive public display of emotions (e.g., embracing casual acquaintances with excessive ardor, sobbing uncontrollably on minor sentimental occasions, having temper tantrums). However, their emotions often seem to be turned on and off too quickly to be deeply felt, which may lead others to accuse the individual of faking these feelings.

Individuals with histrionic personality disorder have a high degree of suggestibility (Criterion 7). Their opinions and feelings are easily influenced by others and by current fads. They may be overly trusting, especially of strong authority figures whom they see as magically solving their problems. They have a tendency to play hunches and to adopt convictions quickly. Individuals with this disorder often consider relationships more intimate than they actually are, describing almost every acquaintance as “my dear, dear friend” or referring to physicians met only once or twice under professional circumstances by their first names (Criterion 8).

Associated Features

Impairment in general tends to be lower in histrionic personality disorder than in many other personality disorders. However, the impairment most associated with histrionic personality disorder appears to be interpersonal in nature. Individuals with histrionic personality disorder have an interpersonal style characterized by social dominance, which can span a spectrum of behaviors that include a “warmer dominance” that can be intrusive in nature (e.g., need to be center of attention; exhibitionistic) to a “colder dominance” that can include arrogant, controlling, and aggressive behaviors. Romantic relationships appear to be particularly impaired, with evidence suggesting that individuals with histrionic personality disorder symptoms are more likely to get divorced or never get married. Individuals with histrionic personality disorder may have difficulty achieving emotional intimacy in romantic or sexual relationships. Individuals